Quantum Mechanics

Leonard Susskind

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Chapter 1

Modern Physics: Quantum Mechanics

This first lecture opens in an unusual way. Instead of beginning with a formal postulate list, Leonard Susskind begins by explaining the style of the course and then turns at once to a deeper claim: quantum mechanics is not classical mechanics with a little noise stirred in. To make that point, he unfolds the subject in the order the lecture itself demands. We begin with classical randomness as a foil, then the two-slit experiment, then reversibility and the cost of looking, then Heisenberg's microscope, and only after that do we take the first abstract step toward complex vector spaces. Where board frames curated through LazyingArt LLC and Video2Book help stabilize equations or layout, they are retained as supporting evidence; the transcript remains the primary source for the sequence and force of the argument.

1.1 Why This Course Starts With Logic, Not Machinery

Susskind begins with the audience. This is not freshman service physics, and it is not a standard undergraduate or graduate course. It is a continuing-education course, and that fact shapes the method. The audience wants the basic ideas quickly, but not cheaply. We are to use equations when necessary, and when we do use them, we are to use the simplest equations that will actually do the job. He even jokes that one might call it quantum mechanics for old people, including himself. The joke matters less than the pedagogical point behind it: we are not here for ornament, but neither are we here for slogans.

That brief preface also explains why the lecture opens with logic rather than machinery. Susskind reminds us that classical mechanics came first in the larger sequence of courses, but quantum mechanics is more fundamental. Classical physics is what we get when quantum mechanics may be approximated by classical logic in the appropriate regime. So instead of starting with Schrödinger's equation or operator formalism, the lecture starts by asking what classical logic gets wrong. That is the right place to begin, because the mathematics introduced later is meant to replace a structure that has already failed.

1.2 Classical Randomness Is Not Quantum Unpredictability

The first task is to separate two ideas that are too easily conflated. Quantum mechanics certainly contains unpredictability. But that does not mean that it is simply classical mechanics plus random

disturbances. To make the distinction sharp, Susskind invents a classical foil.

Imagine an otherwise Newtonian world in which a deity occasionally gives the moon tiny random kicks. Every tenth of a second the dice are thrown; on one outcome the moon is nudged one way, on another it is nudged the other way. This world would be statistically unpredictable, but the unpredictability would be of a familiar classical kind. It would enter by occasional deviations from deterministic law.

Now ask what such randomness would do. It would not merely blur our predictions. It would also jostle conserved quantities. If the kicks are random, then the energy of the system would, step by step, randomly increase or decrease. On average it might drift slowly or perhaps not at all, but exact conservation would be gone. The very form of the randomness would show itself in the energy bookkeeping.

Quantum mechanics does not behave that way. We prepare a system with a precise energy, let it evolve, and then measure again. The later result may be unpredictable in some respects, but the energy comes back exactly the same. No trace remains of a hidden hand knocking the system this way and that. So the lecture's first lesson is already in place: quantum unpredictability is highly special. It is not what one would get by sprinkling classical randomness into Newton's laws.

1.3 The Two-Slit Experiment and the Failure of Additive Probabilities

Susskind's first great example is the two-slit experiment, because it exhibits the weirdness of quantum mechanics in the most direct and least encumbered way. Let us follow the order of the lecture carefully.

We begin with a source of very light particles—photons, electrons, neutrons, anything for which quantum effects are appreciable. They are sent through the apparatus one at a time. This matters. The lecture insists on that sparsity precisely so that no later particle can plausibly be blamed for what happens to the earlier one.

First open only one small hole. A particle passes through and hits the screen in an isolated blip. Repeat the experiment many times and those individual blips build up a smooth profile: a bright central region fading away to the sides. We may call that one-hole distribution $P_1(y)$. Close that hole, open the second one, and repeat. We obtain another profile $P_2(y)$, displaced relative to the first.

At this stage classical logic seems irresistible. If both holes are open, then a particle reaching a point y on the screen must have come either through the first hole or through the second. Therefore the probabilities ought to add:

$$P_{12}(y) = P_1(y) + P_2(y). \tag{1.1}$$

This is not the quantum rule. It is the classical expectation that Susskind wants fully on the table before overturning it.

Now do the real experiment. The particles still come one at a time. The screen still records one blip at a time. But the built-up pattern is no longer the sum of the two one-hole profiles. Instead we get an interference pattern, and at some points the probability vanishes entirely:

$$P_{12}(y) = 0 \qquad \text{even though} \qquad P_1(y) \neq 0, \quad P_2(y) \neq 0. \tag{1.2}$$

This is the shock of destructive interference. With only the first hole open, particles arrive at that point. With only the second hole open, particles also arrive at that point. Open both holes, and no particle ever arrives there.

That is the point where Susskind widens the claim. One could try to invent some elaborate mechanism, some baroque memory hidden in the apparatus, but the phenomenon is too generic for that sort of one-off repair. The difficulty is not a local detail. The difficulty is that classical statistics and classical logic are built on the wrong underlying structure. The additive rule for alternatives has simply failed.

1.4 Reversibility, Information, and the Cost of Looking

Having displayed the failure of additive probabilities, the lecture shifts to a second diagnostic: reversibility. The move is deliberate. Susskind wants a different test, one that is not tied to slit interference, and for that he first backs up to very simple discrete systems.

Consider a two-state system, a coin with states H and T. A deterministic law of motion can take the form

$$H \mapsto H, \qquad T \mapsto T, \tag{1.3}$$

or it can alternate the states,

$$H \mapsto T, \qquad T \mapsto H. \tag{1.4}$$

Now move to a three-state system with heads, tails, and feet:

$$H \mapsto T, \qquad T \mapsto F, \qquad F \mapsto H. \tag{1.5}$$

The important point is not which law we choose. The important point is that in each of these examples the evolution is exact and information-preserving. If we let the system evolve for a long time and then reverse the arrows of the update rule, it retraces its history and returns to the starting state.

This gives Susskind an operational definition of reversibility. We do not need the detailed law; we need only the possibility of reversing it. Evolve for a time t, reverse the rule, evolve for the same time t again. If the system always returns to its starting point, the dynamics is deterministic in the sense he wants.

Now add a tiny bit of classical randomness. Suppose that with some very small probability the system does not follow the prescribed update and instead does something else—for example, it stays put when the rule says move on. Over short times the reversal test may still succeed. But over sufficiently long times a fluctuation will almost certainly occur, and then running backward does not return us uniquely to the initial state. Information has been lost. That is why Susskind says that classical randomness destroys the conservation of information.

He now carries the same test over into quantum mechanics. Here the lecture momentarily wavers in its wording, but the logic is clear: for this thought experiment we no longer need two slits. One slit is enough. Send one electron through the slit. Remove the screen. Let the electron pass through, but do not detect it. Then reverse the law of physics and run the system backward for the same length of time.

What happens? If nothing has looked at the electron, nothing has recorded its location, nothing has disturbed it in the intermediate stage, then the reversal is exact. The electron retraces the reverse trajectory and goes back into the gun. That is already a remarkable statement. The apparent randomness of the one-slit deflection is not the same thing as classical random compounding of errors.

Now alter only one thing. Put in a detector and record where the electron is during the intermediate stage. Then reverse the law and run backward. In that case the exact return fails. The uncertainty does not cancel; it compounds. The reversibility test, which classical gentle measurement would have left untouched, is destroyed by quantum measurement itself.

This is the deeper transition of the lecture. Classical physics allows an arbitrarily gentle observation. One can, in principle, look at a classical system without significantly altering its subsequent behavior. Quantum mechanics does not. Determining the state of the system is never a harmless afterthought.

Susskind then returns to the two-slit experiment and states the same idea in a different form. The interference pattern is present only if nothing in the apparatus or environment records which slit the electron went through. That recorder need not be a human observer. It could be a gas molecule, another electron, or any degree of freedom that remembers afterward which path was taken. Once such path information exists in the environment, the interference pattern disappears and the answer reverts to the classical one: the probabilities add.

1.4.1 Question & Answer

Question. If the particle is deflected by the slit or plate, does the plate not recoil? And if it recoils, does that not already reveal which path the particle took?

Answer. The recoil is real. The issue is whether it is readable. Susskind's answer is qualitative and deliberately anticipates the uncertainty principle before formally deriving it. A detector that is very well localized in position must have a broad momentum distribution. So although an electron gives the detector a small kick, that kick only shifts an already broad momentum distribution by a small amount. Unless the shift is comparable to the width of the distribution, one cannot tell afterward whether the kick occurred.

The lesson is precise enough for our purposes. Recoil does not automatically provide which-path information. It provides readable information only if the shift in detector momentum is large compared with the detector's own intrinsic momentum uncertainty.

1.5 Heisenberg's Microscope and the Momentum of a Photon

At this point Susskind turns explicitly to the uncertainty principle. But again he does not present it first as a formal theorem. He stages it as a demand for physical explanation. Heisenberg may say that position and momentum cannot be simultaneously determined, but Bohr's challenge is: do not merely tell me that x and p fail to commute; show me the physics that forces this obstruction.

The physical setting is measurement by light. To locate a particle under a microscope, we scatter a photon from it, collect the scattered light, and form an image. So the next question is simple: what did Einstein, de Broglie, Bohr, and Heisenberg know about photons that makes this measurement problematic?

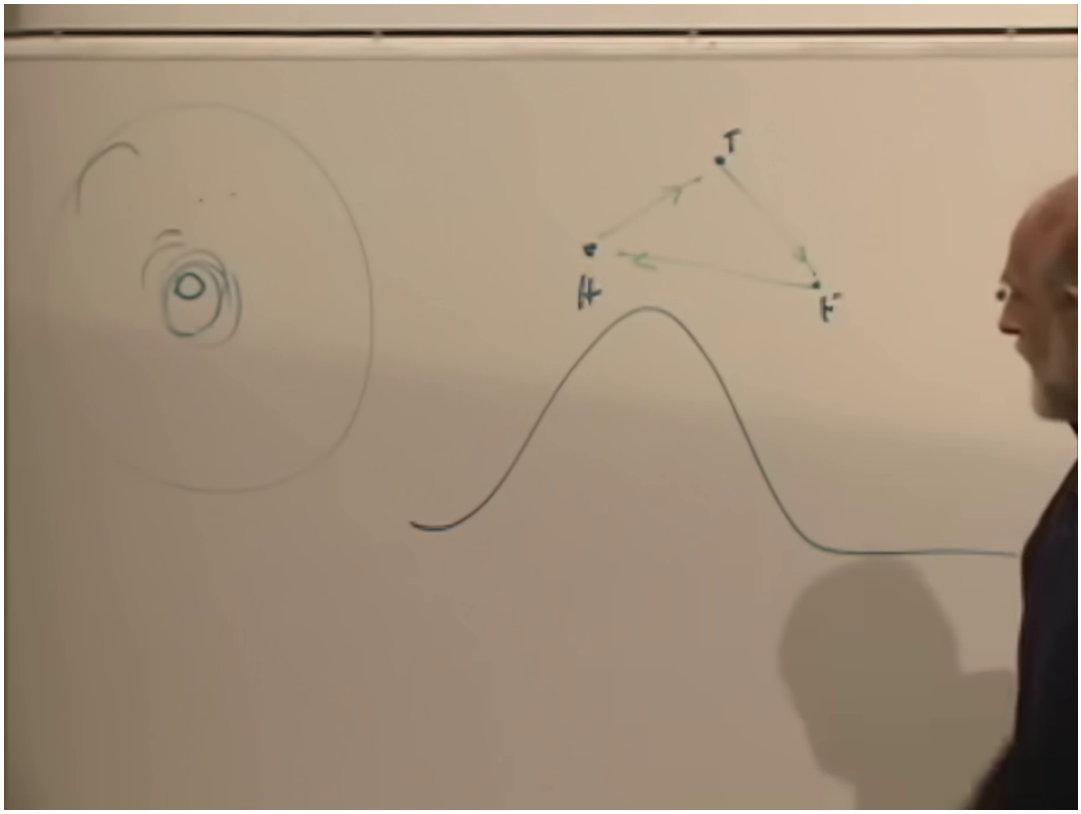


Figure 1.1: Board layout for the recoil argument. The broad central hump is the detector momentum distribution introduced in the answer; the large oval at left and the small triangle-like sketch at upper right are residual board context and should not be overread.

The board snapshot is valuable here because it preserves the cluster of relations as Susskind arranged them.

The first relation is Einstein's photon-energy formula:

$$E = hf = \hbar\omega, \tag{1.6}$$

with

$$\omega = 2\pi f, \qquad \hbar = \frac{h}{2\pi}. \tag{1.7}$$

Susskind then recalls that light carries momentum as well as energy. For ordinary nonrelativistic particles, one has

$$E = \frac{p^2}{2m} = \frac{1}{2}pv, \tag{1.8}$$

so energy and momentum already have the right dimensions to be related by a speed. For light, the relativistic relation is simpler:

$$E = cp, \qquad \text{hence} \qquad p = \frac{E}{c}. \tag{1.9}$$

Now combine this with the photon-energy formula:

$$p = \frac{hf}{c}. \tag{1.10}$$

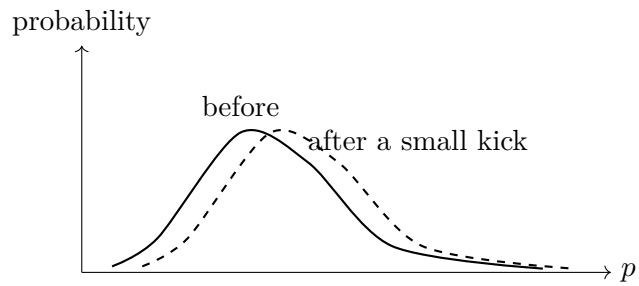


Figure 1.2: Transcript-derived redraw of the detector momentum distribution. The lecture's point is that a small recoil may shift the distribution without making that shift experimentally readable.

To eliminate the frequency we use the kinematics of a wave. If one cycle takes time T, then

$$T = \frac{1}{f}. \tag{1.11}$$

If the wavelength is λ, then in one period the wave advances by one wavelength, so

$$c = \frac{\lambda}{T} = \lambda f, \qquad \text{or equivalently} \qquad f = \frac{c}{\lambda}. \tag{1.12}$$

Substituting into the momentum relation gives the de Broglie formula:

$$p = \frac{hf}{c} = \frac{h}{c}\frac{c}{\lambda} = \frac{h}{\lambda}. \tag{1.13}$$

This is the decisive step: momentum and wavelength are inverse to one another.

The lecture then pauses and leans on the physical meaning of that inverse relation. If we want to form an image that is sharp on a scale Δx, then the wavelength used to form the image must be at most of that order:

$$\lambda \lesssim \Delta x. \tag{1.14}$$

That is not a fancy theorem here; it is ordinary imaging logic. A wavelength much larger than the desired detail necessarily produces a fuzzy image on that larger scale.

But now the trouble appears. If λ must be small, then the photon momentum must be large:

$$p_\gamma = \frac{h}{\lambda}. \tag{1.15}$$

So the very photon that makes a sharp image is also a high-momentum projectile. When it scatters from the particle whose position we are trying to determine, it gives that particle an uncontrollable momentum kick. The lecture does not derive a sharp inequality such as $\Delta x\,\Delta p \geq \hbar/2$, and we should not import one prematurely. What it does justify is the order-of-magnitude statement

$$\Delta p_{\text{kick}} \sim \frac{h}{\lambda}. \tag{1.16}$$

Thus a precise position measurement necessarily produces a large uncertainty in the post-measurement momentum.

Susskind then points out why classical physics seemed to evade this difficulty. In classical physics, light is not quantized. One imagines that one can keep the wavelength fixed while making the

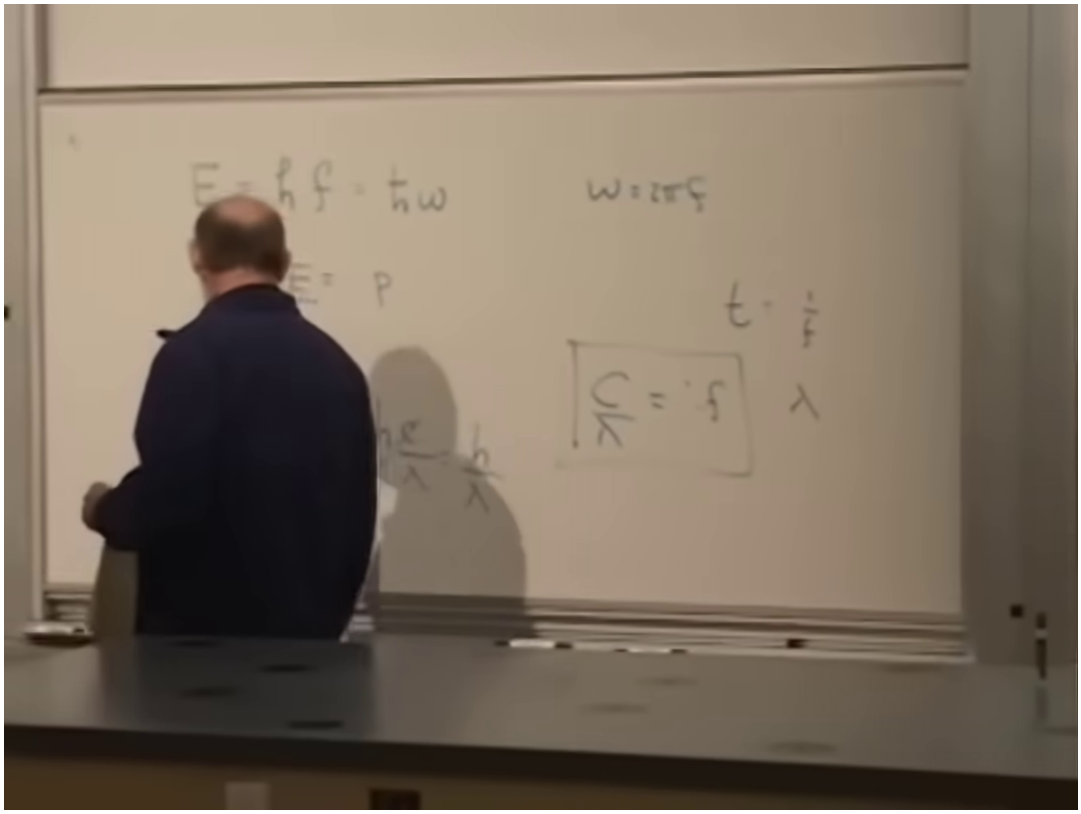


Figure 1.3: Board layout for the photon-energy and wave-kinematics relations leading to the inverse momentum-wavelength law.

intensity arbitrarily small, thereby preserving resolution without delivering a meaningful kick. Quantum mechanics blocks that maneuver. At fixed wavelength there is a minimum packet, namely one photon, and one photon already carries momentum h/λ. There is no infinitely gentle probe of a given short wavelength.

At precisely this stage of the lecture, Susskind remarks that he had erased the key formula and rewrites it. The chalk line itself is partly hidden in the frame, so the reconstruction must be guided by the transcript and the surrounding visible equations.

The physical conclusion is now inescapable. There is no perfectly gentle sharp measurement of position in quantum mechanics. If the measurement is sharp, the momentum is badly disturbed. If the measurement is gentle, the position information is necessarily coarse.

1.5.1 Question & Answer

Question. If a sharp position measurement ruins momentum, how can we measure a particle's velocity at all?

Answer. We measure it gently, and we take our time. The lecture's answer is operational. Measure the position twice, at two different times, and infer the velocity from the displacement divided by the elapsed time. But each position measurement must be gentle, so each must use a long-wavelength

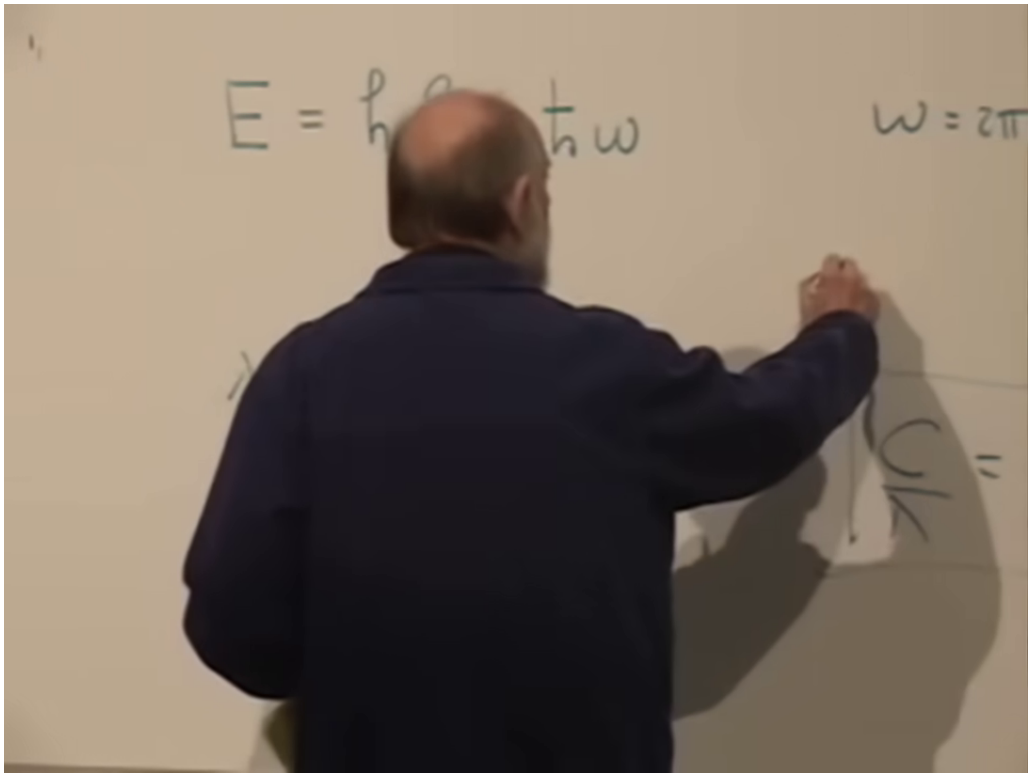


Figure 1.4: Susskind rewriting the erased momentum-wavelength relation. The exact chalk line is partly occluded, but the transcript makes clear that the restored equation is $p = h/\lambda$.

photon.

A long-wavelength photon localizes the particle only coarsely:

$$x_1 = x \pm O(\lambda), \tag{1.17}$$

$$x_2 = x + vt \pm O(\lambda). \tag{1.18}$$

Subtracting the two measurements gives the traveled distance,

$$d = x_2 - x_1 = vt \pm O(\lambda). \tag{1.19}$$

Therefore the inferred velocity has uncertainty

$$\Delta v \sim \frac{\lambda}{t}. \tag{1.20}$$

If we make t very large, then Δv can be made very small even when λ is large. So velocity may indeed be measured accurately. But the price is that the position was never known to better than order λ. This is exactly the complementary tradeoff Susskind wants us to keep in mind. We can know velocity well, but only by giving up precise knowledge of position; or we can localize sharply, but only by badly disturbing the momentum.

1.6 From Classical States to Quantum Vector Spaces

Only after all of these physical examples have done their work does the lecture make its final turn. Susskind says, in effect: now we are ready to ask what is really new in the mathematical language.

In classical mechanics, the state of a system is a point in a set. That already tells us something about the operations available. We can ask whether two sets intersect or unite, but it makes no sense to multiply a point by a number. If the only states are heads and tails, then $3 \times$ heads and $(2 + 4i) \times$ heads are meaningless expressions.

Quantum mechanics will replace this set-like notion of state by a vector-space notion. That should feel radical, and Susskind wants it to feel radical. He even says that at this point it should sound like mumbo jumbo. But the lecture is careful: before asking why physical states should be vectors, it first explains what a vector space is.

Definition 1.1. A complex vector space is a collection of objects for which two operations are defined: any two objects can be added, and any object can be multiplied by an arbitrary complex number, with the result again belonging to the same collection.

Once those operations exist, any linear combination is again an allowed object:

$$\alpha A + \beta B \in V \qquad \text{whenever} \qquad A, B \in V,\ \alpha, \beta \in \mathbb{C}. \tag{1.21}$$

That is the basic algebra the lecture wants us to absorb.

Ordinary geometric pointers provide the familiar analogy. They form a vector space over the real numbers: they may be added, and they may be multiplied by real scalars. But quantum mechanics wants something more general, namely vector spaces over the complex numbers.

The first example Susskind gives is the space of complex-valued functions of a real variable x. Here x itself is an ordinary real variable; what is complex is the value of the function:

$$\psi(x) \in \mathbb{C}. \tag{1.22}$$

If $\psi(x)$ and $\phi(x)$ are such functions, and if α is a complex number, then

$$(\alpha\psi)(x) = \alpha\,\psi(x), \tag{1.23}$$

$$(\psi + \phi)(x) = \psi(x) + \phi(x). \tag{1.24}$$

Both operations produce new complex-valued functions of x. Therefore complex functions of one real variable form a complex vector space. This is the example that will later become central in quantum mechanics.

The lecture also makes an important contrast. Real-valued functions form a vector space over the real numbers, but not over the complex numbers. The reason is immediate: multiplying a real-valued function by i does not keep it real-valued.

The second example is the fixed-length column vector. If we take a column of four arbitrary complex numbers,

$$A = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix}, \qquad B = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \end{pmatrix}, \tag{1.25}$$

then addition is defined entry by entry:

$$A + B = \begin{pmatrix} A_1 + B_1 \\ A_2 + B_2 \\ A_3 + B_3 \\ A_4 + B_4 \end{pmatrix}, \tag{1.26}$$

and multiplication by a complex scalar is equally simple:

$$\alpha A = \begin{pmatrix} \alpha A_1 \\ \alpha A_2 \\ \alpha A_3 \\ \alpha A_4 \end{pmatrix}. \tag{1.27}$$

For a fixed length, these columns form a vector space. But here the lecture is careful about a subtle point: the collection of all columns of all lengths is not a single vector space under these rules, because no addition law has been given between, say, a two-component vector and a three-component one.

Finally, Susskind steps back to the complex numbers themselves. They already form a one-dimensional complex vector space. That makes complex conjugation the next natural operation to inspect. Write

$$z = x + iy, \qquad z^* = x - iy. \tag{1.28}$$

Geometrically, z^* is the reflection of z across the real axis. Algebraically,

$$z^* z = (x + iy)(x - iy) = x^2 + y^2. \tag{1.29}$$

So the product of a number with its conjugate gives the square of its magnitude. Susskind treats this as a basic operation, not a decorative trick, because it is one of the first examples of a mapping that will later be generalized from numbers to vectors.

The closing note of the lecture is important. We are not yet being told in what precise sense physical states are vectors. That comes later. For now we are learning the abstract operations themselves. And that is exactly right: if classical logic is built on sets, then quantum logic will need a new arena, and the lecture ends by introducing the first pieces of that arena.

1.7 Summary

The lecture begins by rejecting the easy idea that quantum mechanics is just classical mechanics with random noise added. Classical random kicks would spoil exact conservation laws and destroy reversibility. Quantum mechanics does neither in that simple way. The two-slit experiment then shows that probabilities do not merely add: with both slits open, there are points where the probability vanishes even though each one-slit experiment separately gives nonzero arrivals there.

The lecture then turns the notion of reversibility into an operational test of determinism. Classical randomness destroys that test; quantum evolution preserves exact reversibility only so long as nothing records intermediate information. That is the deeper lesson behind both the one-slit reversal experiment and the two-slit which-path story: in quantum mechanics, measurement is part of the dynamics.

Heisenberg's microscope then gives the physical mechanism. To resolve position sharply we need short wavelength; short wavelength means large photon momentum; the very photon that makes the image gives the particle an uncontrollable kick. Velocity can still be measured well, but only by very gentle long-wavelength position measurements spread over a long time, and therefore only at the cost of poor position information.

Only after these physical examples have dismantled the classical picture does Susskind make the mathematical pivot. Classical states are points in a set. Quantum states will live in a complex

vector space. Functions, column vectors, and complex numbers themselves are the first examples. That is where the lecture stops: not yet with the full machinery of operators and eigenvalues, but with the clear sense that some new logic is required, and with the first abstract language in which that logic can be expressed.

Chapter 2

Vector Spaces, Superposition, and the First Quantum Postulates

These notes follow Leonard Susskind's second quantum mechanics lecture as a guided argument rather than as a list of facts. The lecture begins by removing a conceptual obstruction left over from the discussion of the uncertainty principle, then turns to the formal language of complex vector spaces, bras and kets, inner products, bases, and finally the first quantum-mechanical postulates in the simple language of coins and dice. The selected lecture frames are kept where the board layout or the chalk geometry matters, but the transcript remains the primary source of order and emphasis. In that spirit we keep explicit credit to Leonard Susskind, together with the lecture-frame curation work of LazyingArt LLC and Video2Book.

2.1 Uncertainty Principle Prelude

The lecture does not begin with vector spaces. It begins by clearing away an objection. Suppose we measure the position of a particle only very poorly, to within a region of size δ, and then wait a very long time T before measuring its position again, again only to within a region of size δ. If the centers of the two fuzzy regions are separated by a distance L, then one seems to know the average velocity very accurately:

$$v_{\text{avg}} \sim \frac{L}{T}, \tag{2.1}$$

$$\Delta v_{\text{avg}} \sim \frac{\delta}{T}. \tag{2.2}$$

Since T may be made arbitrarily large, the uncertainty δ/T may be made arbitrarily small. At first sight this looks like a way of knowing position only to accuracy δ, while knowing velocity essentially exactly.

That, however, is not the quantity that the uncertainty principle is about. What we wanted was the position and momentum of the same state of the particle, or at least the momentum just before the position measurement. The two-measurement trick does not return that momentum.

2.1.1 Question & Answer

Can two imprecise position measurements defeat the uncertainty principle?

No. The first position measurement already disturbs the momentum. To localize a particle to scale δ, one must probe it with radiation of wavelength of order δ or smaller, and this produces a momentum kick of order

$$\Delta p \sim \frac{\hbar}{\delta}. \tag{2.3}$$

So the later motion reveals the velocity after the first measurement, not the original velocity of the undisturbed state. The uncertainty principle is not being evaded; rather, the wrong velocity is being inferred.

The lecture also presses the point of repeatability. One may say, "I think I measured the velocity," but if one tries to check that velocity one changes it again by an amount of order $\hbar/\delta$. In classical physics one imagines measuring position arbitrarily gently and leaving the velocity alone. In quantum mechanics, not so.

With that conceptual obstacle removed, the lecture pivots very explicitly: now we are ready to begin.

2.2 Complex Numbers as the Simplest Complex Vector Space

The next step is intentionally modest. Susskind says he is not trying to teach complex numbers; he wants only a very quick review, just for notational reasons. The subject of linear vector spaces may be dry, he says, but the good news is that it is also easy. So we begin with the easiest complex vector space of all: the complex numbers themselves.

A complex number is

$$z = x + iy, \qquad x, y \in \mathbb{R}, \tag{2.4}$$

and its complex conjugate is

$$z^* = x - iy. \tag{2.5}$$

Geometrically, z^* is the reflection of z across the real axis in the complex plane. The lecture briefly reminds us not to confuse that plane with the vector space we ultimately care about. The point of the example is structural: complex numbers can be added, and they can be multiplied by complex numbers, so they already form a one-dimensional complex vector space.

At this stage the bra-ket notation is introduced in embryo. In the one-dimensional case one may think of z as a ket and z^* as the corresponding bra. This is not yet deep; it is just a simple rehearsal for what will happen in more complicated vector spaces.

The lecture then isolates one algebraic fact that will return later in a more abstract form. If

$$z_1 = x_1 + iy_1, \qquad z_2 = x_2 + iy_2, \tag{2.6}$$

then

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2). \tag{2.7}$$

Complex conjugation flips only the sign of the imaginary part, and therefore

$$(z_1 z_2)^* = z_1^* z_2^*. \tag{2.8}$$

A related identity, written in the notation that foreshadows bras and kets, is

$$(z_1^* z_2)^* = z_2^* z_1. \tag{2.9}$$

For ordinary numbers the order is harmless, since the factors commute. But the lecture lingers on this order-swapping because it prepares the listener for the later statement that interchanging bra and ket corresponds to complex conjugation.

Only after this warm-up does the lecture state the abstract idea. A complex vector space is made of objects that can be added and multiplied by arbitrary complex numbers, and that stay within the space under those operations. In particular, if A and B are vectors and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha A + \beta B \tag{2.10}$$

is again a vector of the same space.

That definition is deliberately spare. Vectors are not defined as arrows in physical space. They are defined by the operations we are allowed to perform on them. Ordinary arrows form only a real vector space, because multiplication by a general complex number is not defined for them. The real numbers form a real vector space, but not a complex one. Even the positive real numbers fail to form a real vector space, because multiplication by a negative number takes us out of the set. This operational point of view is the real beginning of the mathematics.

2.3 Complex Vector Spaces: Examples, Dimension, and Abstract Notation

Once the definition is on the table, the lecture immediately asks for examples. The first important example is the space of complex-valued functions of a real variable:

$$\psi(x) = \psi_{\mathrm{Re}}(x) + i\,\psi_{\mathrm{Im}}(x). \tag{2.11}$$

Here x is real, but the value of $\psi(x)$ is complex. If we multiply $\psi(x)$ by a complex number, we get another complex-valued function. If we add two such functions, we get another. In this mathematical sense, functions are vectors.

The second example is the column vector. A column vector is really a function of a discrete index. In the four-component case we may write

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}, \tag{2.12}$$

and define the vector-space operations component by component:

$$|a\rangle + |b\rangle = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \\ a_4 + b_4 \end{pmatrix}, \tag{2.13}$$

$$\alpha|a\rangle = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \alpha a_3 \\ \alpha a_4 \end{pmatrix}. \tag{2.14}$$

Thus column vectors are simply the discrete analog of complex functions. The lecture is careful here: if one is used to thinking of vectors only as arrows, one may not be used to thinking of functions as vectors. But that unfamiliarity is psychological, not mathematical.

Having supplied examples, the lecture turns to dimension. The dimension of a vector space is defined as the minimum number of vectors needed so that every vector in the space can be written as a linear combination of them. On a line, one nonzero vector suffices. On the blackboard plane, two non-collinear vectors suffice. This is the point at which the lecture draws the spanning picture.

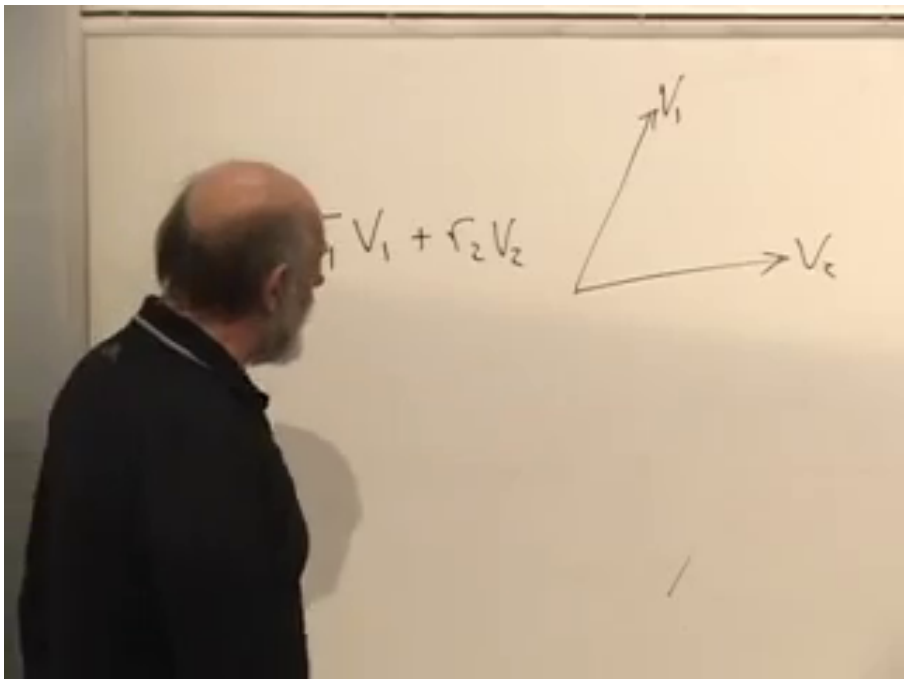


Figure 2.1: Linear combination from two spanning vectors. The chalk labels look closer to uppercase V_1, V_2, while the notes normalize the mathematics to lowercase v_1, v_2.

The chalk expression at the left of the board is partially obscured, but the clean mathematical statement is straightforward: if v_1 and v_2 are not on the same line, then every vector in the plane can be written as

$$r_1 v_1 + r_2 v_2. \tag{2.15}$$

A third vector v_3 drawn in the wedge between them would be redundant, because it can already be expressed in terms of the first two:

$$v_3 = r_1 v_1 + r_2 v_2. \tag{2.16}$$

The selected frame does not actually show v_3; that vector comes from the spoken argument and is drawn only in the cleaned reconstruction below.

From here the lecture runs through a sequence of dimensions. A four-component column space is four-dimensional. A one-component column space is just the one-dimensional complex vector space of ordinary complex numbers. Smooth functions of a real variable form an infinite-dimensional vector space, because no finite set of functions can be combined to generate every smooth function.

Only after that progression does the lecture shift notation. The point is to detach the reader from any one representation and move to the abstract vector itself.

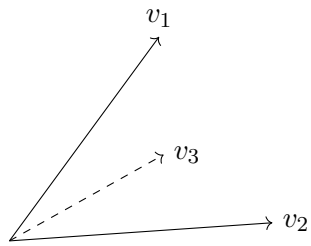


Figure 2.2: Clean reconstruction of the spanning argument. The dashed vector v_3 is reconstructed from the lecture discussion and is not visibly completed in the screenshot.

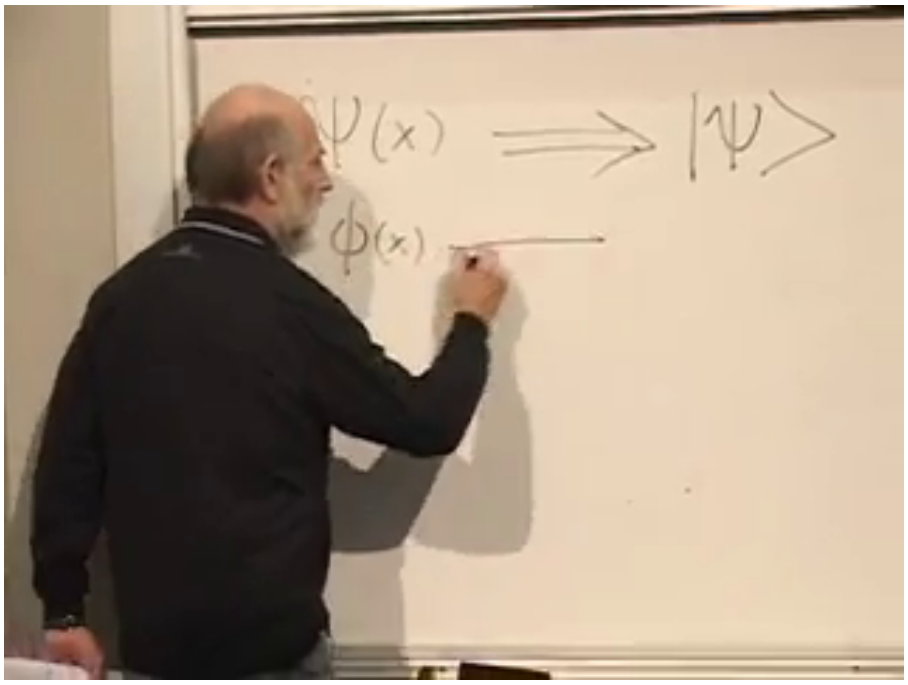


Figure 2.3: Mapping $\psi(x)$ to ket notation. The upper relation is fully visible on the board; the lower line is only partially visible and is completed cautiously in the text.

The board shows the completed line

$$\psi(x) \Rightarrow |\psi\rangle \tag{2.17}$$

and begins a second analogous line. In the text we complete that second line cautiously as

$$\phi(x) \mapsto |\phi\rangle. \tag{2.18}$$

The arrow here is not a strict equality sign. It is a change of viewpoint. We are now to think of the function as a vector in an abstract complex vector space. In the same way, a column of entries may be thought of as the abstract vector determined by those entries. This is the point at which the lecture's working notation settles into place.

2.4 Dual Vectors and Inner Products

Once abstract vectors are in place, the lecture introduces the dual vector space. For the purposes of this course, the dual space is the complex-conjugate partner of the original ket space. For every ket $|A\rangle$ there is a corresponding bra $\langle A|$. Mathematicians can define duals more generally, but the physical case under discussion is the complex-conjugate one.

In the case of functions, the bra corresponding to $|\psi\rangle$ is represented by the complex-conjugate function. In the case of column vectors, the corresponding bra is a row vector whose entries are the complex conjugates of the entries of the column. The lecture emphasizes that some of this is notation, but notation chosen to keep track of which objects live in the ket space and which in the bra space.

The first nontrivial rule concerns scalar multiplication. If $\langle A|$ corresponds to $|A\rangle$, then the bra associated with $\alpha|A\rangle$ is

$$\alpha^*\langle A|, \tag{2.19}$$

not $\alpha\langle A|$. This is just the familiar rule that complex conjugation of a product conjugates both factors. The lecture even pauses over a student's question about whether the α^* has to be written on the right; for ordinary numbers it does not matter, though later it will matter once operators are introduced.

The next construction is the inner product. Susskind introduces it by analogy with the ordinary dot product of real vectors. For ordinary vectors,

$$a \cdot b = a_x b_x + a_y b_y + a_z b_z + \cdots . \tag{2.20}$$

The general inner product $\langle A|B\rangle$ is the complex analogue of this idea, but now one must decide which vector is the bra and which is the ket. Once that choice is made, the inner product is required to satisfy the following properties:

$$\langle A|(\beta B)\rangle = \beta\langle A|B\rangle, \tag{2.21}$$

$$\langle A|(B+C)\rangle = \langle A|B\rangle + \langle A|C\rangle, \tag{2.22}$$

$$\langle A|B\rangle = \langle B|A\rangle^*, \tag{2.23}$$

$$\langle A|A\rangle > 0 \qquad \text{for } |A\rangle \neq 0. \tag{2.24}$$

From the third relation we learn immediately that $\langle A|A\rangle$ is real, since it equals its own complex conjugate. But the lecture insists on something subtler: positivity does *not* follow from the other properties. It is an additional part of the definition. Susskind even mentions relativity as an example where a bilinear form on a real vector space need not be positive.

The scalar model for positivity is the elementary identity

$$z^* z = (x - iy)(x + iy) = x^2 + y^2, \tag{2.25}$$

which is real and nonnegative. That is the pattern we carry into vector spaces.

The lecture then gives the concrete inner products we will actually use. For complex-valued functions of a real variable,

$$\langle\phi|\psi\rangle = \int dx\, \phi^*(x)\psi(x). \tag{2.26}$$

The lecture allows that functions of several real variables would simply require several integrations, but keeps the arguments real and the example simple.

For column vectors, if

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \qquad \langle b| = \begin{pmatrix} b_1^* & b_2^* & b_3^* \end{pmatrix}, \tag{2.27}$$

then

$$\langle b|a\rangle = b_1^* a_1 + b_2^* a_2 + b_3^* a_3 = \sum_i b_i^* a_i. \tag{2.28}$$

The lecture briefly corrects itself on the blackboard here, because one must remember that the row entries are the conjugates. In the final notes it is best simply to present the corrected form.

2.5 Orthonormal Bases and Expansion Coefficients

After the break the lecture becomes constructive again. We now introduce basis vectors, and again the order of ideas matters. First come unit length, then orthogonality, then orthonormality, and only then the practical use of the basis.

Let $\{|b_i\rangle\}_{i=1}^d$ be a collection of vectors in a d-dimensional vector space. They are normalized if

$$\langle b_i|b_i\rangle = 1 \qquad \text{for each } i, \tag{2.29}$$

with no sum on i. They are orthogonal if

$$\langle b_i|b_j\rangle = 0 \qquad \text{for } i \neq j. \tag{2.30}$$

Together these two conditions are summarized by

$$\langle b_i|b_j\rangle = \delta_{ij}, \tag{2.31}$$

where the Kronecker delta is defined by

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{2.32}$$

A set of vectors satisfying this relation is orthonormal, and if there are d of them in a d-dimensional space, they form a basis.

The lecture illustrates this with the standard basis for a four-component column space. In cleaned notation we may write

$$|b_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |b_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |b_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |b_4\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{2.33}$$

The corresponding bras are the conjugate row vectors. These vectors are normalized because each has only one nonzero entry, and they are orthogonal because two distinct basis vectors never overlap in the same slot. The lecture corrects a bra-versus-ket slip while discussing this example; here we simply present the corrected basis cleanly.

Now comes the practical payoff. Any vector $|v\rangle$ in a d-dimensional space can be expanded as

$$|v\rangle = \sum_{i=1}^{d} v_i |b_i\rangle. \tag{2.34}$$

The coefficients v_i are determined by taking inner products with the basis vectors. If we multiply on the left by $\langle b_j|$, we obtain

$$\langle b_j|v\rangle = \sum_{i=1}^{d} v_i \langle b_j|b_i\rangle \tag{2.35}$$

$$= \sum_{i=1}^{d} v_i \delta_{ji} \tag{2.36}$$

$$= v_j. \tag{2.37}$$

So the expansion coefficients are

$$v_j = \langle b_j|v\rangle. \tag{2.38}$$

This is the sequence the lecture wants us to see: first expand, then take the inner product, then use orthonormality, then discover the coefficients. Substituting back gives the compact formula that will recur throughout the course:

$$|v\rangle = \sum_{i=1}^{d} |b_i\rangle\langle b_i|v\rangle. \tag{2.39}$$

In the lecture the local blackboard derivation is a little messy; in the notes it is best presented in this cleaned form while preserving the same logical order.

2.6 Quantum States as Vectors: Coin, Die, Superposition, and Normalization

Only after the mathematics has been assembled does the lecture turn to physical meaning. The contrast with classical mechanics is drawn sharply. In classical mechanics, the space of states is a set. One speaks of unions and intersections, of AND and OR. In quantum mechanics, states are vectors, and the algebra is altogether different.

The lecture begins with the coin. Classically there are two states, heads and tails. One may imagine labeling them by symbols, but there would be no point in taking linear combinations of those labels. In quantum mechanics, by contrast, we assign to the two distinguishable configurations two orthogonal basis vectors, which we denote by $|H\rangle$ and $|T\rangle$. These are not lengths $+1$ and -1; they are simply names for two basis states. Their mathematical properties are

$$\langle H|T\rangle = 0, \qquad \langle H|H\rangle = \langle T|T\rangle = 1. \tag{2.40}$$

The orthogonality is not arbitrary. It is a postulate with physical meaning: states that can be distinguished by a single clean measurement are represented by orthogonal vectors. For the coin, the clean measurement is simply to look and ask whether the outcome is heads or tails. The same principle later applies to spin up and spin down, which is the real quantum-mechanical example lurking behind the coin.

Once the basis states are in place, the new quantum feature appears. The expression

$$|cc\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle \tag{2.41}$$

is meaningful. Susskind calls this the state of a "confused coin." In classical physics the linear combination would mean nothing at all. In quantum mechanics it is a genuine state, different from both $|H\rangle$ and $|T\rangle$. The same idea applies to the die:

$$|cd\rangle = \sum_{i=1}^{6} \alpha_i|i\rangle, \tag{2.42}$$

where $|1\rangle, \ldots, |6\rangle$ are six orthonormal basis states corresponding to the six distinguishable outcomes of a die. Again, the labels $1, \ldots, 6$ are names, not vector lengths.

2.6.1 Question & Answer

How do the amplitudes become probabilities?

At this stage the lecture treats this as a postulate, not as something derived. If the confused coin is in the state

$$|cc\rangle = \alpha_H|H\rangle + \alpha_T|T\rangle, \tag{2.43}$$

then the probability of finding heads is

$$p_H = \alpha_H^*\alpha_H = |\alpha_H|^2, \tag{2.44}$$

and the probability of finding tails is

$$p_T = \alpha_T^*\alpha_T = |\alpha_T|^2. \tag{2.45}$$

This is the Born rule in its simplest discrete form.

The lecture is very careful here. This is not said to be the whole meaning of the amplitudes α_H and α_T. There is more in them than probabilities. But one feature of their meaning is already fixed: the squared moduli are the probabilities for measurement outcomes in the chosen orthonormal basis.

Operationally, these probabilities are understood from large ensembles of identically prepared systems. One prepares many copies of the same state $|cc\rangle$, performs the same measurement on each copy, and reads probabilities from the frequencies of outcomes. That is the level at which the postulate meets experiment.

Normalization now becomes unavoidable:

$$|\alpha_H|^2 + |\alpha_T|^2 = 1. \tag{2.46}$$

In bra-ket language this is simply

$$\langle cc|cc\rangle = 1. \tag{2.47}$$

Likewise, for the confused die,

$$\sum_{i=1}^{6} |\alpha_i|^2 = 1, \qquad \langle cd|cd\rangle = 1. \tag{2.48}$$

Thus the lecture arrives at another postulate: physically admissible states are normalized.

A useful explicit example is the equal-probability state. If the confused coin gives heads and tails with equal probability, then

$$|\alpha_H| = |\alpha_T| = \frac{1}{\sqrt{2}}. \tag{2.49}$$

But the amplitudes are complex, so the general equal-probability form is

$$\alpha_H = \frac{e^{i\theta_H}}{\sqrt{2}}, \qquad \alpha_T = \frac{e^{i\theta_T}}{\sqrt{2}}. \tag{2.50}$$

Similarly, an equal-probability die state may be written as

$$\alpha_i = \frac{e^{i\theta_i}}{\sqrt{6}}. \tag{2.51}$$

The lecture stresses exactly this point: equal probability does not mean the amplitudes are literally $1/\sqrt{2}$ or $1/\sqrt{6}$; it means they have those moduli, up to unit complex phases.

By the end of this section the first postulates have been stated in the language of simple discrete systems:

1. Clearly distinguishable configurations correspond to orthogonal vectors.
2. Physical states are normalized.
3. In a basis of distinguishable states, probabilities are given by squared moduli of amplitudes.

The deeper meaning of superposition is not yet exhausted, but the lecture has now established its first operational consequences.

2.7 Operators as the Next Mathematical Step

There is one more turn before the lecture ends. Susskind says he would like to get to operators, and he does, but only as an opening move. The subject is introduced first through ordinary blackboard vectors, before any full physical interpretation is attached to it.

The sketch is intentionally elementary: a vertical guide and two ordinary vectors. The question is what kinds of operations one can do on all vectors in such a space. One may rotate every vector by the same angle, double every vector while keeping its direction fixed, or reflect every vector about an axis. These are genuine operations on the vector space.

Not every operation is linear. The lecture gives a deliberately simple counterexample: replace every vector of length 2 by one of length 4, every vector of length 3 by one of length 9, and so on. That is certainly an operation, but it is not linear.

A linear operator, denoted with a hat, is characterized by two properties:

$$\hat{L}(\alpha a) = \alpha\, \hat{L}(a), \tag{2.52}$$

$$\hat{L}(a + b) = \hat{L}(a) + \hat{L}(b). \tag{2.53}$$

The lecture checks these properties on the simplest geometric example. If R denotes rotation by a fixed angle, then

$$R(2a) = 2R(a), \tag{2.54}$$

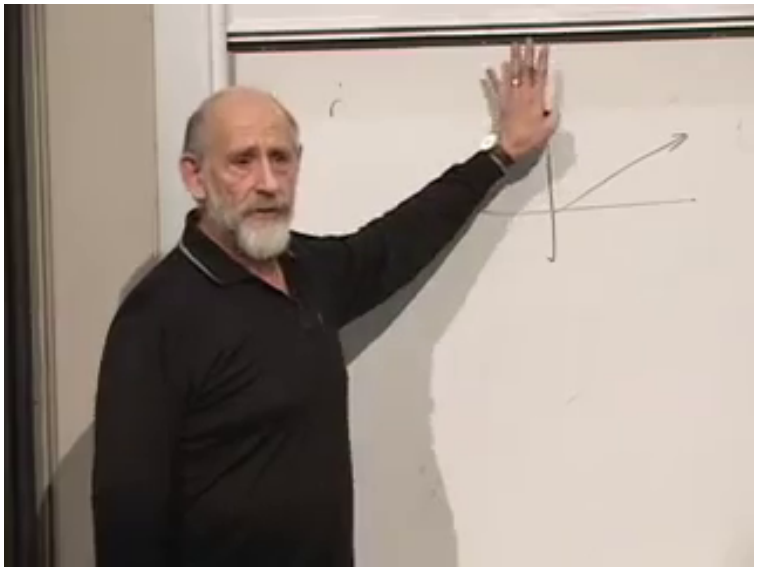


Figure 2.4: Ordinary vectors drawn on the blackboard. The figure is diagrammatic rather than symbolic, and serves as evidence for the geometric setup used to introduce operators.

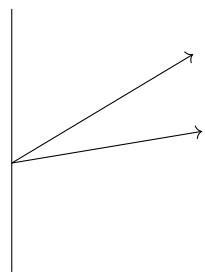


Figure 2.5: Clean redraw of the ordinary-vector sketch used in the lecture's first discussion of operations on vectors.

and if $a+b$ is drawn as the diagonal of the usual parallelogram, then rotating the whole parallelogram shows that

$$R(a+b) = R(a) + R(b). \tag{2.55}$$

So fixed-angle rotation is linear. Uniform doubling is linear. Reflection is linear. Squaring the length is not.

This final section is explicitly transitional. The lecture reaches just far enough to state the physical slogan: states are represented by vectors, but measurable quantities are represented by linear operators. In that sense observables such as the "headsness" or "tailsness" of the coin belong on the operator side of the formalism. The actual identification of observables with operators is postponed to the next lecture.

2.8 Summary

This lecture unfolds in two layers. First it clears away a conceptual misunderstanding about the uncertainty principle. Then it builds the mathematical language in which the first quantum

postulates can be stated.

We began by seeing why two poor position measurements far apart in time do not reveal the original momentum of a particle: the first position measurement itself changes the momentum by an amount of order $\hbar/\delta$. From there the lecture reset and moved through complex numbers, complex vector spaces, concrete examples in functions and column vectors, the notion of dimension, the shift to abstract kets, the dual space, inner products, orthonormal bases, and the extraction of expansion coefficients.

Only after that apparatus was in place did the lecture turn to physics. Classical states form a set; quantum states form a vector space. Distinguishable configurations correspond to orthogonal vectors, superpositions are new states, normalization encodes total probability one, and squared amplitudes give measurement probabilities in a basis of distinguishable states. Finally, the lecture opened the next mathematical subject, operators, just far enough to show the direction of travel. The formalism is still abstract here, but the chapter ends with the same forward pressure as the lecture itself: we are now very close to the point where these rules begin to make concrete predictions.

Chapter 3

Linear Operators, Hermitian Observables, and a Particle on a Line

We begin where the lecture begins: not with a new subject, but with a review. The mathematics of quantum mechanics is linear algebra, and the first order of business is to keep two ideas distinct. States are vectors. What we measure are not states, but observables, and observables are represented by Hermitian linear operators acting on those vectors. The lecture then unfolds in a precise order: first the language of operators and bases, then the passage to matrices, then Hermitian operators and eigenvectors, then the postulates, and only after that the first truly infinite-dimensional example, a particle moving on a line. A few board frames are preserved here as visual evidence, with credit to Leonard Susskind, LazyingArt LLC, and Video2Book.

3.1 Linear Operators, Matrix Elements, and Bases

The opening insistence is worth preserving: quantum mechanics is linear algebra. We start with a vector space, because states will live in that space, and we introduce linear operators, because measurable quantities will eventually be represented by a special class of them.

A linear operator is denoted by a capital letter with a hat. It acts on a ket and produces another ket:

$$\hat{K}|A\rangle = |C\rangle. \tag{3.1}$$

This much is still abstract, but the inner product immediately allows us to turn the action of an operator into an ordinary number. If we let $\hat{K}$ act on $|A\rangle$, and then take the inner product with $\langle B|$, we obtain the matrix element

$$\langle B|\hat{K}|A\rangle. \tag{3.2}$$

The notation is meant to be read in that order: first $\hat{K}$ acts on $|A\rangle$, then we take the inner product with $\langle B|$.

Now choose an orthonormal basis $\{|n\rangle\}$. If the vector space has dimension D, then the basis contains D independent vectors, mutually orthogonal and of unit norm. The lecture pauses here to draw an ordinary three-dimensional picture, not because the quantum vector space is literally $\mathbb{R}^3$, but because complex vector spaces are hard to draw and one still wants a geometric picture of what a basis is. There can be many different bases, tilted relative to one another, and that visual fact will later become physically important.

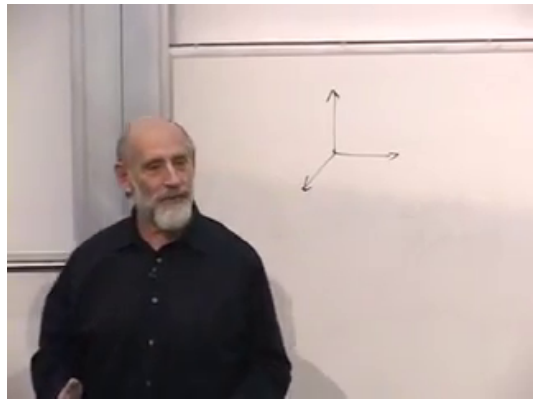


Figure 3.1: Ordinary three-vector basis sketch on the board.

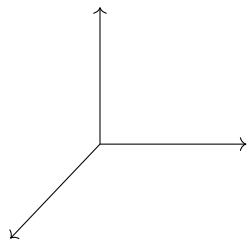


Figure 3.2: Transcript-based redraw of the basis sketch. The point is not the literal geometry, but the existence of distinct orthogonal basis directions.

Once a basis has been chosen, we can tabulate the operator by its matrix elements between basis vectors. The board isolates this notation very clearly:

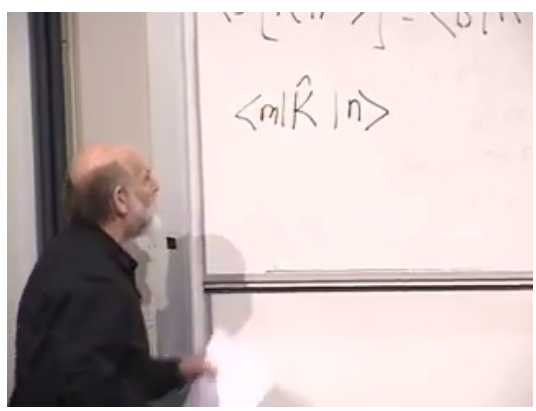


Figure 3.3: Matrix element between basis vectors.

$$\langle m|\hat{K}|n\rangle. \tag{3.3}$$

If we define

$$K_{mn} := \langle m|\hat{K}|n\rangle, \tag{3.4}$$

then the operator is represented, in that basis, by a square matrix of numbers. The lecture emphasizes the simple counting: if there are D basis vectors, there are D^2 such matrix elements. Once the basis is fixed, these entries describe the operator completely.

3.2 From Abstract Kets to Matrix Multiplication

The next step is to turn vectors themselves into components. If $\{|n\rangle\}$ is a basis, any vector $|A\rangle$ can be expanded as

$$|A\rangle = \sum_n A_n|n\rangle, \qquad A_n = \langle n|A\rangle. \tag{3.5}$$

So the coefficients are not mysterious new objects; they are just inner products with basis vectors.

Substituting the second formula into the first gives

$$|A\rangle = \sum_n |n\rangle\langle n|A\rangle. \tag{3.6}$$

The lecture describes the operator in the middle as something that "does nothing." In compact notation we write

$$\sum_n |n\rangle\langle n| = \mathbb{I}. \tag{3.7}$$

This is the completeness relation. The lecture itself does not pause to write $\mathbb{I}$, but that is the clean modern way to say what is meant by those basis vectors "nose to nose" under a summation sign.

Now we go back to the operator $\hat{K}$ acting on an arbitrary vector $|A\rangle$. The goal is not merely to write $\hat{K}|A\rangle$ abstractly, but to compute its components. The n-th component is

$$\langle n|\hat{K}|A\rangle. \tag{3.8}$$

Insert the expansion of $|A\rangle$:

$$\langle n|\hat{K}|A\rangle = \left\langle n \middle| \hat{K}\sum_m A_m|m\rangle \right\rangle \tag{3.9}$$

$$= \sum_m \langle n|\hat{K}|m\rangle A_m \tag{3.10}$$

$$= \sum_m K_{nm}A_m. \tag{3.11}$$

This is the exact point at which the lecture moves from abstract bra-ket notation to ordinary matrix language.

The upper line on the board is the bra-ket statement

$$\langle n|\hat{K}|A\rangle = \sum_m \langle n|\hat{K}|m\rangle\langle m|A\rangle, \tag{3.12}$$

and the lower line compresses the same content into the indexed form

$$(\hat{K}A)_n = \sum_m K_{nm}A_m. \tag{3.13}$$

That is just matrix multiplication: the matrix K_{nm} acts on the column of components A_m.

The lecture then pushes this one step further. If two operators $\hat{K}$ and $\hat{L}$ are multiplied, their product means that $\hat{L}$ acts first and $\hat{K}$ acts second. Since operators are not numbers, one must not assume that $\hat{K}\hat{L} = \hat{L}\hat{K}$. In general they do not commute. The matrix elements of the product are

$$(KL)_{nm} = \langle n|\hat{K}\hat{L}|m\rangle \tag{3.14}$$

$$= \sum_r \langle n|\hat{K}|r\rangle\langle r|\hat{L}|m\rangle \tag{3.15}$$

$$= \sum_r K_{nr}L_{rm}. \tag{3.16}$$

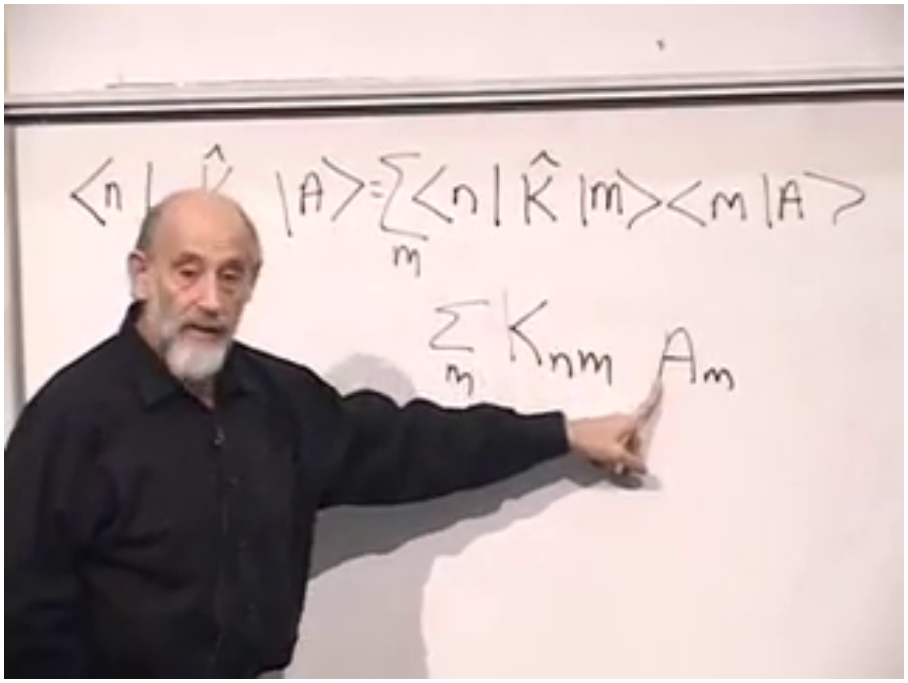


Figure 3.4: Operator action rewritten in components.

So the abstract rule "do one thing and then another" becomes the concrete rule of matrix multiplication. This is the last piece of the linear-algebra toolkit the lecture wants in place before turning to the operators of physical interest.

3.3 Hermitian Operators and the Eigenvalue Theorem

The next idea is the decisive one. Hermitian operators play, in operator theory, the role that real numbers play among ordinary numbers. They are the operators appropriate to measurable quantities, because measurements return real values.

The first chalkboard statement of the Hermitian definition is garbled in the transcript, but the later repetitions make the intended rule clear. We therefore state the standard form.

Definition 3.1. An operator $\hat{H}$ is Hermitian if

$$\langle B|\hat{H}|A\rangle = \langle A|\hat{H}|B\rangle^*. \tag{3.17}$$

Equivalently, for every vector $|A\rangle$,

$$\langle A|\hat{H}|A\rangle \in \mathbb{R}. \tag{3.18}$$

This is the precise sense in which Hermitian operators are "real." If we lived in a purely real vector space, the analogue would simply be symmetry: the matrix elements would satisfy $H_{mn} = H_{nm}$.

To appreciate the significance of Hermitian operators, the lecture turns immediately to eigenvectors and eigenvalues. In general an operator changes the direction of a vector. But there may be special directions that it leaves unchanged. If $|\lambda\rangle$ is such a vector, then

$$\hat{H}|\lambda\rangle = \lambda|\lambda\rangle. \tag{3.19}$$

The vector is an eigenvector; the number λ is the corresponding eigenvalue.

Theorem 3.2 (Fundamental facts about Hermitian operators)**.** *If $\hat{H}$ is Hermitian, then:*

1. *every eigenvalue of $\hat{H}$ is real;*
2. *eigenvectors corresponding to distinct eigenvalues are orthogonal;*
3. *the eigenvectors of $\hat{H}$ form a complete orthonormal basis of the vector space. In the degenerate case, one can choose orthogonal eigenvectors within each degenerate eigenspace.*

The lecture does not stop to prove the full theorem, but it does emphasize that the first two facts follow very quickly from the definition. Let us record them.

Proof. Suppose

$$\hat{H}|\lambda\rangle = \lambda|\lambda\rangle. \tag{3.20}$$

Then

$$\langle\lambda|\hat{H}|\lambda\rangle = \lambda\langle\lambda|\lambda\rangle. \tag{3.21}$$

The left-hand side is real, because $\hat{H}$ is Hermitian. The factor $\langle\lambda|\lambda\rangle$ is positive for a nonzero vector. Hence λ must be real.

Now suppose

$$\hat{H}|\lambda_1\rangle = \lambda_1|\lambda_1\rangle, \qquad \hat{H}|\lambda_2\rangle = \lambda_2|\lambda_2\rangle. \tag{3.22}$$

Then

$$\langle\lambda_1|\hat{H}|\lambda_2\rangle = \lambda_2\langle\lambda_1|\lambda_2\rangle. \tag{3.23}$$

On the other hand, Hermiticity gives

$$\langle\lambda_1|\hat{H}|\lambda_2\rangle = \langle\hat{H}\lambda_1|\lambda_2\rangle = \lambda_1\langle\lambda_1|\lambda_2\rangle, \tag{3.24}$$

where we used the reality of λ_1. Therefore

$$(\lambda_2 - \lambda_1)\langle\lambda_1|\lambda_2\rangle = 0. \tag{3.25}$$

So if $\lambda_1 \neq \lambda_2$, then $\langle\lambda_1|\lambda_2\rangle = 0$. □

The third statement is the lecture's real payoff. Hermitian operators do not merely have a few privileged vectors; they have enough of them to form a basis. And because an eigenvector may be multiplied by any nonzero complex number without ceasing to be an eigenvector, we can always rescale it to unit norm. That is how the orthogonal eigenvectors become an orthonormal basis.

The lecture then returns to the concrete matrix picture. The board frame at this point shows the matrix and column-vector layout explicitly, though only the upper-left corner of the matrix is fully legible.

Remark 3.3. The frame shows the matrix entries $H_{11}, H_{12}, H_{21}, \ldots$ and the first components $A_1, A_2, \ldots$ of the column vector. The fully typeset array below is a cautious completion of that visible pattern, preserving the lecture's local use of H_{ij} rather than forcing everything back to K_{ij}.

$$\begin{pmatrix} H_{11} & H_{12} & \cdots \\ H_{21} & H_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ \vdots \end{pmatrix}. \tag{3.26}$$

If the column happens to be an eigenvector of the matrix, then the result is simply λ times the same column vector. That is the concrete matrix form of the eigenvalue equation.

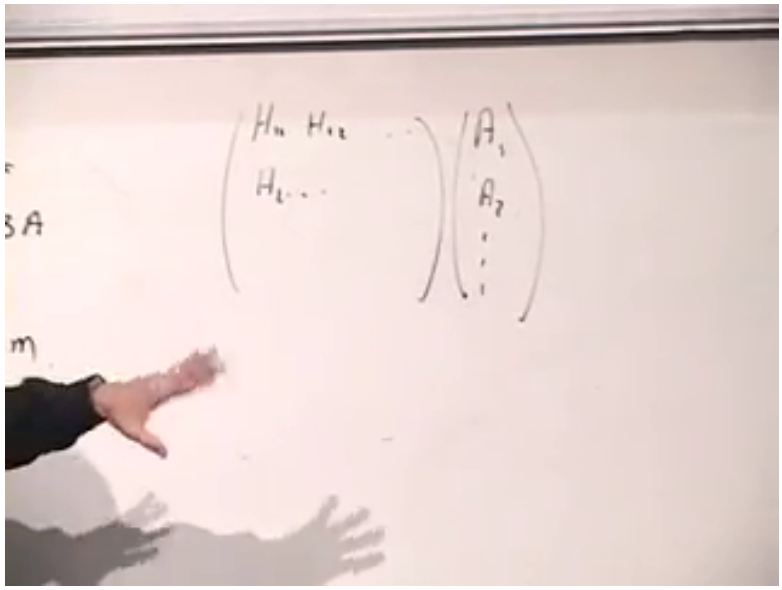


Figure 3.5: Concrete matrix-and-column-vector layout in the Hermitian-operator discussion.

3.3.1 Question & Answer

Do all operators have eigenvectors? No. The lecture makes a point of pausing here because this is exactly where one might overgeneralize. A generic linear operator need not have eigenvectors in the useful sense, and it certainly need not have a complete basis of them. Hermitian operators are special. They do.

This is also why the exchange about rotations matters. A rotation is not Hermitian. In a real vector space the corresponding matrices are antisymmetric, and in the complex setting one is led naturally to anti-Hermitian generators. So the theorem is not a fact about arbitrary operators. It is a fact about the Hermitian ones.

3.4 The Five Postulates of Quantum Mechanics

Only now does the lecture say that we are ready for the postulates of quantum mechanics. They are not presented as the smallest logically independent set. They are presented in a slightly expanded form because, pedagogically, it is better to expose the ideas than to be clever.

1. **States are vectors.** Every state of a quantum system is represented by a ket vector in a complex vector space.
2. **Observables correspond to Hermitian operators.** The things we measure in experiments are represented by Hermitian operators acting on the state space.
3. **Possible measurement outcomes are eigenvalues.** If an observable corresponds to $\hat{H}$, then the only values that can be obtained in measurement are the eigenvalues of $\hat{H}$.
4. **Eigenvectors are states of definite value.** If the system is in an eigenvector $|\lambda_n\rangle$ of $\hat{H}$, then a measurement of the observable yields the value λ_n with certainty.

5. **Arbitrary states give probabilities.** If the system is in an arbitrary state $|A\rangle$, then the probability of measuring λ_n is
$$P(\lambda_n|A) = |\langle\lambda_n|A\rangle|^2. \tag{3.27}$$

The lecture keeps correcting the language here, and those corrections are part of the substance. One does not directly observe the state of a system. One observes observables and records real numbers. The state is what determines the probabilities for those numbers.

3.4.1 Question & Answer

What does it mean for an observable to be *definite*, and is it right to say that we observe eigenvectors? In the lecture, a quantity is definite when there is no statistical spread in the outcome. If the system has been prepared in an eigenvector of $\hat{H}$, then repeated measurements of H always yield the same eigenvalue. The observable is definite in that state.

But it is still better not to say that we observe eigenvectors. The cleaner statement is that we measure an observable and obtain one of its eigenvalues. From that result we may infer something about the state, but the thing recorded by the measuring apparatus is a number.

The lecture illustrates this with spin. Prepare an electron in a spin eigenstate along one axis and measure spin along that same axis: the result is definite. Measure along a different axis: now the result is only probabilistic. The state has not ceased to exist; rather, it is not an eigenstate of the second observable.

Remark 3.4. At this stage the lecture has stated the probability rule, but it has not yet given an account of the measurement process itself. The effect of measurement on the state, and the issue of collapse, are explicitly postponed.

3.5 A Particle on a Line and the Position Basis

The lecture now makes a deliberate jump. Instead of returning to coins, dice, or other finite-state systems, it goes directly to a particle moving on a line. This is where the state space becomes genuinely infinite-dimensional.

Classically, a particle on a line is described by a coordinate x, perhaps depending on time. Quantum mechanically, the state is described instead by a vector in the space of complex functions of x. We write that vector as $\psi(x)$. This is not a metaphor. The set of such functions forms a vector space: we can add functions, multiply them by complex numbers, and define an inner product

$$\langle\phi|\psi\rangle = \int dx\, \phi^*(x)\psi(x). \tag{3.28}$$

The lecture also pauses here to head off a possible confusion. This inner product is not real in general. What is guaranteed real is the inner product of a vector with itself:

$$\langle\psi|\psi\rangle = \int dx\, |\psi(x)|^2. \tag{3.29}$$

Now we introduce the first operator in this function space: the position operator $\hat{x}$. Its action is as simple as possible:

$$(\hat{x}\psi)(x) = x\psi(x). \tag{3.30}$$

The lecture checks Hermiticity by using the equivalent condition that $\langle\psi|\hat{x}|\psi\rangle$ be real for every ψ:

$$\langle\psi|\hat{x}|\psi\rangle = \int dx\, \psi^*(x)\, x\, \psi(x) = \int dx\, x\, |\psi(x)|^2. \tag{3.31}$$

Because x and dx are real, this quantity is real. So $\hat{x}$ is Hermitian.

That means it must have eigenvectors. The eigenvalue equation is

$$\hat{x}|\lambda\rangle = \lambda|\lambda\rangle, \tag{3.32}$$

which in the function representation becomes

$$x\, \psi(x) = \lambda\psi(x), \tag{3.33}$$

or

$$(x - \lambda)\psi(x) = 0. \tag{3.34}$$

Taken literally, this looks absurd. For every $x \neq \lambda$, the factor $x - \lambda$ is nonzero, so $\psi(x)$ must vanish. The only place where it can fail to vanish is at one point. As an ordinary square-integrable function, that is impossible.

The lecture does not hide the difficulty. Instead it introduces the Dirac delta function heuristically. One imagines a narrow spike of width ϵ centered at $x = \lambda$, of height $1/\epsilon$, so that the area under it is one. In standard notation, the cleaned-up form of the lecture's spike picture is

$$\langle x|\lambda\rangle = \delta(x - \lambda). \tag{3.35}$$

This is not an ordinary normalized wavefunction. It is an idealized generalized eigenfunction.

The lecture also insists on another distinction that is easy to blur: these position eigenfunctions are orthogonal, but not normalized in the usual sense. If $\lambda \neq \lambda'$, then the corresponding spikes do not overlap, so their inner product vanishes. But the inner product of such a spike with itself is not a nice finite unit norm. In the lecture's language, it is "ortho, but not normal."

There is one such eigenfunction for every real value of λ. The spectrum of $\hat{x}$ is therefore the entire real axis.

The real payoff comes when we ask what the wavefunction $\psi(x)$ means. Project the state onto the position eigenvector at λ:

$$\langle\lambda|\psi\rangle = \int dx\, \delta(x - \lambda)\psi(x). \tag{3.36}$$

Now use the spike picture. Over the narrow support of the delta function, $\psi(x)$ varies negligibly, so we may replace it by $\psi(\lambda)$:

$$\langle\lambda|\psi\rangle \approx \psi(\lambda) \int dx\, \delta(x - \lambda) \tag{3.37}$$

$$= \psi(\lambda). \tag{3.38}$$

In the standard distributional reading, this is the familiar identity. Relabel the eigenvalue λ by the position variable itself, and we obtain

$$\langle x|\psi\rangle = \psi(x). \tag{3.39}$$

So the wavefunction is the component of the state vector along the position basis. It is not an extra object attached to the ket. It is the ket, expressed in a particular basis.

Now the fifth postulate becomes concrete. The probability is the square of the projection onto the relevant eigenvector. In the position basis that means

$$\rho(x) = |\psi(x)|^2. \tag{3.40}$$

The lecture phrases this informally as the probability to detect the particle at position x. Mathematically, for a continuous spectrum, it is better to say that $|\psi(x)|^2$ is the position probability density.

3.5.1 Question & Answer

How can position eigenvectors be represented if normalizable wavefunctions must decay at infinity? The lecture raises this difficulty directly and then refuses to overtidy it. Exact position eigenvectors are not ordinary normalizable Hilbert-space vectors. They are idealized objects, represented heuristically by delta spikes.

What one can do physically is approximate them arbitrarily well by sharply localized wave packets. That is enough for the lecture's immediate purpose. The normalization issue is real, but it is not the issue the lecture wants to settle at this moment.

3.6 The k-Operator, Plane Waves, and the Momentum Hint

At this point the lecture asks a very natural question. If all we care about is the position probability density $|\psi(x)|^2$, why go through the trouble of introducing a complex wavefunction $\psi(x)$ at all? Why not simply use a probability density on the line?

The answer is that position is not the only observable. We need a single state vector from which the probabilities for many different observables can be extracted. That is the reason the full wavefunction is needed.

The next operator is differentiation:

$$\frac{d}{dx}. \tag{3.41}$$

It is linear, but it is not Hermitian. The lecture's first blackboard definition of anti-Hermitian is also garbled, so here again we state the corrected standard form:

$$\langle B|\hat{A}|A\rangle = -\langle A|\hat{A}|B\rangle^*. \tag{3.42}$$

Under this rule, multiplying an anti-Hermitian operator by i produces a Hermitian one.

For differentiation, the cleaned-up operator statement is

$$\left\langle \phi \left| \frac{d}{dx} \right| \psi \right\rangle = -\left\langle \psi \left| \frac{d}{dx} \right| \phi \right\rangle^*, \tag{3.43}$$

provided the boundary terms vanish at infinity. The lecture then settles on the Hermitian combination

$$\hat{K} = -i\,\frac{d}{dx}. \tag{3.44}$$

There is a brief sign wobble in the live discussion, but $-i\,d/dx$ is where the lecture comes to rest, and that is the convention we keep.

The lecture chooses the equivalent Hermiticity check $\langle\psi|\hat{K}|\psi\rangle \in \mathbb{R}$. Let us follow that calculation.

Proof. We compute

$$\langle\psi|\hat{K}|\psi\rangle = \int dx\, \psi^*(x) \left(-i\, \frac{d\psi}{dx}\right). \tag{3.45}$$

Integrating by parts and dropping the boundary term—for the moment assuming that $\psi(x) \to 0$ sufficiently fast at infinity—gives

$$\int dx\, \psi^*(x) \left(-i\, \frac{d\psi}{dx}\right) = i \int dx\, \frac{d\psi^*}{dx}\, \psi(x). \tag{3.46}$$

But the right-hand side is the complex conjugate of the left-hand side. Therefore the original expression is equal to its own complex conjugate and is real. Hence $\hat{K}$ is Hermitian. □

The lecture is already aware of the tension. This proof assumes decay at infinity, but the eigenfunctions we are about to write down do not decay. That is not a mistake to be hidden. It is one of the reasons the discussion remains heuristic at this stage.

Now solve the eigenvalue equation

$$-i\, \frac{d\psi}{dx} = k\psi. \tag{3.47}$$

The solution is

$$\psi_k(x) \propto e^{ikx} = \cos(kx) + i \sin(kx). \tag{3.48}$$

The lecture does not belabor the solution method; it simply checks it:

$$\frac{d}{dx} e^{ikx} = ik\, e^{ikx}, \tag{3.49}$$

so multiplying by $-i$ yields $k\, e^{ikx}$, exactly as required.

These eigenfunctions are radically different from those of position. The position eigenfunctions are sharply peaked. The k-eigenfunctions are spread across the whole line. Moreover,

$$\psi_k^*(x)\psi_k(x) = \text{const.}, \tag{3.50}$$

so the corresponding position probability density is uniform. There is no information here about where the particle is.

The lecture then lingers on the oscillatory character of e^{ikx}. As x changes, the complex number e^{ikx} moves around the unit circle. Small k means slow winding, hence long wavelength. Large k means rapid winding, hence short wavelength. One full period occurs when kx changes by 2π, so

$$kL = 2\pi, \qquad L = \frac{2\pi}{k}. \tag{3.51}$$

Only after this mathematical work does the lecture attach a physical name. It does so cautiously. Using the pre-quantum intuition of de Broglie and Einstein, one recalls that a wave of wavelength L is associated with momentum

$$p = \frac{h}{L}. \tag{3.52}$$

Substituting $L = 2\pi/k$ gives

$$p = \frac{h}{L} = \frac{h}{2\pi} k = \hbar k. \tag{3.53}$$

So the operator $\hat{K}$ is provisionally interpreted as momentum in units of $\hbar$.

The lecture is explicit about the status of this statement. It is a name, not yet a derivation from first principles. To show that $\hbar k$ really behaves like classical momentum, one must study time evolution, wave packets, and the classical limit. That is postponed.

Even before that deeper analysis, however, the contrast with position is already stark. The position eigenbasis consists of sharply localized states. The k-basis consists of delocalized oscillatory states. No position eigenfunction is also a k-eigenfunction, and no k-eigenfunction is also a position eigenfunction. The lecture rightly hears, in that clash, the first unmistakable note of the uncertainty principle.

3.7 Summary

The lecture begins by rebuilding the machinery: operators act on kets, matrix elements turn operator action into numbers, and a basis turns vectors into components. That bridge to components is what lets linear operators become matrices and operator products become matrix multiplication.

From there the lecture narrows to the physically significant class of Hermitian operators, introduces eigenvalues and eigenvectors, and states the fundamental theorem that Hermitian operators have complete orthonormal eigenbases. Only then does it state the postulates of quantum mechanics and move to the first serious continuous example, a particle on a line.

In that example we learn two things at once. First, $\psi(x)$ is the component of the state along the position basis, and $|\psi(x)|^2$ is the position probability density. Second, there are other observables, represented by different Hermitian operators. The operator $\hat{K} = -i\, d/dx$ leads to plane-wave eigenfunctions e^{ikx}, to the wavelength relation $L = 2\pi/k$, and, provisionally, to the momentum identification $p = \hbar k$. What remains for later is exactly what the lecture says remains for later: the probability distribution for momentum, the role of measurement in changing the state, and the time-dependent route by which these quantum constructions recover the classical notion of momentum.

Chapter 4

From Discrete Observables to Interference

This chapter follows Leonard Susskind's fourth lecture in the quantum mechanics sequence and keeps its order and emphasis: review first, replacement second, technical detour third, then the return to probabilities, then a new system, then the question of compatibility, and finally the two-slit experiment. The lecture itself is Leonard Susskind's; these notes also preserve selected blackboard evidence, with curation support acknowledged to LazyingArt LLC and Video2Book. The governing idea is that the postulates of quantum mechanics do not change when the observable becomes continuous, but the mathematical objects that express those postulates do.

4.1 Review: Discrete Observables, Probabilities, and Basis States

Let us begin exactly where the lecture begins: with a measurable quantity that can take discrete values. Call those values $n = 1, 2, 3, \ldots$. If the system is in some state $|\psi\rangle$, then a measurement returns one of these values with probability $P(n)$, and the first rule is simply

$$\sum_n P(n) = 1. \tag{4.1}$$

The second rule is that the basis vectors associated with the different values of the observable may be taken to form an orthonormal basis:

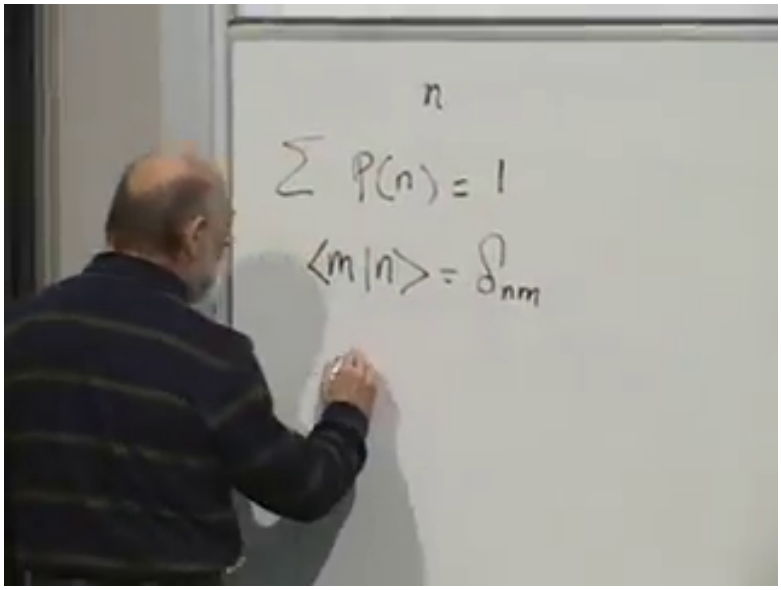
$$\langle m|n\rangle = \delta_{nm}. \tag{4.2}$$

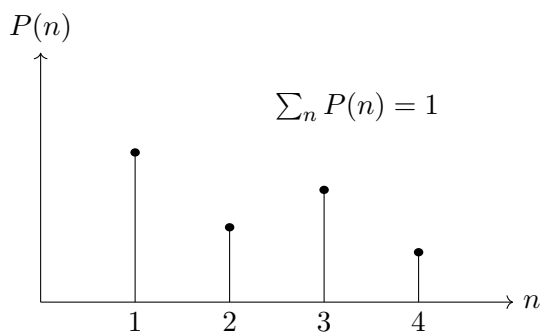
Here δ_{nm} is the Kronecker delta, equal to 1 when $m = n$ and 0 otherwise. Susskind pauses to note that one may think of it as depending only on the difference $n - m$; it is really the vanishing or nonvanishing of that difference that matters.

Probabilities are governed by amplitudes. In the discrete case the amplitude for obtaining the value n is $\langle n|\psi\rangle$, and the probability is the squared magnitude of that amplitude:

$$P(n) = |\langle n|\psi\rangle|^2. \tag{4.3}$$

So already, before any continuum enters, we have the full pattern: a basis of states, an inner product, amplitudes, and the rule that probabilities are obtained by squaring amplitudes.





The blackboard here still has the character of a review. The normalization and orthonormality relations are written cleanly at the top, while the probability picture is only beginning below. That is faithful to the lecture's rhythm: the picture is secondary; the formal rules are being re-established first.

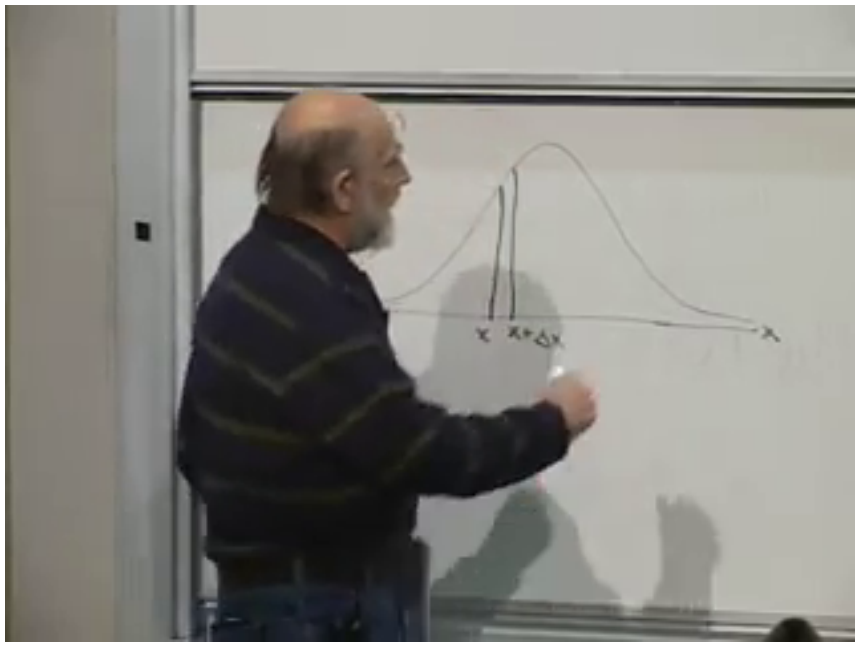
If we plot the probability as a function of the discrete variable n, the values live only on the integers. The picture is therefore not a continuous curve but a collection of heights, and it is those heights that add up to 1.

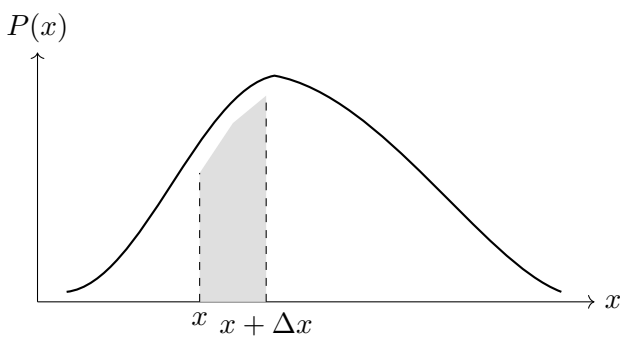
This review is not detachable background. It is the template for everything that follows. Once the observable becomes continuous, sums will be replaced by integrals and Kronecker deltas by Dirac deltas, but the structure of the argument remains the same.

4.2 From Discrete Probabilities to Continuous Probability Density

Now let the observable be the position of a particle on a line. The first change is conceptual rather than algebraic. It no longer makes sense to ask for the probability that the coordinate is exactly equal to one particular real number. A single point has measure zero, and the probability of one exact value is therefore zero.

What does make sense is the probability that the particle lies in an interval. Susskind temporarily





continues to use the symbol p for probability density, but to keep it separate from the later use of p for momentum, we will write the density as $P(x)$. The probability that the particle is found between x and $x + \Delta x$ is

$$P(x \in [x, x + \Delta x]) = \int_x^{x+\Delta x} P(x')\, dx'. \tag{4.4}$$

Accordingly, the normalization rule becomes

$$\int P(x)\, dx = 1. \tag{4.5}$$

This is the first replacement. In the discrete case the probabilities themselves add to 1; in the continuous case the area under the density curve adds to 1.

The surviving frame is strong evidence for the geometry of the argument. The lecture is not yet doing something sophisticated here. It is making a simple point carefully: for a continuous variable, the physically meaningful probability is associated with a small interval and therefore with a small area under the graph.

At this stage one replacement has been made: discrete probabilities have become probability densities. But a second replacement is now forced on us. If the basis is labeled by a continuous

variable, what replaces $\langle m|n\rangle = \delta_{nm}$? Before returning to probabilities, the lecture deliberately steps aside to introduce the Dirac delta.

4.3 The Dirac Delta and the Particle on a Line

The Dirac delta is introduced as the continuous analogue of the Kronecker delta. Susskind presents it first in the intuitive way and only then in the operational one. Intuitively, $\delta(x-y)$ is zero whenever $x \neq y$, but at $x = y$ it is concentrated into an infinitely narrow, infinitely tall spike of unit area:

$$\delta(x-y) = 0 \quad \text{for } x \neq y, \qquad \int \delta(x-y)\,dx = 1. \tag{4.6}$$

As he says, no honest ordinary function behaves this way. But the spike picture is a good guide to how the object works.

The more careful statement is that the delta is defined by what it does under integration:

$$\int \delta(x-y) f(x)\,dx = f(y). \tag{4.7}$$

Only the value of the function at $x = y$ survives. The heuristic picture and the operational definition are not rivals here. In the lecture they play different roles. The spike picture carries the intuition; the integral rule carries the mathematics.

With that in hand, we can now write the continuum analogue of orthogonality:

$$\langle x|y\rangle = \delta(x-y). \tag{4.8}$$

This is the natural replacement of $\langle m|n\rangle = \delta_{nm}$ when the basis is labeled by position. If $x \neq y$, the two position states are orthogonal; if $x = y$, the overlap is concentrated at one point, exactly as the delta function prescribes.

4.3.1 Question & Answer

Question. How can $\langle x|y\rangle$ be a delta function, and in what sense can a position eigenvector have infinite norm?

Answer. The lecture answers this by not pretending the continuum is innocent. Instead, it backs up to a dense discrete approximation. Imagine replacing the line by a set of lattice sites separated by a tiny spacing a, and let $|n\rangle$ be the corresponding orthonormal basis:

$$\langle n|m\rangle = \delta_{nm}.$$

Now define the continuum-normalized position state by stretching the discrete state:

$$|x\rangle = \frac{1}{\sqrt{a}}\,|n\rangle.$$

If x corresponds to site n and y to site m, then

$$\langle x|y\rangle = \frac{1}{a}\,\delta_{nm}. \tag{4.9}$$

This object is zero unless the two sites coincide. When they do coincide, its height is $1/a$. As a becomes very small, we get exactly the picture of a spike of width a and height $1/a$, that is, the Dirac delta.

So yes, the continuum-normalized position states have divergent norm. But that is not a defect; it is the normalization convention adapted to continuum integration. One almost never needs $\langle x|x\rangle$ by itself. What one actually uses is the delta function inside integrals, where it behaves perfectly well. The lecture's point is that the Dirac delta is not mysterious once one sees it as the rescaled limit of the Kronecker delta.

4.4 Born's Rule in the Continuum

Having inserted the delta-function detour, the lecture now returns, very explicitly, to the rules of probability. This return matters. Susskind is not content merely to quote $P(y) = |\psi(y)|^2$. He wants to derive it from the same postulates used in the discrete case.

The states are now wavefunctions, that is, complex functions of a real variable x, and the inner product on this vector space is

$$\langle\phi|\psi\rangle = \int \phi^*(x)\psi(x)\,dx. \tag{4.10}$$

A particle known to be localized at the point y is represented in the x-representation by the wavefunction $\delta(x-y)$. Therefore the amplitude to detect the particle at y in the state $|\psi\rangle$ is

$$\langle y|\psi\rangle = \int \delta(x-y)\psi(x)\,dx \tag{4.11}$$

$$= \psi(y). \tag{4.12}$$

This is the formal version of a statement that is often introduced heuristically: the wavefunction evaluated at y is the amplitude to find the particle at y.

Now we simply apply the probability postulate in its continuum form:

$$P(y) = |\langle y|\psi\rangle|^2. \tag{4.13}$$

Using the previous result,

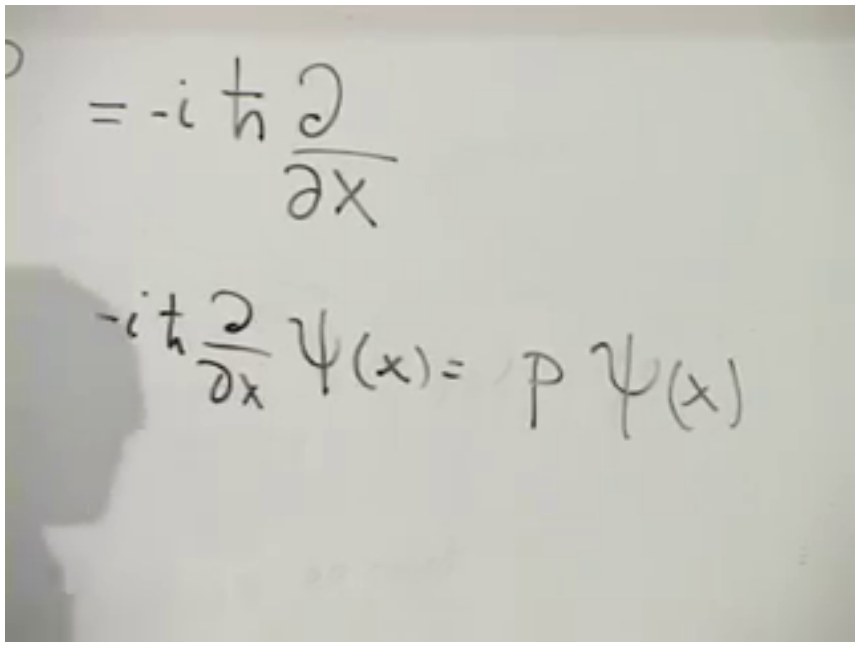
$$P(y) = |\psi(y)|^2 = \psi^*(y)\psi(y). \tag{4.14}$$

Susskind remarks that one could have begun with this formula directly. But the point of the lecture is different. The point is to show that nothing new has been added. We have simply taken the discrete postulates, replaced sums by integrals and Kronecker deltas by Dirac deltas, and then carried the same formal logic through.

The wavefunction of a particle on a line is therefore a probability amplitude in precisely the same sense that $\langle n|\psi\rangle$ is a probability amplitude in the discrete case. The difference is only that the label on the basis is now continuous.

4.5 Particle on a Circle: Periodicity, Momentum Eigenfunctions, and Quantization

The next transition in the lecture is from one system to another, not from one formalism to another. We now study a particle moving on a circle of radius R. To connect it with the line notation, the



circle is cut and laid out as the interval

$$0 \leq x \leq 2\pi R.$$

Everything looks the same as for a particle on a line until one remembers that this interval really came from a circle. That memory is encoded in one extra condition:

$$\psi(x + 2\pi R) = \psi(x). \tag{4.15}$$

The allowed state space is therefore the vector space of periodic complex functions on the interval. The lecture stops to note that this is indeed a vector space: multiplying a periodic function by any complex number keeps it periodic, and the sum of two periodic functions is periodic as well.

Now Susskind returns to momentum. In the x-representation, the momentum operator acts as

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}. \tag{4.16}$$

The frame below is the strongest board witness for the eigenvalue equation itself; the operator line above is only partly visible in the frame, but the transcript settles its standard form.

The momentum eigenvalue equation is

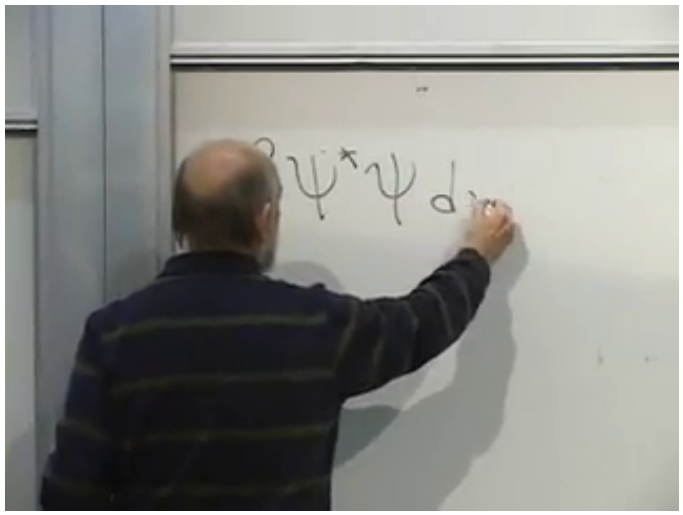
$$-i\hbar \frac{\partial}{\partial x}\psi(x) = p\,\psi(x). \tag{4.17}$$

Solving it is immediate:

$$\frac{d\psi}{dx} = \frac{ip}{\hbar}\,\psi, \tag{4.18}$$

$$\psi_p(x) = C\,e^{ipx/\hbar}. \tag{4.19}$$

So the momentum eigenfunctions are plane waves. This is one of the first places where the wave language of quantum mechanics becomes literal.



The lecture then normalizes the eigenfunctions on the circle.

The board shows the integrand $\psi^*\psi$ and the measure dx; the equality to 1 comes from the transcript rather than the frame. The normalization condition is

$$\int_0^{2\pi R} \psi_p^*(x)\psi_p(x)\,dx = 1. \tag{4.20}$$

Since $\psi_p(x) = Ce^{ipx/\hbar}$, we have $\psi_p^*\psi_p = |C|^2$. Therefore

$$1 = |C|^2 \int_0^{2\pi R} dx = 2\pi R\,|C|^2. \tag{4.21}$$

Hence

$$C = \frac{1}{\sqrt{2\pi R}} \tag{4.22}$$

up to an irrelevant phase, and the normalized eigenfunction is

$$\psi_p(x) = \frac{1}{\sqrt{2\pi R}}\,e^{ipx/\hbar}. \tag{4.23}$$

For different eigenvalues $p \neq q$, the corresponding eigenfunctions are orthogonal. One can check this directly, but the lecture prefers to emphasize the general reason: eigenvectors of a Hermitian operator with different eigenvalues are orthogonal. Thus

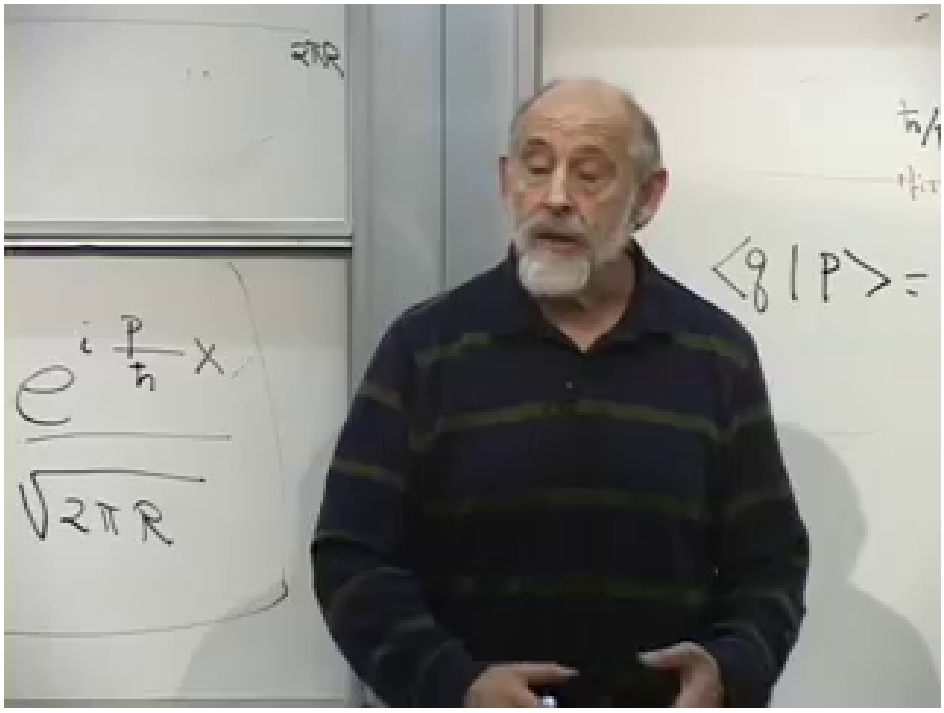
$$\int_0^{2\pi R} \psi_p^*(x)\psi_q(x)\,dx = 0 \qquad (p \neq q). \tag{4.24}$$

The next frame is less legible mathematically, but it is valuable because it preserves the board layout: the explicit plane-wave form on one side and the bra-ket representation on the other.

The exact bra-ket label in the screenshot is partly obscured, so the following is a cautious transcript-based reconstruction:

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi R}}\,e^{ipx/\hbar}. \tag{4.25}$$

That is, a momentum eigenvector, when written in the position basis, is a normalized plane wave.



Now comes the crucial new step. Not every plane wave is allowed, because the state must be periodic. Imposing periodicity gives

$$e^{ip(x+2\pi R)/\hbar} = e^{ipx/\hbar}. \tag{4.26}$$

Cancelling the common factor $e^{ipx/\hbar}$, we obtain

$$e^{i2\pi Rp/\hbar} = 1. \tag{4.27}$$

This happens precisely when the exponent is an integer multiple of $2\pi i$:

$$\frac{2\pi Rp}{\hbar} = 2\pi n, \qquad n \in \mathbb{Z}. \tag{4.28}$$

Therefore the allowed momenta are

$$p_n = \frac{n\hbar}{R}. \tag{4.29}$$

The position on the circle is continuous, but the momentum is discrete. The lecture insists on the origin of this discreteness: it comes from periodicity of the wavefunction.

Now let L be the angular momentum. Classically,

$$L = Rp. \tag{4.30}$$

Substituting the allowed momenta, we find

$$L_n = Rp_n = n\hbar. \tag{4.31}$$

The factor of R disappears. Angular momentum is quantized in units of $\hbar$, independent of the size of the circle.

4.5.1 Question & Answer

Question. Does the large-R limit give us classical mechanics, or only a dense momentum spectrum?

Answer. Only a dense spectrum. As $R \to \infty$, the spacing

$$\Delta p = \frac{\hbar}{R}$$

between neighboring momentum values goes to zero. The discrete spectrum becomes dense, and the lecture says that it is then convenient to pass from Kronecker-delta normalization to Dirac-delta normalization in momentum space. But nothing essentially classical has happened yet.

Classical mechanics would require that a particle could have a sharply defined position and a sharply defined momentum simultaneously. The large-R limit does not provide that. It merely makes the momentum eigenvalues closely packed. The real obstruction lies in the incompatibility of the position and momentum operators, which is the next topic.

4.6 Compatibility, Common Eigenvectors, and the Commutator $[x, p]$

The lecture now answers the classical-limit question by shifting from eigenvalues to states. A dense spectrum is not enough. The issue is whether the system can be in a state that is simultaneously an eigenstate of position and momentum.

Suppose two observables are represented by operators A and B. They are compatible if there exists a complete basis of vectors that are eigenvectors of both at once. Write

$$A|n\rangle = \alpha_n|n\rangle, \qquad B|n\rangle = \beta_n|n\rangle. \tag{4.32}$$

Then

$$AB|n\rangle = \alpha_n\beta_n|n\rangle, \qquad BA|n\rangle = \beta_n\alpha_n|n\rangle. \tag{4.33}$$

Since ordinary numbers commute, these two results agree. This leads to the criterion

$$[A, B] \equiv AB - BA = 0. \tag{4.34}$$

The lecture states, without proving it, that this is both necessary and sufficient: two observables are compatible precisely when their commutator vanishes.

Position and momentum do not satisfy this condition. Position eigenstates are sharply localized; momentum eigenstates are plane waves. On the circle, each momentum eigenstate has constant magnitude around the entire circle; on the infinite line, it is delocalized over the whole axis. So the lecture's physical point is immediate even before the algebra: these are not the same kind of states, and there are no common eigenvectors.

The algebra confirms this. Let x act by multiplication and p act as $-i\hbar\, d/dx$. To determine the commutator, apply it to an arbitrary wavefunction $\psi(x)$:

$$(xp - px)\psi(x) = x\left(-i\hbar\frac{d\psi}{dx}\right) - \left(-i\hbar\frac{d}{dx}\right)(x\psi(x)) \tag{4.35}$$

$$= -i\hbar\, x\frac{d\psi}{dx} + i\hbar\,\frac{d}{dx}(x\psi(x)). \tag{4.36}$$

Now use the product rule,

$$\frac{d}{dx}(x\psi(x)) = \psi(x) + x\frac{d\psi}{dx}, \tag{4.37}$$

so that

$$(xp - px)\psi(x) = -i\hbar\, x\frac{d\psi}{dx} + i\hbar\left(\psi(x) + x\frac{d\psi}{dx}\right) \tag{4.38}$$

$$= i\hbar\,\psi(x). \tag{4.39}$$

Because this is true for every $\psi(x)$, we conclude that

$$[x, p] = xp - px = i\hbar. \tag{4.40}$$

This is the formal hinge of the lecture. As a statement about ordinary numbers it would be nonsense, because $xp = px$. But as a statement about operators it is exact. Susskind emphasizes that the right-hand side is just a number, and that is about as strong a failure of commutativity as one could have. There are no common eigenvectors at all.

The lecture also addresses a natural confusion here. Multiplication by x does not turn an arbitrary function into a constant multiple of itself. That is why generic functions are not eigenfunctions of the position operator. Only delta functions have the property that multiplication by x picks out one specific numerical value.

4.7 Two-Slit Interference as Superposition of States

The lecture now cashes out all of this formalism in the two-slit experiment. The central idea is not diffraction theory in detail. It is superposition. Quantum states can be added, and the probability is computed only after the amplitudes have been added.

Start with one hole open. The wave emerging from that hole spreads out and reaches the screen with a varying phase. Over a small region of the screen one may approximate the vertical dependence by something like e^{ipy}, but Susskind is careful not to pretend that this is globally exact. A better heuristic description is

$$\psi_{\text{1-hole}}(y) \sim e^{i\varphi(y)}\rho(y), \tag{4.41}$$

where $\rho(y)$ is a slowly varying envelope and $\varphi(y)$ is a rapidly varying phase. The lecture also gives a more geometric outgoing-wave model:

$$\psi(r) \sim \frac{e^{ipr}}{r}, \qquad r = \sqrt{\ell^2 + y^2}. \tag{4.42}$$

The point of this formula is not to launch a full diffraction calculation. It is to explain why the local vertical frequency varies with y. The phase is pr, not simply py, so as one moves away from the center the apparent oscillation in the vertical direction speeds up. In physical language, the particle must have picked up more vertical momentum.

4.7.1 Question & Answer

Question. Why does one hole give a smooth blob, while two holes can create dark fringes where no particle ever lands?

Answer. For one hole, the phase may oscillate rapidly, but the probability is computed by multiplying the amplitude by its complex conjugate:

$$|\psi_{\text{1-hole}}(y)|^2 \sim \big(e^{i\varphi(y)}\rho(y)\big)\big(e^{-i\varphi(y)}\rho(y)\big) = \rho(y)^2. \tag{4.43}$$

The oscillating phase cancels. What remains is the smooth envelope. That is why a single hole can produce a broad blob even though the amplitude itself carries rapid phase variation.

With two holes open, however, the total state is the sum of two amplitudes:

$$\psi = \psi_1 + \psi_2.$$

When we square that sum, cross-terms survive. Those cross-terms contain the relative phase, and that relative phase produces the interference pattern. So one hole gives a smooth probability because its phase disappears in $|\psi|^2$; two holes give fringes because the relative phase does not disappear in $|\psi_1 + \psi_2|^2$.

To see the algebra in the form used in the lecture, let us work locally on the screen and choose units in which $\hbar = 1$. Approximate the two contributions by

$$\psi_1(y) \approx e^{ipy}, \qquad \psi_2(y) \approx e^{iqy}. \tag{4.44}$$

Each one separately has constant magnitude. Hence each one separately would give a uniform probability over that small interval. But with both holes open,

$$|\psi_1 + \psi_2|^2 = (e^{ipy} + e^{iqy})(e^{-ipy} + e^{-iqy}) \tag{4.45}$$

$$= 1 + 1 + e^{i(p-q)y} + e^{i(q-p)y} \tag{4.46}$$

$$= 2 + 2\cos\big((p-q)y\big). \tag{4.47}$$

This is not constant. It oscillates, and for those values of y where the cosine is -1, the probability is zero. These are the dark fringes.

That is the lecture's real payoff. Open one hole and a particle can arrive at a given point. Open the other hole and it can arrive there as well. Open both holes together and there are points that no particle reaches in the idealized limit. This is not because probabilities were added incorrectly; it is because amplitudes were added first, and only then squared.

The lecture repeatedly returns to the particle character of the phenomenon. One sends particles through one at a time. Each event is just a spot on the screen. But after many repetitions the spots build up the probability distribution. The distribution is always nonnegative, because it comes from an amplitude times its complex conjugate, but it need not be strictly positive. In the ideal limit it can vanish.

The dependence on slit separation now reads like ordinary beats. If the two holes are very close together, the local vertical frequencies p and q are almost the same, so $p - q$ is small and the cosine varies slowly. The fringe spacing is therefore large. If the holes are farther apart, $p - q$ is larger and the fringes become more closely spaced.

The lecture also adds a useful caution. A single aperture with sharp edges can itself produce diffraction structure because each edge behaves as a secondary source. That is a different effect. The point here is the distinct and stronger interference that arises from the superposition of the states associated with two separate holes.

Finally, the lecture stops at exactly the right place. It refuses, for the time being, to answer the which-path detector question. If a detector tells us which slit the particle went through, the interference pattern is destroyed, but to explain that properly one needs composite systems and entanglement. The same is true of the reversibility question that comes up at the end. Those topics are deferred until Hamiltonians, evolution, and tensor-product structure are in place.

4.8 Summary

The lecture unfolds as one deliberate chain of replacements and payoffs. We begin with the discrete case: probabilities $P(n)$, orthonormal basis states, amplitudes, and the Kronecker delta. We then pass to continuous variables: exact point-probabilities give way to interval probabilities, sums become integrals, and the Kronecker delta becomes the Dirac delta. With those replacements in hand, the continuum Born rule follows from the same abstract postulates:

$$P(y) = \psi^*(y)\psi(y).$$

The move from the line to the circle changes not the formalism but the allowed state space, and periodicity then forces momentum and angular momentum to be quantized. The student's natural question about the large-R limit pushes the lecture to the deeper issue of compatibility, culminating in the operator identity

$$[x, p] = i\hbar.$$

That identity explains why position and momentum are not simultaneously sharp. The two-slit experiment then gives the physical payoff: amplitudes add, probabilities do not, and the cross-term in

$$|\psi_1 + \psi_2|^2$$

creates an interference pattern with genuine dark fringes. The lecture ends by leaving the detector question open, not because it is unimportant, but because it belongs to the next layer of the subject.

Chapter 5

Periodic Line, Fourier Duality, and Polarization

We begin not with the infinite line, but with a particle on a line whose ends are identified. That is the deliberate opening move of the lecture. One first solves a tame problem in which momentum is discretized and normalization is honest, and only then lets the size of the box go to infinity. From there the lecture unfolds in a very characteristic sequence: orthogonality, completeness, Fourier transforms, two equivalent representations of the same state, and then the incompatibility of position and momentum. After the break the same logic is repeated in miniature for the polarization of a photon. These notes follow Leonard Susskind's progression closely, with companion-note curation indebted to LazyingArt LLC and Video2Book.

5.1 A Particle on a Periodic Line

A periodic line can be pictured in two ways. One may imagine a particle moving on a literal circle, but one may also imagine an ordinary line segment together with the rule that when the particle exits one end, it reappears at the other. The second picture is the one the lecture emphasizes: the geometry itself is not sacred; what matters is the identification rule. The infinite line will later be recovered as the limit in which the length grows without bound.

The quantum-mechanical translation of that rule is immediate. If the line has length L, then the wave function must satisfy

$$\psi(x) = \psi(x + L). \tag{5.1}$$

This is the first mathematical statement of the lecture.

The board here is useful evidence: the equation is written together with a finite line segment labeled L, so the notation and the geometry arrive as one package. It is worth keeping that pairing visible.

At this point the lecture backs up one step and says, in effect: let us write

$$L = 2\pi r. \tag{5.2}$$

This is introduced purely for notational convenience. One may think of the periodic line as a circle of circumference $2\pi r$, but the point is not to insist that the particle is really moving on a circle. The point is to have a compact notation that makes the periodicity of plane waves easy to read.

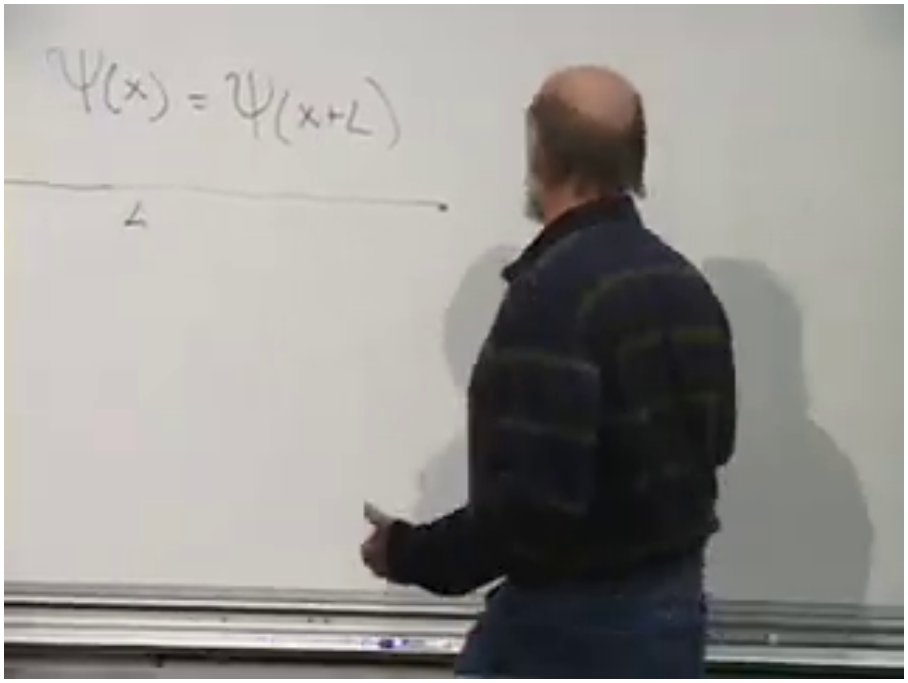


Figure 5.1: Periodicity condition on a line of length L.

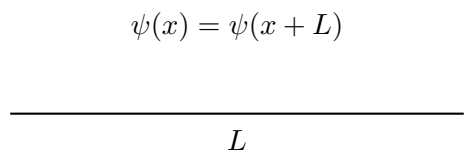


Figure 5.2: Cleaned redraw of the periodic-line schematic.

Now the periodicity condition has implications for momentum. The wave function for a momentum eigenstate is first written in the naive form e^{ipx}, and then immediately corrected:

$$e^{ipx/\hbar} = e^{ikx}, \qquad k = \frac{p}{\hbar}. \tag{5.3}$$

Susskind then switches to k-notation and stays with it, except for occasional lapses when $\hbar$ is tacitly set to 1.

The plane wave, however, is not yet normalized. On the interval $0 \leq x \leq 2\pi r$, the normalized momentum eigenfunction is

$$\psi_k(x) = \frac{e^{ipx/\hbar}}{\sqrt{2\pi r}} = \frac{e^{ikx}}{\sqrt{2\pi r}}. \tag{5.4}$$

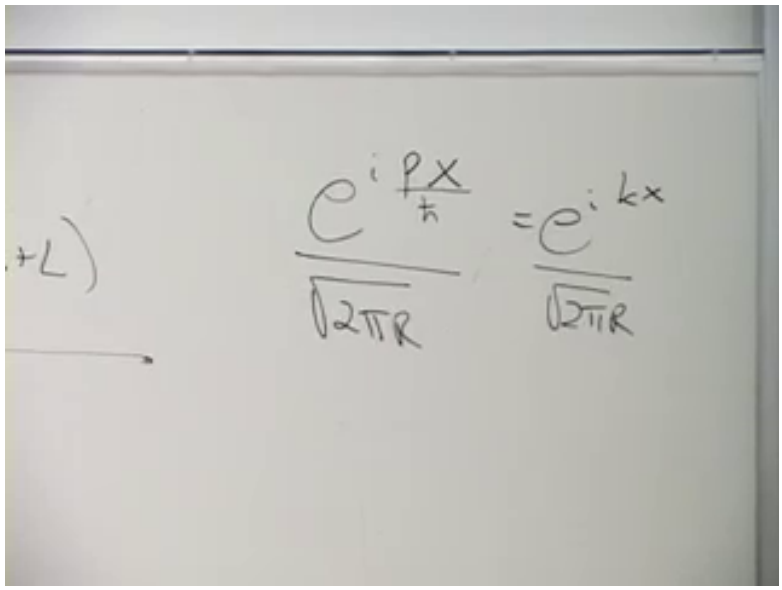


Figure 5.3: Normalized plane wave in p- and k-notation.

The frame is useful because it records the final board form: the normalized expression is shown side by side in $p/\hbar$- and k-notation, precisely at the moment when the lecture is explaining why the square root appears in the denominator.

5.1.1 Question & Answer

Why is there a square root in the denominator? Because the normalization condition is imposed on $\psi^*\psi$, not on ψ itself. For the bare plane wave e^{ikx} we have

$$|e^{ikx}|^2 = 1, \tag{5.5}$$

and so over one full period,

$$\int_0^{2\pi r} |e^{ikx}|^2 \, dx = 2\pi r. \tag{5.6}$$

If we divide the wave function by $\sqrt{2\pi r}$, then the square of that denominator is exactly $2\pi r$, and the normalization condition becomes

$$\int_0^{2\pi r} |\psi_k(x)|^2 \, dx = 1. \tag{5.7}$$

That is the whole reason for the square root. As Susskind remarks in class, the factor is not mysterious; it is simply where all those later 2π's come from.

5.2 Allowed Momenta and Discrete Orthogonality

Once the normalized plane wave has been written down, the periodicity condition can be imposed on the eigenfunction itself. This is where the periodic box begins to earn its keep. We require

$$e^{ik(x+2\pi r)} = e^{ikx}. \tag{5.8}$$

Dividing out the common factor e^{ikx}, we obtain

$$e^{i2\pi rk} = 1, \tag{5.9}$$

and therefore the phase accumulated in one trip around the circle must be an integer multiple of 2π:

$$2\pi r\, k = 2\pi n, \qquad n \in \mathbb{Z}. \tag{5.10}$$

Thus

$$rk = n, \qquad k = \frac{n}{r}. \tag{5.11}$$

The allowed values of k are therefore discrete, with spacing

$$\Delta k = \frac{1}{r}. \tag{5.12}$$

This is the staging ground for the continuum. At finite r the momentum spectrum is a lattice of evenly spaced values. As $r \to \infty$, the spacing goes to zero, the lattice becomes dense, and we are led back toward the infinite line.

Before taking that limit, however, the lecture pauses and says: before we do that, notice something first. These functions are eigenfunctions of a Hermitian operator, and therefore different eigenvalues must correspond to orthogonal eigenfunctions.

Theorem 5.1. *If two eigenvectors of a Hermitian operator have distinct eigenvalues, then they are orthogonal.*

The lecture then checks the statement explicitly for the periodic plane waves. For allowed values $k = n/r$ and $k' = n'/r$,

$$\frac{1}{2\pi r}\int_0^{2\pi r} e^{ikx}e^{-ik'x}\,dx = \delta_{nn'}. \tag{5.13}$$

If $k = k'$, the exponentials cancel and the integrand is 1, so the integral gives 1. If $k \neq k'$, the product oscillates, with positive contributions canceling negative ones over a full period, and the integral gives 0.

This is exactly the point where the lecture stops being merely about a list of allowed momenta. These normalized eigenfunctions are already an orthonormal set; in other words, they are already behaving like a basis of vectors.

5.2.1 Question & Answer

Is the orthogonality relation a definition? No. This is one of the places where the lecture deliberately slows down. The integral is something we calculate. The deeper statement is the theorem: eigenvectors of a Hermitian operator with different eigenvalues are orthogonal. One may view the explicit integral as a confirmation, in this example, of the general theorem. But the theorem is prior.

5.3 From the Box to the Continuum

Now the lecture takes the next step: the transition from the finite periodic box to the continuum. This is also the point where the spoken derivation becomes a little tangled, so it is worth writing the clean version carefully.

We begin with the box-normalized states

$$\psi_k(x) = \frac{e^{ikx}}{\sqrt{2\pi r}}. \tag{5.14}$$

At finite r, these are normalized to 1, and their inner products give Kronecker deltas on the integer labels n.

To prepare the continuum limit, Susskind rescales the states by $\sqrt{r}$. Define

$$\phi_k(x) = \sqrt{r}\,\psi_k(x) = \frac{e^{ikx}}{\sqrt{2\pi}}. \tag{5.15}$$

These are no longer normalized to 1 on the finite circle. That is intentional. They are being prepared for a different normalization, one suitable to a continuous spectrum.

What does this rescaling do? For $k \neq k'$, nothing essential changes: the inner product is still 0. But for $k = k'$, the inner product, which used to be 1, becomes r. As r becomes very large, the equal-k spike grows while the spacing $\Delta k = 1/r$ shrinks. This is the mechanism by which the Kronecker delta becomes a Dirac delta. In the continuum notation we write

$$\langle k|k'\rangle = \delta(k - k'). \tag{5.16}$$

At the same time, the momentum eigenfunction in the position basis becomes

$$\langle x|k\rangle = \frac{e^{ikx}}{\sqrt{2\pi}}, \qquad \langle k|x\rangle = \frac{e^{-ikx}}{\sqrt{2\pi}}. \tag{5.17}$$

The second equation is the complex conjugate of the first. The notation is now adapted to a continuum basis, and the r has disappeared from the kernel.

The lecture does not attempt a rigorous theory of distributions here. Its real point is more practical: once the rescaling has been performed, one may pass comfortably to the continuum notation in which sums become integrals and Kronecker deltas become delta functions.

5.4 Dyads, Completeness, and Fourier's Trick

Before going on to Fourier transforms, the lecture inserts a simple but indispensable algebraic fact. The point is to make the coming manipulations lawful. We are not going to pull Fourier transforms out of thin air; we are going to insert the identity operator in a useful way.

The inner product $\langle a|b\rangle$ is a number. The outer product

$$|b\rangle\langle a| \tag{5.18}$$

is an operator. Its action on a vector $|c\rangle$ is

$$(|b\rangle\langle a|)|c\rangle = |b\rangle\,\langle a|c\rangle. \tag{5.19}$$

The bra $\langle a|$ first contracts with $|c\rangle$ to produce a number, and that number multiplies the vector $|b\rangle$. This is the dyad.

Now let $\{|n\rangle\}$ be an orthonormal basis. Any vector may be expanded as

$$|v\rangle = \sum_n v_n|n\rangle, \qquad v_n = \langle n|v\rangle. \tag{5.20}$$

Substituting the second formula into the first gives

$$|v\rangle = \sum_n |n\rangle\langle n|v\rangle. \tag{5.21}$$

Since this holds for every vector $|v\rangle$, the operator in front must be the identity:

$$I = \sum_n |n\rangle\langle n|. \tag{5.22}$$

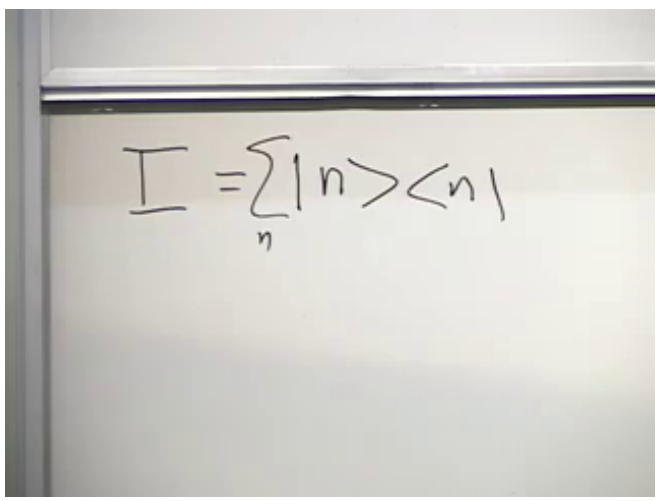


Figure 5.4: Resolution of the identity in a discrete basis.

This is exactly the centered board formula captured in the retained frame. In the continuum, the same statement becomes

$$I = \int dk\,|k\rangle\langle k|, \qquad I = \int dx\,|x\rangle\langle x|. \tag{5.23}$$

The notation changes, but the idea does not. A complete basis resolves the identity.

This is the trick the lecture now uses over and over. If a calculation looks hard in one basis, insert the identity written in another basis.

5.5 Position and Momentum as Two Representations

Let $|\psi\rangle$ be the state of a particle on a line. The position-space wave function is defined by

$$\psi(x) = \langle x|\psi\rangle. \tag{5.24}$$

The momentum-space wave function is defined by

$$\tilde{\psi}(k) = \langle k|\psi\rangle. \tag{5.25}$$

These are not two different states. They are two different descriptions of the same state vector.

The first key observation is that the wave function of a definite momentum state $|k\rangle$ in the position basis is just

$$\langle x|k\rangle = \frac{e^{ikx}}{\sqrt{2\pi}}. \tag{5.26}$$

This is already a strong hint that position and momentum are going to be related by Fourier theory.

To compute $\tilde{\psi}(k)$ from $\psi(x)$, insert the identity in the position basis:

$$\begin{aligned}
\tilde{\psi}(k) &= \langle k|\psi\rangle && (5.27)\\
&= \int dx\, \langle k|x\rangle\langle x|\psi\rangle && (5.28)\\
&= \frac{1}{\sqrt{2\pi}} \int dx\, e^{-ikx}\psi(x). && (5.29)
\end{aligned}$$

That is the Fourier transform. The minus sign in the exponential appears because $\langle k|x\rangle$ is the complex conjugate of $\langle x|k\rangle$.

Now go in the other direction. Insert the identity in the momentum basis:

$$\begin{aligned}
\psi(x) &= \langle x|\psi\rangle && (5.30)\\
&= \int dk\, \langle x|k\rangle\langle k|\psi\rangle && (5.31)\\
&= \frac{1}{\sqrt{2\pi}} \int dk\, e^{ikx}\tilde{\psi}(k). && (5.32)
\end{aligned}$$

This is the inverse Fourier transform. The asymmetry between plus and minus signs is the familiar asymmetry of the Fourier-transform pair.

These formulas are, of course, older than quantum mechanics. The lecture explicitly notes that. But quantum mechanics gives them a new interpretation: they relate two descriptions of one and the same physical state. If somebody hands us $\psi(x)$, we can immediately compute $\tilde{\psi}(k)$; if somebody hands us $\tilde{\psi}(k)$, we can recover $\psi(x)$. The information is the same.

It is also worth recording the operator actions in the two representations. In the position representation,

$$\begin{aligned}
\hat{X}\,\psi(x) &= x\,\psi(x), && (5.33)\\
\hat{k}\,\psi(x) &= -i\frac{\partial}{\partial x}\psi(x). && (5.34)
\end{aligned}$$

In the momentum representation,

$$\hat{k}\,\tilde{\psi}(k) = k\,\tilde{\psi}(k), \tag{5.35}$$

$$\hat{X}\,\tilde{\psi}(k) = i\frac{\partial}{\partial k}\tilde{\psi}(k). \tag{5.36}$$

The board equation for $\hat{k}\,\psi(x)$ is partly occluded in the retained frame, but the transcript makes the intended form unmistakable: $-i\partial_x$ is the operator being identified with momentum in the x-representation.

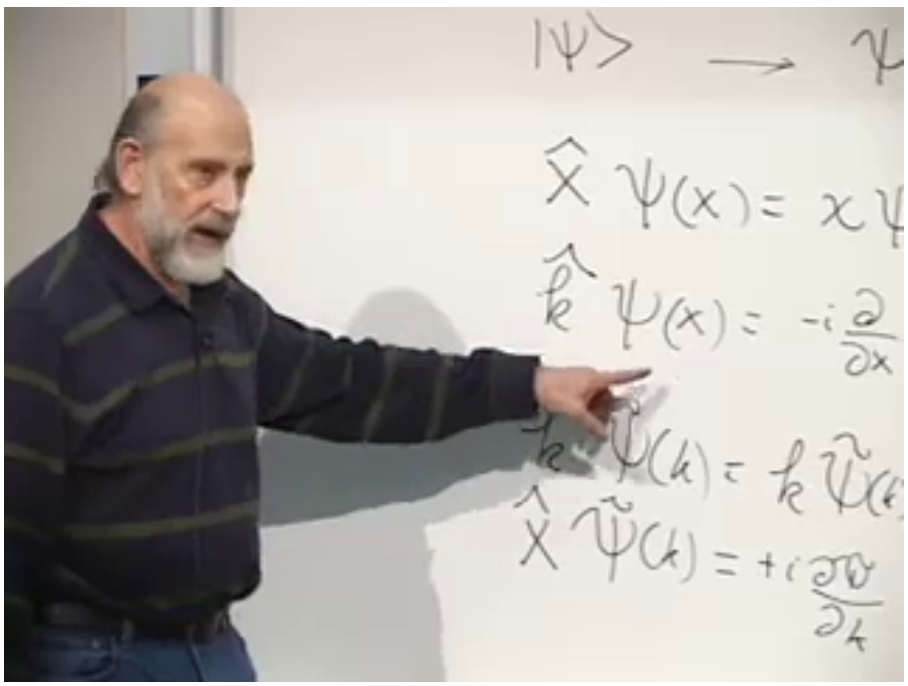


Figure 5.5: Operator actions in position and momentum space.

The top line of the board, mapping the abstract state to a wave function, is only partly visible, but the intended pattern is clear enough: the lecture is laying the x-space and k-space rules side by side so that the symmetry can be seen rather than merely asserted.

Two remarks from the lecture are worth preserving here. First, k can certainly be negative; that simply corresponds to momentum in the opposite direction. Second, the symmetry between x and k is not something classical mechanics makes obvious at first sight. It looks strange until one has learned to think in terms of conjugate quantities.

5.5.1 Question & Answer

Why do we represent the state as $\psi(x)$ or as $\tilde{\psi}(k)$, but not as a joint function of x and k? Because the two variables correspond to incompatible observables. No vector is simultaneously an eigenvector of $\hat{X}$ and $\hat{k}$. A state of definite momentum is completely spread out in position, and by the same symmetry a state of definite position is completely spread out in momentum. One may pass back and forth between the two representations, but one does not assign a single wave function of x and k as though both quantities were simultaneously sharp. This is one of the places where quantum mechanics replaces “and” by “or.”

5.6 Wave Packets and the Deferred Classical Limit

At this point the lecture pauses and refuses to claim too much. We have not yet established a full bridge to classical mechanics. We have only seen that the quantum quantities called position and momentum are represented by the operators $\hat{X}$ and $\hat{k}$, and that they are incompatible. So why should they deserve the classical names at all?

Susskind's answer is a promise rather than a proof. The bridge is provided by wave packets. A wave packet is a state whose position-space wave function is concentrated in some finite region, and whose momentum-space wave function is also concentrated, though of course not infinitely sharply, around some finite range of k-values.

A convenient schematic form is

$$\psi(x) \sim e^{ik_0 x} f(x), \tag{5.37}$$

where $f(x)$ is a localized envelope. The factor $f(x)$ keeps the state concentrated in position space, while the oscillatory factor $e^{ik_0 x}$ makes it look locally like a plane wave, and hence gives it an approximate momentum k_0.

The lecture stresses three points. First, the envelope need not be exactly Gaussian, though a Gaussian is the standard example. Second, the packet is as sharp in both position and momentum as the uncertainty principle allows, but never sharper. Third, for sufficiently smooth potentials, such packets tend to propagate in a way that resembles classical motion: the center of the packet in position space and the center of the packet in momentum space move approximately as a classical particle would move.

But all of that is postponed as a theorem to be proved later. Here the packet appears only as a preview, the first quantum object that looks even vaguely classical.

5.7 Photon Polarization: The Same Logic in Two Dimensions

After the break, the lecture makes a brief return to operator language. If A and B are complex vectors, then the inner product can be complex, and the outer product $|A\rangle\langle B|$ need not be Hermitian. One should think in terms of Hermitian conjugation, not merely ordinary complex conjugation. This remark is short and somewhat noisy in the transcript, but its function is clear: it reopens the operator formalism just before the lecture pivots to a simpler system.

That simpler system is the polarization of a photon. The physical setup is concrete. A polarizer transmits one polarization and blocks, reflects, or absorbs the orthogonal one. For a single photon the result is discrete: the photon either gets through or it does not. The apparatus may therefore be viewed in two ways. It prepares a state, because the transmitted photon emerges in a definite polarization. And it measures a property, because the next polarizer can test what that polarization is.

In the lecture the classical electromagnetic-wave picture is used only long enough to define what is meant by polarization. The electric field oscillates in some transverse direction, and that direction labels the polarization. But once the single-photon description begins, the crucial point is that the outcome is indivisible. One photon does not partly pass and partly reflect. It does one or the other.

Now ignore the motion of the photon and focus only on its polarization state. Choose two orthogonal

directions, x and y, and denote the corresponding states by

$$|x\rangle \leftrightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |y\rangle \leftrightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{5.38}$$

These are orthonormal basis vectors:

$$\langle x|x\rangle = 1, \qquad \langle y|y\rangle = 1, \qquad \langle x|y\rangle = 0. \tag{5.39}$$

The orthogonality is physically motivated. A single experiment can distinguish the two configurations, and so the corresponding vectors must be orthogonal.

Now define an observable associated with an xy-oriented polarizer. Susskind chooses a convenient convention: assign $+1$ to one outcome and -1 to the other. The numbers themselves are arbitrary; the apparatus and the notebook convention are what matter. In the chapter we follow the lecture's mathematical content and write

$$\hat{P}_{xy}|x\rangle = +|x\rangle, \qquad \hat{P}_{xy}|y\rangle = -|y\rangle. \tag{5.40}$$

This completely specifies the operator. In matrix form,

$$\hat{P}_{xy} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{5.41}$$

Equivalently, in the dyad language already introduced earlier,

$$\hat{P}_{xy} = |x\rangle\langle x| - |y\rangle\langle y|. \tag{5.42}$$

That is a useful formula, because it ties the two-state polarization example back to the earlier discussion of outer products and completeness.

5.7.1 Question & Answer

What state comes out of a 45° polarizer? The lecture handles this in exactly the right spirit: not by starting with a matrix, but by asking what the state ought to do. A photon emerging from a 45° polarizer is neither purely x-polarized nor purely y-polarized. But if we send it into an x- or y-aligned polarizer, we expect equal probabilities for the two outcomes. The natural guess is therefore the equal-amplitude superposition

$$|+\rangle = \frac{|x\rangle + |y\rangle}{\sqrt{2}}. \tag{5.43}$$

Its orthogonal partner is

$$|-\rangle = \frac{|x\rangle - |y\rangle}{\sqrt{2}}. \tag{5.44}$$

In column-vector form,

$$|+\rangle \leftrightarrow \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}, \qquad |-\rangle \leftrightarrow \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}. \tag{5.45}$$

The lecture then insists on doing the obvious probability calculation in the formalism, just to show that the machinery really works. Suppose the photon is in the state $|+\rangle$, and we ask for the probability that it passes a y-polarizer. The amplitude is

$$\langle y|+\rangle = \left\langle y \middle| \frac{|x\rangle + |y\rangle}{\sqrt{2}} \right\rangle \tag{5.46}$$

$$= \frac{1}{\sqrt{2}}(\langle y|x\rangle + \langle y|y\rangle) \tag{5.47}$$

$$= \frac{1}{\sqrt{2}}. \tag{5.48}$$

Therefore

$$|\langle y|+\rangle|^2 = \frac{1}{2}. \tag{5.49}$$

Likewise,

$$|\langle x|+\rangle|^2 = \frac{1}{2}, \qquad |\langle x|-\rangle|^2 = \frac{1}{2}, \qquad |\langle y|-\rangle|^2 = \frac{1}{2}. \tag{5.50}$$

So the diagonal states are exactly the states whose x- and y-probabilities are equal.

Now define the observable associated with the 45° basis. Let $|+\rangle$ have eigenvalue $+1$ and $|-\rangle$ have eigenvalue -1:

$$\hat{P}_{45}|+\rangle = +|+\rangle, \qquad \hat{P}_{45}|-\rangle = -|-\rangle. \tag{5.51}$$

The simplest matrix that does this, in the $(|x\rangle, |y\rangle)$ basis, is

$$\hat{P}_{45} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{5.52}$$

One checks it immediately:

$$\hat{P}_{45}\begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}, \qquad \hat{P}_{45}\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = -\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}. \tag{5.53}$$

Equivalently,

$$\hat{P}_{45} = |+\rangle\langle+| - |-\rangle\langle-|. \tag{5.54}$$

Now the lecture arrives at the real point. The eigenvectors of $\hat{P}_{xy}$ are $|x\rangle$ and $|y\rangle$. The eigenvectors of $\hat{P}_{45}$ are $|+\rangle$ and $|-\rangle$. There are no common eigenvectors. Therefore the two observables are incompatible. This is exactly the same logic as before, but now in a two-dimensional vector space rather than on the infinite line.

5.7.2 Question & Answer

Why is probability not itself an observable? Because an observable, in the technical sense used in the lecture, is something that gives a definite outcome in a single experiment. A photon sent through a polarizer yields a definite record in the notebook: through or not through, or equivalently $+1$ or -1. Probability does not appear in a single run. It is inferred only after many repeated runs.

The lecture's late discussion is worth recording in the cleaner dyad language. Define the projectors

$$\Pi_x = |x\rangle\langle x|, \qquad \Pi_y = |y\rangle\langle y|. \tag{5.55}$$

Then for a state $|\psi\rangle$,

$$\langle\psi|\Pi_x|\psi\rangle = \langle\psi|x\rangle\langle x|\psi\rangle = |\langle x|\psi\rangle|^2, \tag{5.56}$$

and similarly

$$\langle\psi|\Pi_y|\psi\rangle = |\langle y|\psi\rangle|^2. \tag{5.57}$$

These expectation values are probabilities. But the output of the single-shot experiment associated with Π_x is 1 or 0, not the probability. The same distinction is what underlies the lecture's final insistence that an observable is tied to a particular apparatus and to the little number one writes down event by event after the system has interacted with that apparatus.

The lecture closes by sharpening the language further: all observables are Hermitian operators, and all Hermitian operators are observables. But the observable is never the abstract probability itself. The observable is the single-run result associated with a definite measurement setup.

5.8 Summary

The first half of the lecture takes a particle on a periodic line, normalizes the plane waves, quantizes the allowed momenta, and then carefully stretches the finite-box normalization into continuum normalization. With dyads and completeness in hand, the lecture derives the Fourier-transform relation between the position and momentum representations and uses that relation to sharpen the incompatibility of $\hat{X}$ and $\hat{k}$.

The second half rebuilds the same logic in a much simpler system. A polarizer prepares states and measures outcomes; the orthogonal states $|x\rangle$ and $|y\rangle$ form a basis; rotated states such as $|+\rangle$ and $|-\rangle$ define new observables; and the absence of common eigenvectors expresses incompatibility once again.

What runs through the whole lecture is the same structural lesson. A state may be described in more than one representation, but incompatible observables do not fit into one simultaneous sharp picture. And an observable is not whatever number we infer from an ensemble after the fact; it is the definite outcome recorded in a single experiment by a particular apparatus.

Chapter 6

Photon Polarization and the Logic of a Two-State System

Quantum mechanics has already looked odd through interference and uncertainty, but in this lecture Susskind turns to a setting in which the oddness can be isolated with almost no technical overhead: the polarization of a photon. These notes follow Leonard Susskind's lecture closely, with explicit gratitude to Leonard Susskind, LazyingArt LLC, and Video2Book for the preserved lecture record from which the mathematical order and the blackboard evidence are reconstructed. The progression is the lecture's own: first the physical picture, then the symmetry puzzle, then the two-dimensional state space, then the enlargement from horizontal and vertical polarization to arbitrary angle, complex phase, and finally average value.

6.1 Why Polarization Is the Right Starting Point

The lecture opens by narrowing the photon to its simplest essential content. Let us imagine that the photon moves in a definite direction, say along the z-axis, with definite momentum. Then we are not discussing its motion at all; we are discussing only the remaining internal degree of freedom, namely polarization. In Susskind's language, the photon carries a little flag, and that flag lives in the plane perpendicular to the motion, the xy-plane.

This is a physical picture, not yet a quantum formalism. It is useful precisely because it is so economical. We do not need the full machinery of particle motion; we need only one clean degree of

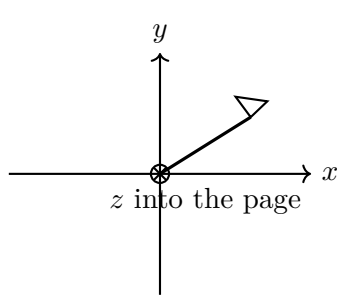


Figure 6.1: A photon moving along the z-axis carries a transverse polarization "flag" in the xy-plane.

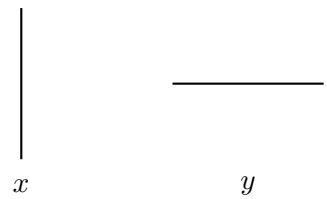


Figure 6.2: The horizontal and vertical polarizers that give the first two basis experiments.

freedom. The reward is that polarization gives us, in miniature, the logic of quantum mechanics itself.

At the same time Susskind is careful not to let the picture harden into naive classical language. We may say provisionally that the little flag points in some direction in the xy-plane, but the exact meaning of that sentence will only become clear after we translate the story into state vectors and observables. Before that translation, we first need to see how polarizers prepare and test the states.

6.2 Classical Polarizers and the Quantum Tension

A wire-grid polarizer is introduced only as far as the lecture needs it. The grid can support electric current in one direction but not in the orthogonal one. As a result, one component of the electric field is reflected and the perpendicular component passes through. Thus the grid prepares a transmitted wave polarized along a definite axis. It is that transmitted axis, not the grid lines themselves, that we use to label the polarizer.

Now send photons through one at a time. A horizontal polarizer prepares a photon in the horizontal state. If we place a second horizontal polarizer behind it, the photon goes through again. If we instead put a vertical polarizer behind the first, the photon does not go through. So a polarizer acts both as a preparer and as a measuring apparatus. It prepares a state when the photon passes, and it measures by pass-or-fail when the photon is tested by a later apparatus.

Up to this point there is nothing especially nonclassical. One could imagine bullets with little flags on them, together with an apparatus that lets one flag orientation through and blocks the other. The setup is deterministic. If the polarization matches the analyzer, the photon passes. If it does not, the photon is blocked, reflected, or absorbed. End of story.

But that is only the calm before the real difficulty. The lecture's explicit pivot is symmetry. If a polarizer may be held horizontally, and it may be held vertically, then rotational symmetry says it may be held at any intermediate angle. A photon can therefore be prepared with polarization axis at any direction in the xy-plane. Yet a fixed analyzer still produces only two outcomes: pass or fail. The photon is never chopped into pieces, partly transmitted and partly blocked. The outcomes are discrete even though the preparation angle varies continuously.

6.2.1 Question & Answer

Question. If polarization can point in any direction, why does a fixed measurement still return only two outcomes?

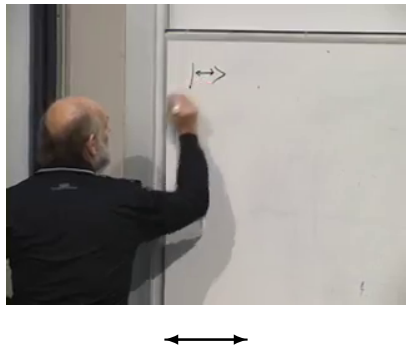


Figure 6.3: Bidirectional mark for a polarization axis.

Answer. That is the first real quantum question of the lecture. Continuity belongs to the space of possible preparations. Discreteness belongs to the outcome of a particular measurement. The classical picture tries to force both into the same mold, as if every intermediate polarization already carried a preexisting yes-or-no answer for every analyzer. Quantum mechanics does not do that. It assigns amplitudes to states, and a chosen analyzer converts those amplitudes into probabilities for its two possible outcomes.

This is why Susskind says the mystery begins here. We have a continuum of possible polarization states, but only two outcomes for any fixed polarizer. The tension between symmetry and discreteness is exactly what the two-dimensional quantum formalism must capture.

6.3 The x-y Basis and the First Observable

The lecture now restarts in a more formal register. We choose the simplest basis: horizontal and vertical polarization. Susskind labels them x and y. Before he writes vectors, however, he pauses over a notation choice that matters conceptually. There is no distinction between horizontal polarization "to the right" and horizontal polarization "to the left." A polarization axis is an axis, not a signed arrow. That is why he draws a double arrow.

With that convention in hand, we define

$$|x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{6.1}$$

They are normalized and orthogonal:

$$\langle x|x\rangle = 1, \qquad \langle y|y\rangle = 1, \qquad \langle x|y\rangle = 0. \tag{6.2}$$

The space of states is therefore two-dimensional. That does not mean there are only two possible pure states; it means that every polarization state will be built from two basis vectors.

The blackboard fragment above is useful precisely because it preserves the board layout: orthogonality at upper left, operator action in the main field. The visible multiplication is

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \tag{6.3}$$

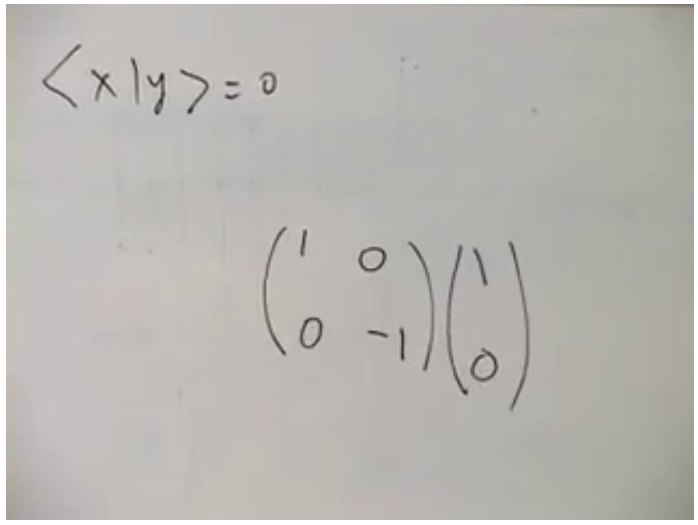


Figure 6.4: Orthogonality and matrix action in the x, y basis.

and the standard continuation of that displayed step is

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{6.4}$$

The first observable is built directly from the measuring procedure. Send the photon through an x-polarizer. Record $+1$ if it passes and -1 if it does not. Those numbers are conventional; Susskind emphasizes that we could have chosen others. But ± 1 are natural. The operator is defined by

$$\hat{P}_{xy}|x\rangle = +|x\rangle, \qquad \hat{P}_{xy}|y\rangle = -|y\rangle. \tag{6.5}$$

In the $(|x\rangle, |y\rangle)$ basis this means

$$P_{xy} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{6.6}$$

This is the first concrete example of the general postulate. An observable is represented by an operator. Its eigenvectors are the states in which the observable is definite, and its eigenvalues are the definite outcomes. The language is abstract, but the content here is still very physical: pass gives $+1$, fail gives -1.

6.4 From 45° to Arbitrary Angles

The lecture now extends the basis in the most controlled way possible. What if we rotate the polarizer by 45°? The outgoing state is neither $|x\rangle$ nor $|y\rangle$. Since it lies halfway between the two axes, it is natural that it have equal probabilities to pass an x-analyzer and a y-analyzer. This motivates

$$|/\rangle = \frac{1}{\sqrt{2}}(|x\rangle + |y\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{6.7}$$

Its orthogonal companion may be taken as

$$|\backslash\rangle = \frac{1}{\sqrt{2}}(|x\rangle - |y\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{6.8}$$

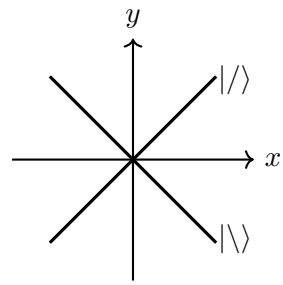


Figure 6.5: The slash and backslash polarization axes, rotated by 45° with respect to the x, y basis.

The overall sign is inessential; one could equally well put the minus sign on the first component. What matters is orthogonality:

$$\langle /|\backslash\rangle = \frac{1}{2} - \frac{1}{2} = 0. \tag{6.9}$$

The measurement rule now shows its first genuinely quantum face. The amplitude to find the photon in the x-state is

$$\langle x|/\rangle = \frac{1}{\sqrt{2}}, \tag{6.10}$$

so the probability that a slash-polarized photon passes an x-polarizer is

$$|\langle x|/\rangle|^2 = \frac{1}{2}. \tag{6.11}$$

Likewise

$$|\langle y|/\rangle|^2 = \frac{1}{2}. \tag{6.12}$$

Now we may ask for the observable that is definite in the slash/backslash basis. The transcript is visibly garbled at the moment the matrix is written, so the clean matrix must be presented cautiously. The natural reconstruction, and the unique symmetric matrix consistent with the stated eigenvalue equations, is

$$P_{/\backslash} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{6.13}$$

since

$$P_{/\backslash}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad P_{/\backslash}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = -\begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{6.14}$$

This is also the moment at which Susskind cleans up a conceptual point that is easy to lose. A state does not tell us what is observable; everything represented by an observable is observable. A state tells us what is definite. In the slash state, the slash/backslash observable is definite, but the original x, y observable is only probabilistic.

From here the lecture proceeds to an arbitrary angle θ. If the polarization axis is rotated by θ from horizontal, the corresponding unit vector has components $\cos\theta$ and $\sin\theta$. Thus

$$|\theta\rangle = \cos\theta\,|x\rangle + \sin\theta\,|y\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}. \tag{6.15}$$

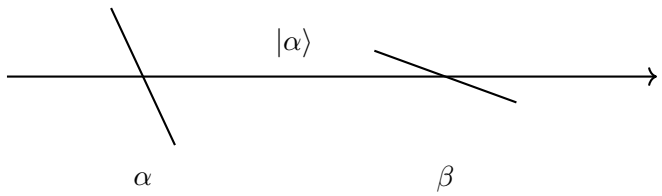


Figure 6.6: A photon prepared by an α-polarizer and tested by a second polarizer at angle β.

The orthogonal companion is the state at $\theta + \pi/2$:

$$|\theta + \pi/2\rangle = -\sin\theta\,|x\rangle + \cos\theta\,|y\rangle = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}. \tag{6.16}$$

These are normalized because $\cos^2\theta + \sin^2\theta = 1$, and orthogonal because

$$\langle\theta|\theta + \pi/2\rangle = \cos\theta(-\sin\theta) + \sin\theta\cos\theta = 0. \tag{6.17}$$

Now we ask the experimental question that quantum mechanics is built to answer. A photon emerges from a θ-polarizer. What is the probability that it passes an x-polarizer? The amplitude is

$$\langle x|\theta\rangle = \cos\theta, \tag{6.18}$$

so

$$P(\theta \to x) = |\langle x|\theta\rangle|^2 = \cos^2\theta. \tag{6.19}$$

Likewise

$$P(\theta \to y) = |\langle y|\theta\rangle|^2 = \sin^2\theta, \tag{6.20}$$

and the total probability is 1.

Trigonometric interlude. At this point the lecture pauses for a small table of trigonometric identities, not as a digression but because the general two-polarizer experiment now demands them:

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta, \tag{6.21}$$
$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta, \tag{6.22}$$
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha, \tag{6.23}$$
$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha, \tag{6.24}$$
$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha, \tag{6.25}$$
$$\sin 2\alpha = 2\sin\alpha\cos\alpha. \tag{6.26}$$

If the first polarizer prepares

$$|\alpha\rangle = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix}, \qquad \langle\beta| = \begin{pmatrix} \cos\beta & \sin\beta \end{pmatrix}, \tag{6.27}$$

then the amplitude to pass the second polarizer is

$$\langle\beta|\alpha\rangle = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta). \tag{6.28}$$

This is not yet the probability. The lecture makes a point of this: the quantity can be negative, so it cannot itself be a probability. The transmission probability is the square of the amplitude,

$$P(\alpha \to \beta) = |\langle\beta|\alpha\rangle|^2 = \cos^2(\alpha - \beta). \tag{6.29}$$

The answer depends only on the relative angle, as symmetry says it must, and yet the actual outcome of each trial remains one of two discrete possibilities. This is exactly the pattern the lecture set out to display.

6.5 Intermediate Polarizers and the Observable P_θ

After the break the lecture reopens not with a new theorem but with a puzzle raised by students. That is the right rhythm for the notes as well.

6.5.1 Question & Answer

Question. How can adding another polarizer increase the chance of transmission?

Answer. Suppose a photon has passed a θ-polarizer, so it is in the state $|\theta\rangle$. If we test it immediately with a $\theta + \pi/2$ polarizer, the probability of transmission is

$$P(\theta \to \theta + \pi/2) = \cos^2(\pi/2) = 0. \tag{6.30}$$

The direct channel is completely closed.

Now insert a horizontal polarizer in between. First the photon must pass the middle polarizer:

$$P(\theta \to x) = \cos^2\theta. \tag{6.31}$$

But if it does pass, the state has changed. It is no longer $|\theta\rangle$; it has been prepared anew as $|x\rangle$. The probability that this new state passes the final polarizer is therefore

$$P(x \to \theta + \pi/2) = \sin^2\theta. \tag{6.32}$$

Hence the total probability is not an addition but a product of successive conditional probabilities:

$$P(\theta \to x \to \theta + \pi/2) = \cos^2\theta\,\sin^2\theta. \tag{6.33}$$

The surprise is not that extra apparatus creates photons that were not there. The surprise is that measurement changes the state. By inserting a polarizer in between, we do not merely diminish an existing channel; we alter the state on which the next analyzer acts. That is why Susskind calls this one of the mildly freaky interference phenomena.

The lecture then returns to the general observable $\hat{P}_\theta$. Here Susskind is candid: solving for the matrix component by component is not very illuminating, but seeing the answer and checking it is reassuring. We want an operator with

$$\hat{P}_\theta|\theta\rangle = +|\theta\rangle, \qquad \hat{P}_\theta|\theta + \pi/2\rangle = -|\theta + \pi/2\rangle. \tag{6.34}$$

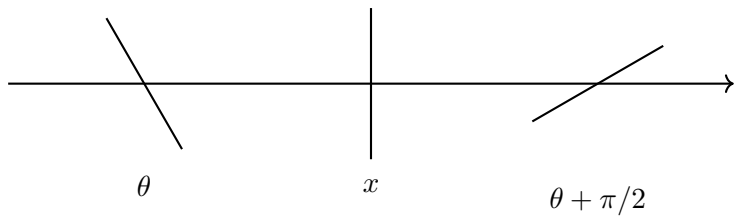


Figure 6.7: The inserted horizontal polarizer prepares a new state. That is why the forbidden direct channel can reopen with nonzero probability.

The transcript is partially garbled where the matrix is first written, but the cleaned result is standard and fully consistent with the stated eigenvalue equations:

$$P_\theta = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}. \tag{6.35}$$

To check it, set

$$c = \cos\theta, \qquad s = \sin\theta, \qquad |\theta\rangle = \begin{pmatrix} c \\ s \end{pmatrix}. \tag{6.36}$$

Then

$$P_\theta|\theta\rangle = \begin{pmatrix} c^2 - s^2 & 2sc \\ 2sc & s^2 - c^2 \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \tag{6.37}$$

$$= \begin{pmatrix} (c^2 - s^2)c + 2sc\,s \\ 2sc\,c + (s^2 - c^2)s \end{pmatrix} \tag{6.38}$$

$$= \begin{pmatrix} c(c^2 + s^2) \\ s(c^2 + s^2) \end{pmatrix} = \begin{pmatrix} c \\ s \end{pmatrix} = |\theta\rangle. \tag{6.39}$$

The companion vector $|\theta + \pi/2\rangle$ is similarly sent to minus itself. Thus the matrix has exactly the required eigenvectors and eigenvalues.

The 90° rotation now has a clean operator statement:

$$P_{\theta+\pi/2} = -P_\theta. \tag{6.40}$$

That is just the exchange of the positive and negative eigenvalues. The orthogonal analyzer does not introduce a wholly different observable; it flips the sign convention for the same pair of eigenstates.

6.6 Complex Coefficients and Circular Polarization

At this point Susskind widens the state space one more time. We have not yet used complex numbers, and quantum mechanics is built from them. So we introduce a new state,

$$|c_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \tag{6.41}$$

together with its orthogonal partner

$$|c_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}. \tag{6.42}$$

Normalization requires complex conjugation:

$$\langle c_+|c_+\rangle = \frac{1}{2}(1\cdot 1 + (-i)\cdot i) = 1. \tag{6.43}$$

Orthogonality does too:

$$\langle c_-|c_+\rangle = \frac{1}{2}(1\cdot 1 + i\cdot i) = 0. \tag{6.44}$$

This is a point the lecture emphasizes strongly. If we forget the bra and simply dot the columns together, we get the wrong answer. The bra is already the complex-conjugated object.

From the wave point of view, these states correspond to a phase difference between the x- and y-components of the electric field. If

$$E_y = \cos(z-ct), \qquad E_x = \cos(z-ct), \tag{6.45}$$

then the two components are in phase and the result is just linear polarization at 45°. Circular polarization appears when the components are out of phase by 90°:

$$E_y = \cos(z-ct), \qquad E_x = \sin(z-ct), \tag{6.46}$$

or with the opposite sign in the second equation. At a fixed point in space the electric field rotates in time. That is the wave picture behind the complex state vector.

The lecture informally calls the two states left and right circular polarization, but Susskind explicitly remarks that he may have the handedness labels mixed up. The important thing for us is the pair of states $\frac{1}{\sqrt{2}}(1,\pm i)^T$, not the convention for the names.

Now test a circularly polarized photon with a linear polarizer at angle θ. Since

$$|\theta\rangle = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}, \tag{6.47}$$

the amplitude is

$$\langle\theta|c_+\rangle = \frac{1}{\sqrt{2}}(\cos\theta + i\sin\theta). \tag{6.48}$$

The probability is the modulus squared,

$$|\langle\theta|c_+\rangle|^2 = \frac{1}{2}(\cos\theta + i\sin\theta)(\cos\theta - i\sin\theta) \tag{6.49}$$

$$= \frac{1}{2}(\cos^2\theta + \sin^2\theta) = \frac{1}{2}. \tag{6.50}$$

So a circularly polarized photon passes any linearly polarized analyzer with probability 1/2. That is the precise fact the lecture uses. Beyond that lies the wider family of elliptically polarized states, but Susskind only gestures toward them here and postpones the corresponding observables.

A quarter-wave plate is mentioned as the device that produces circular polarization by shifting the two linear components out of phase. The details are left for the next lecture, which is exactly right: the mathematical point needed here is only that complex phase opens new states which are not visible in the real two-parameter picture of linear polarization alone.

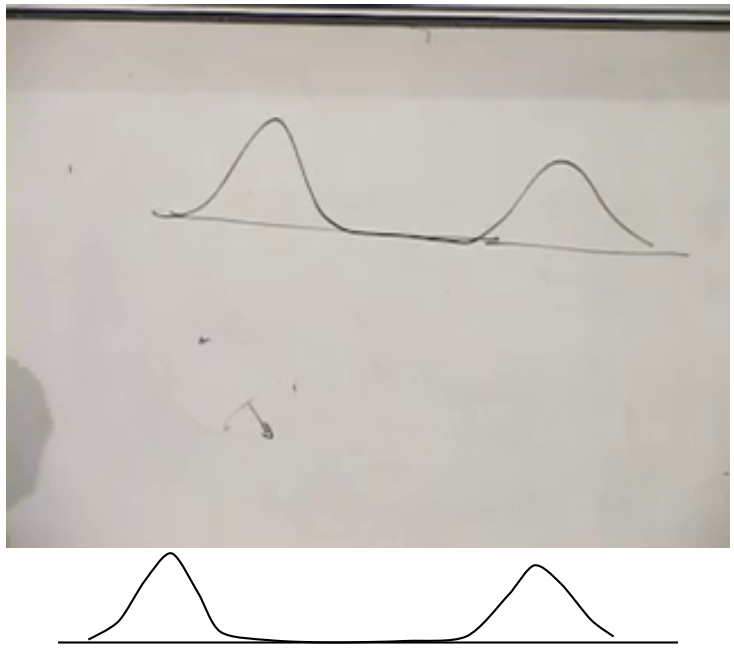


Figure 6.8: Symmetric two-lump wavefunction profile.

6.7 Incompatible Observables and Average Value

By now we have a whole family of polarization observables: horizontal/vertical, slash/backslash, the general P_θ, and a circular observable left as homework. The common structural fact is that different polarization observables generally do not commute. The reason is already visible at the level of eigenvectors. Horizontal and vertical states are one basis. Slash and backslash are another. Circular states are a third. In general they do not share eigenvectors, so they are not simultaneously definite.

The special case is the orthogonal pair θ and $\theta + \pi/2$. Their observables are just negatives of each other:

$$P_{\theta+\pi/2} = -P_\theta. \tag{6.51}$$

Therefore they commute. Measuring one automatically tells us the other, since the eigenvalue has simply changed sign. But for genuinely different angles, or for linear versus circular polarization, the observables are incompatible. One cannot have a polarizer that prepares definite x-polarization and definite θ-polarization at the same time.

The lecture now makes its final pivot. We have been calculating probabilities. What about average value? Susskind immediately objects to the standard terminology. “Expectation value” is, he says, a poor choice of words, because it often names the last value one should expect to see in a single trial.

The sketch above carries no algebra on the board, but it carries the geometry of the example. A symmetric wavefunction with one lump on the left and one on the right has

$$\langle x \rangle = 0, \tag{6.52}$$

yet the particle is least likely to be found at the middle. So the expectation value is not the value one should expect to observe on a single shot. It is an average value.

Now let K be any Hermitian observable with eigenvectors $|n\rangle$ and eigenvalues λ_n:

$$K|n\rangle = \lambda_n|n\rangle. \tag{6.53}$$

If $|\psi\rangle$ is any state, define the average value by the sandwich

$$\langle K\rangle_\psi = \langle\psi|K|\psi\rangle. \tag{6.54}$$

There is no extra star on K; the bra already carries the complex conjugation. Insert the identity in the form

$$\sum_n |n\rangle\langle n| = I. \tag{6.55}$$

Then

$$\begin{aligned}
\langle K\rangle_\psi &= \langle\psi|K|\psi\rangle && (6.56)\\
&= \left\langle \psi \middle| K\left(\sum_n |n\rangle\langle n|\right)\middle| \psi \right\rangle && (6.57)\\
&= \sum_n \langle\psi|K|n\rangle\langle n|\psi\rangle && (6.58)\\
&= \sum_n \lambda_n \,\langle\psi|n\rangle\langle n|\psi\rangle && (6.59)\\
&= \sum_n \lambda_n \,|\langle n|\psi\rangle|^2. && (6.60)
\end{aligned}$$

This is exactly the ordinary weighted average of all possible outcomes: eigenvalue times probability, summed over the basis of outcomes.

For polarization the simplest example uses the original operator P_{xy} in the state $|\theta\rangle$:

$$\begin{aligned}
\langle P_{xy}\rangle_\theta &= \langle\theta|P_{xy}|\theta\rangle && (6.61)\\
&= \begin{pmatrix}\cos\theta & \sin\theta\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} && (6.62)\\
&= \cos^2\theta - \sin^2\theta && (6.63)\\
&= \cos 2\theta. && (6.64)
\end{aligned}$$

This lies between $+1$ and -1, as it should. But again, it is not a new measurement outcome. The actual outcomes are still $+1$ and -1. The quantity $\cos 2\theta$ is their average over many repetitions.

6.8 Summary

Polarization is the right starting point because it is small enough to see clearly and rich enough to contain the full logical structure of quantum mechanics. We begin with a photon moving along z, carrying a transverse flag in the xy-plane. We then discover the central tension: symmetry allows a continuum of preparation directions, but every fixed analyzer still returns only two outcomes. From that point the lecture unfolds methodically. We choose a basis, construct observables, rotate the basis, extract amplitudes and probabilities, see how measurements update the state, admit complex phase and circular polarization, and finally learn to distinguish a probability from an average value. The system is minimal, but the logic is already complete.

Chapter 7

Phase, Polarization, and Unitary Evolution

This lecture begins with unfinished business and ends with the basic structure of quantum dynamics. We start by returning to photon polarization, because that two-state system is still the cleanest laboratory for the ideas on the board. From there the lecture pauses to ask what state-vectors are actually for, why multiplying a state by an overall phase changes nothing observable, how many parameters are needed to specify the polarization of a photon, and finally what kind of transformations can move states forward in time without disturbing their physical relations. What follows keeps close to that progression. The original lecture is due to Leonard Susskind, and the supporting curation pipeline is credited to LazyingArt LLC and Video2Book.

7.1 Polarization Observables Revisited

The lecture opens by asking for the homework answer: what operator has the circular polarization states as its eigenvectors? Before going on, Susskind quickly reconstructs the three polarization observables that have been developed in the preceding lectures. This is not merely review. He wants the two-state example back on the board before any more formalism is introduced.

For horizontal and vertical polarization we take

$$\hat{P}_{xy} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad |x\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{7.1}$$

Thus $|x\rangle$ and $|y\rangle$ are eigenvectors of $\hat{P}_{xy}$ with eigenvalues $+1$ and -1.

For linear polarization along an axis making angle θ with the x-direction, the state is written

$$|\theta\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}. \tag{7.2}$$

An orthogonal partner may be chosen as

$$|\theta + \pi/2\rangle = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}, \tag{7.3}$$

or with the opposite overall sign. The lecture explicitly remarks that the sign choice does not matter. That is a local comment at first, but by the end of the lecture it becomes a special case of the general irrelevance of overall phase.

The special 45° states are therefore

$$|45^\circ, +\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad |45^\circ, -\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \tag{7.4}$$

and the corresponding observable is

$$\hat{P}_{45} = \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}. \tag{7.5}$$

Then comes the genuinely new case, the first place in this story where complex numbers are unavoidable:

$$|C_+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}, \qquad |C_-\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}. \tag{7.6}$$

These are again orthogonal, provided we remember the rule that inner products involve complex conjugation. The circular-polarization observable is

$$\hat{P}_{\text{circ}} = \begin{pmatrix}0 & -i\\i & 0\end{pmatrix}, \tag{7.7}$$

and one checks directly that

$$\hat{P}_{\text{circ}}|C_\pm\rangle = \pm|C_\pm\rangle. \tag{7.8}$$

Already the lecture is stressing an operational reading of the mathematics. Orthogonality is not being introduced as a decorative algebraic relation. Two polarization states are orthogonal when they are distinguishable in a single ideal experiment.

7.2 What Quantum States Are For

Now the lecture steps back and asks a broader question: where are we relative to the first discussion of classical mechanics? In the classical setting of a coin, a die, or any finite collection of states, a state was just a point in a set. An observable was merely a numerical label assigned to each point. Time evolution then carried one point to another.

Quantum mechanics has already gone beyond that. The space of states is now a vector space, and observables are operators on that space. But we are not quite ready for dynamics. First we need to say clearly what physically interesting things one does with states once one has them.

There are, in this lecture, two central uses.

First, if the system is in the state $|\psi\rangle$ and we are interested in an observable $\hat{K}$, we form the expectation value

$$\langle\hat{K}\rangle_\psi = \langle\psi|\hat{K}|\psi\rangle. \tag{7.9}$$

This is the quantity we would obtain by preparing the same state over and over and averaging many measurements of K.

Second, suppose $|a\rangle$ and $|b\rangle$ are eigenvectors of two different observables:

$$\hat{K}|a\rangle = \alpha|a\rangle, \qquad \hat{L}|b\rangle = \beta|b\rangle. \tag{7.10}$$

If we prepare the system with a definite value $K = \alpha$ and then measure L, the probability to find $L = \beta$ is

$$P(a \to b) = |\langle b|a\rangle|^2 = \langle a|b\rangle\langle b|a\rangle. \tag{7.11}$$

The lecture notes that the same number is obtained if we reverse the order of preparation and measurement. At this level the two great questions are ensemble averages and transition probabilities.

This is the bridge to the phase discussion. If some change in the state-vector does not alter expectation values or transition probabilities, then it has no physical significance.

7.3 Overall Phase Has No Physical Meaning

Before discussing phase, Susskind inserts the normalization rule. Physical state-vectors are taken to be unit vectors:

$$\langle\psi|\psi\rangle = 1. \tag{7.12}$$

That fixes the total probability to be one. But it does not yet exhaust the redundancies in the description.

To make the next point precise, the lecture briefly reviews complex numbers. A complex number can be written in rectangular form as $x + iy$ and in polar form as

$$z = re^{i\theta}. \tag{7.13}$$

Its complex conjugate is

$$z^* = re^{-i\theta}. \tag{7.14}$$

A *pure phase* is a complex number of unit modulus, so $r = 1$ and

$$z = e^{i\theta}, \qquad z^* = e^{-i\theta}, \qquad zz^* = 1. \tag{7.15}$$

These are the points on the unit circle in the complex plane.

On the board this geometric picture is paired with a comparison of two vectors,

$$|A\rangle, \qquad e^{i\theta}|A\rangle. \tag{7.16}$$

Mathematically they are different vectors. The second is the first multiplied by a complex number. But physically the lecture insists that they describe the same state. The point is not asserted as a slogan; it is checked against the two things states are used for.

Let

$$|\psi'\rangle = e^{i\chi}|\psi\rangle. \tag{7.17}$$

Then the corresponding bra is

$$\langle\psi'| = e^{-i\chi}\langle\psi|. \tag{7.18}$$

So the expectation value of any observable is unchanged:

$$\langle\psi'|\hat{K}|\psi'\rangle = e^{-i\chi}\langle\psi|\hat{K}\, e^{i\chi}|\psi\rangle \tag{7.19}$$

$$= e^{-i\chi}e^{i\chi}\langle\psi|\hat{K}|\psi\rangle \tag{7.20}$$

$$= \langle\psi|\hat{K}|\psi\rangle. \tag{7.21}$$

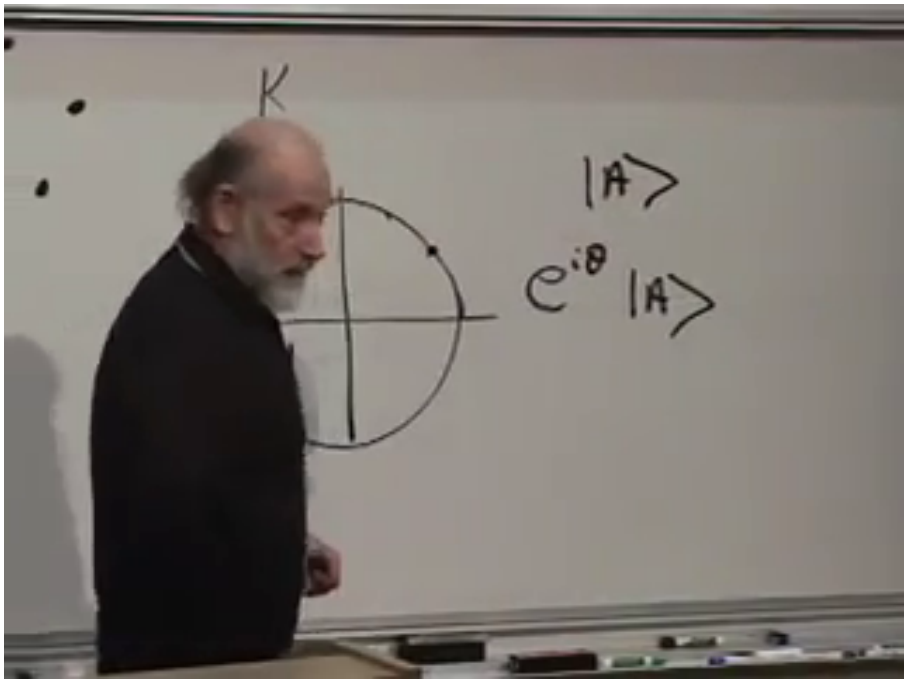


Figure 7.1: Phase-related state vectors and unit-circle sketch from the lecture board.

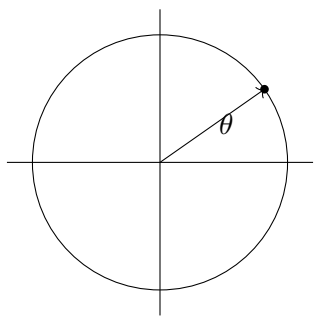


Figure 7.2: Clean reconstruction of the phase-circle picture used in the lecture.

The phase cancels because it is only an ordinary number.

The same happens for transition probabilities. If, for example, we replace $|b\rangle$ by $e^{i\chi}|b\rangle$, then

$$|\langle e^{i\chi}b|a\rangle|^2 = \langle a|e^{i\chi}b\rangle\langle e^{i\chi}b|a\rangle \tag{7.22}$$

$$= e^{i\chi}e^{-i\chi}\langle a|b\rangle\langle b|a\rangle \tag{7.23}$$

$$= |\langle b|a\rangle|^2. \tag{7.24}$$

Again the phase disappears.

This is why the lecture is willing to say that there is no physical significance to the overall phase of the wave function. The physical state is not a single vector but an equivalence class under multiplication by a pure phase.

A simple example is already available:

$$|y\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad -|y\rangle = \begin{pmatrix} 0 \\ -1 \end{pmatrix}. \tag{7.25}$$

The second vector is $e^{i\pi}|y\rangle$. It is not equal to the first one, but it has the same physical significance. Likewise

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix} \qquad \text{and} \qquad \frac{1}{\sqrt{2}}\begin{pmatrix}i\\ -1\end{pmatrix} \tag{7.26}$$

differ by the phase $i = e^{i\pi/2}$ and therefore represent the same circular polarization state.

7.3.1 Question & Answer

If $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are mathematically different vectors, why are they physically the same?

Because all the physical questions under discussion in this lecture are built from bra-ket sandwiches and squared overlaps. In every such expression an overall phase meets its complex conjugate and cancels. So the distinction is real mathematically but empty physically. The lecture briefly experiments with an ad hoc symbol for this equivalence; in the notes it is safer simply to say that the two vectors have the same physical significance.

7.4 Counting the Parameters of Photon Polarization

At this point phase indifference begins to do real work. It is not merely that phase happens to cancel from a few formulas; we may now divide by that redundancy and count the truly physical parameters of photon polarization.

Start from the most general two-component state,

$$|\psi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}, \tag{7.27}$$

with α and β complex. That is four real parameters. Normalization imposes one condition,

$$\alpha^*\alpha + \beta^*\beta = 1, \tag{7.28}$$

so we are left with three.

Now use the fact that overall phase carries no physical information. By multiplying the whole state by a suitable phase, we can always choose the upper component to be real. Call it $a \in \mathbb{R}$. Then normalization becomes

$$a^2 + \beta^*\beta = 1. \tag{7.29}$$

Hence

$$\beta^*\beta = 1 - a^2, \tag{7.30}$$

and so the lower component may be written as

$$\beta = \sqrt{1-a^2}\, e^{i\phi}. \tag{7.31}$$

Here the lecture repairs the notation: the phase is now renamed ϕ so that θ may continue to denote a linear-polarization angle.

Thus the general physical polarization state may be written in the gauge-fixed form

$$|\psi\rangle = \begin{pmatrix} a \\ \sqrt{1-a^2}\, e^{i\phi}\end{pmatrix}. \tag{7.32}$$

There are exactly two real parameters left: the real number a and the phase angle ϕ.

The important point is not merely the counting itself. The lecture immediately asks what those two parameters mean physically. That question is what carries us from state-vector coordinates back to the actual geometry of electromagnetic waves.

7.5 From State Vectors to Elliptical Polarization Geometry

The lecture now shifts language. Up to this point we have been talking in the coordinates of a two-dimensional complex vector space. But photon polarization is not only an algebraic construction. It is also the motion of the electric field of an electromagnetic wave in the plane transverse to propagation.

A plane-polarized wave has its electric field along a single axis. If the wave is moving into the board, the direction of that axis is one parameter. So purely linear polarization gives only a one-parameter family.

If we combine two linear polarizations whose components are in phase, the result is still plane polarization, just along a different axis. But if the two components are out of phase, the electric field need no longer move on a line. It may move on a circle, and more generally on an ellipse.

The lecture gives simple examples. If

$$E_x = \cos(x - ct), \qquad E_y = \sin(x - ct), \tag{7.33}$$

then at fixed x, say $x = 0$, the tip of the electric field moves on a circle. That is circular polarization. If instead we take

$$E_x = \cos(x - ct), \qquad E_y = \frac{1}{2}\sin(x - ct), \tag{7.34}$$

then at fixed x the field traces an ellipse, in this case with major axis along the x-direction.

So the geometry carries two physical parameters:

1. the direction of the major axis,
2. the shape of the ellipse, described in the lecture by its eccentricity or aspect-ratio parameter.

This matches the algebraic count perfectly. The two real parameters in the state-vector are not literally the angle and the eccentricity, but there is a one-to-one correspondence: some functions of them are the angle of the long axis and the shape parameter of the ellipse.

The lecture also pauses over the limiting cases. When the ellipse collapses to a line, we recover plane polarization. When it becomes a circle, the major axis ceases to matter because no axis is distinguished.

7.5.1 Question & Answer

Where did clockwise versus counterclockwise polarization go if there are only two real parameters?

The lecture answers this in a deliberately geometric way. Start from circular polarization and squeeze the circle into an ellipse. Continue squeezing until the ellipse collapses to a line. If one keeps going past that point, the upper and lower branches exchange roles and the sense of rotation reverses.

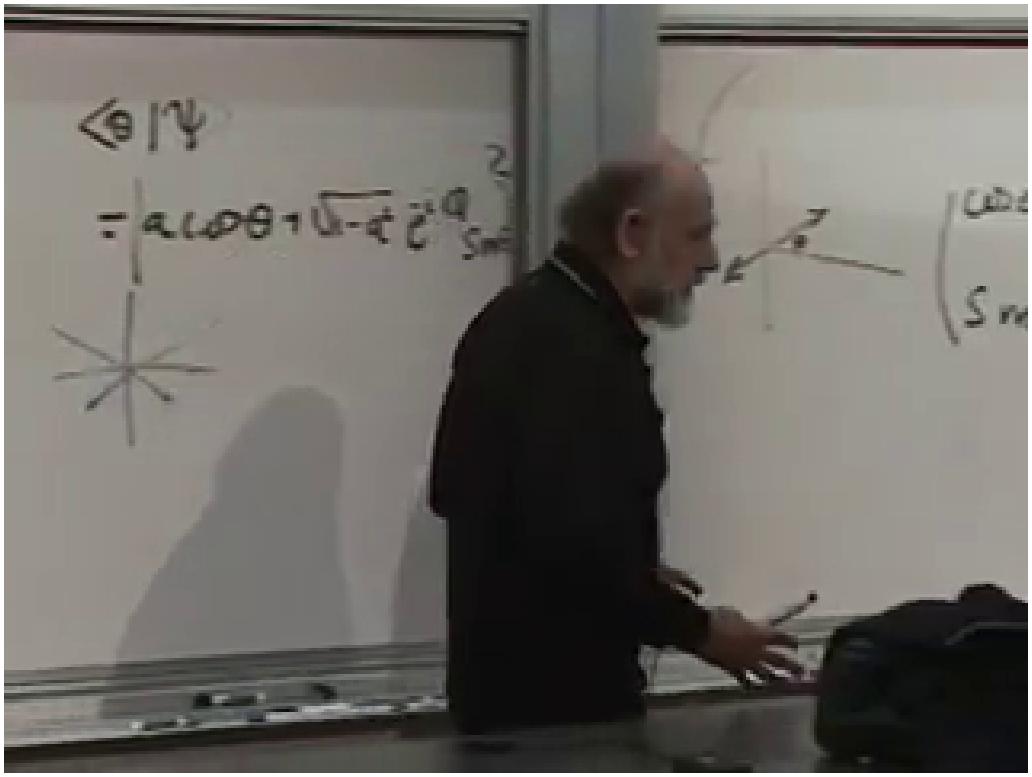


Figure 7.3: State-vector overlap and geometric sketches as laid out across the lecture boards.

So the single shape parameter is extended through zero and into negative values. In that lecture language, one may think of the second sense of rotation as being encoded by *negative eccentricity*. It is a heuristic description, but it is exactly the one used to show that two real parameters are enough.

7.6 Overlap With Linear Polarization and the Major Axis

The lecture now asks a more practical question. Suppose a photon is in a general elliptically polarized state. How do we recover the direction of its major axis from measurement probabilities?

We compare the elliptic state with the linearly polarized state at analyzer angle θ:

$$|\theta\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad |\psi\rangle = \begin{pmatrix} a \\ \sqrt{1-a^2}\, e^{i\phi} \end{pmatrix}. \tag{7.35}$$

Operationally this means: put a linear polarizer at angle θ in the path of the photon and ask for the probability that the photon passes through.

On the board the bra-ket order is partly obscured, and in the spoken derivation Susskind corrects himself by remembering that one factor must be complex conjugated. In a cleaned form consistent with that correction, we may record the overlap as

$$\langle\psi|\theta\rangle = a\cos\theta + \sqrt{1-a^2}\, e^{-i\phi}\sin\theta. \tag{7.36}$$

If one writes the overlap in the opposite order, the phase is conjugated accordingly; the probability is the same either way.

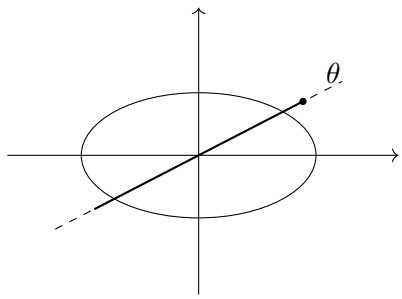


Figure 7.4: Conservative reconstruction of the geometric guide used in the lecture: vary the analyzer direction until the overlap with the elliptic state is maximal.

The transmission probability is therefore

$$P_\theta(\psi) = |\langle\psi|\theta\rangle|^2. \tag{7.37}$$

Here the lecture deliberately stops short of a fully polished computation. Susskind does not expand the modulus squared on the board; instead he assigns that algebra as homework. But the next step is perfectly clear. We keep ϕ fixed, regard θ as the variable analyzer angle, and maximize:

$$\frac{d}{d\theta}P_\theta(\psi) = 0. \tag{7.38}$$

The value of θ that maximizes $P_\theta(\psi)$ gives the direction of the major axis.

The lecture says much less about the ellipse parameter itself. The suggestion is qualitative rather than finished: compare the probabilities for transmission along the major and minor directions, and their difference measures the asymmetry of the ellipse. Since the lecture does not derive a closed formula here, the notes should not invent one.

7.7 Time Evolution, Hermitian Operators, and the Unitary Pivot

Up to this point the states have been standing still on the board. The next step is the big one: how do quantum states change with time? The lecture begins from the classical analogy and then formalizes the quantum answer.

7.7.1 Relations Between States

In classical mechanics with finitely many states, time evolution is essentially a permutation of those states. Different initial states remain different. If two distinct states both evolved into the same final state, information about the initial condition would be lost.

Quantum mechanics has a richer notion of relation between states. Two normalized states may be the same, orthogonal, or somewhere in between. The operational example remains polarization. A vertically polarized photon does not pass a horizontal polarizer, so

$$\langle y|x\rangle = 0 \tag{7.39}$$

expresses a definite physical distinction. By contrast, for normalized states,

$$\langle B|A\rangle = 1 \tag{7.40}$$

means that the two states are the same state, up to the phase distinction that we have already agreed to ignore physically.

So the lecture identifies the basic relation between states with the inner product. Zero means orthogonal, one means identical normalized states, and intermediate values measure partial overlap and therefore partial distinguishability.

The quantum analog of classical distinction-preservation is then that inner products remain fixed in time. If

$$|A\rangle \to |A'\rangle, \qquad |B\rangle \to |B'\rangle, \tag{7.41}$$

then

$$\langle B'|A'\rangle = \langle B|A\rangle. \tag{7.42}$$

In particular, states that start orthogonal stay orthogonal.

This is presented in the lecture as a physical principle rather than a theorem derived from something more primitive. It is taken as an observed fact, one that fits together with deeper conservation ideas and, above all, with the requirement that quantum time evolution not destroy distinctions between states.

7.7.2 Hermitian Conjugation And Hermitian Operators

To implement that idea we need more mathematics. The lecture now reviews linear operators with some care.

Let L be a linear operator defined by its action on kets. If

$$L|A\rangle = |C\rangle, \tag{7.43}$$

then we also want a consistent way to let such an operator appear in bra-ket sandwiches. In a complete orthonormal basis,

$$\sum_n |n\rangle\langle n| = I. \tag{7.44}$$

This allows us to define matrix elements

$$\langle B|L|A\rangle, \tag{7.45}$$

which the lecture emphasizes can be interpreted either by first acting with L on the ket and then taking the inner product with $\langle B|$, or by using the corresponding action on bras and then pairing with $|A\rangle$.

From there the lecture introduces the Hermitian conjugate. Cleanly stated, the rule is

$$\langle n|L^\dagger|m\rangle = \langle m|L|n\rangle^*, \tag{7.46}$$

or in matrix notation,

$$(L^\dagger)_{nm} = L^*_{mn}. \tag{7.47}$$

So Hermitian conjugation is obtained by transposing and then complex conjugating.

An operator is Hermitian when

$$L = L^\dagger. \tag{7.48}$$

This immediately implies two componentwise facts:

$$L_{ii} = L_{ii}^*, \tag{7.49}$$

so the diagonal entries are real, and

$$L_{ij} = (L_{ji})^*, \tag{7.50}$$

so entries reflected across the diagonal are complex conjugates of one another.

The lecture then checks the polarization observables one by one:

$$\hat{P}_{xy} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \hat{P}_{45} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{P}_{\text{circ}} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{7.51}$$

All three are plainly Hermitian.

At this point the lecture recalls the standard physical content of Hermitian operators. Their expectation values are real, their eigenvalues are real, and their eigenvectors form a complete orthonormal basis. That is why observables are represented by Hermitian operators.

The lecture also remarks that real quantum systems are usually infinite-dimensional. The finite-dimensional matrix language is being used because it displays the structure clearly, and because in practice even infinite-dimensional problems are often approximated by very large matrices.

7.7.3 Unitary Operators

But Hermitian operators are not the transformations we need for time evolution. Observables tell us what can be measured. Time evolution requires operators that preserve relations between states.

Let U be such a transformation:

$$U|A\rangle = |A'\rangle, \qquad U|B\rangle = |B'\rangle. \tag{7.52}$$

Then the corresponding bras transform with $U^\dagger$:

$$\langle A'| = \langle A|U^\dagger, \qquad \langle B'| = \langle B|U^\dagger. \tag{7.53}$$

Now impose the physical requirement that the inner product be preserved:

$$\langle B'|A'\rangle = \langle B|U^\dagger U|A\rangle \tag{7.54}$$

$$= \langle B|A\rangle. \tag{7.55}$$

If this is true for every pair of states $|A\rangle$ and $|B\rangle$, then the operator between them must be the identity:

$$U^\dagger U = I. \tag{7.56}$$

Equivalently,

$$U^\dagger = U^{-1}. \tag{7.57}$$

That is the defining property of a unitary operator.

The lecture insists on the right reading of this equation. A unitary operator is not itself the identity. It can move vectors around the state space, rotate polarization directions, or represent time evolution. What makes it unitary is that it moves vectors around without altering the inner products that encode physical relations and probabilities.

This is the closing message of the lecture. Systems change with time by unitary transformations. The detailed properties of unitary eigenvalues are left for the next lecture, together with the introduction of Hamiltonians. Here we stop at the threshold: the correct mathematical form of quantum evolution has now been identified.

7.8 Summary

The lecture begins by tying up the polarization story and ends by opening the door to quantum dynamics. Along the way we see three ideas lock together. First, the physically meaningful uses of a state-vector are expectation values and transition probabilities. Second, both are insensitive to an overall phase, so the physical state is represented by a normalized vector only up to multiplication by a pure phase. Third, once the relation between states is recognized as the inner product, the natural transformations of state space are those that preserve that inner product, namely the unitary operators. Photon polarization remains the anchor throughout, but the lesson is much broader: the geometry of state space, not just the list of possible outcomes, is what controls the way quantum mechanics evolves in time.

Chapter 8

Time Evolution, Interference, and the Schrödinger Equation

This lecture sets out to do one thing: to begin the serious study of how quantum systems change with time. But Susskind does not begin by writing down the Schrödinger equation. He first backs up, gives a very compressed reminder of classical evolution, and isolates the one structural fact that matters for what follows: the Hamiltonian generates motion. From there the lecture moves in a tight sequence. We pass from preserved relations between states to unitary evolution, from infinitesimal evolution to a Hermitian generator, from that generator to the abstract Schrödinger equation, and then immediately into examples where the phases produced by time evolution become measurable through interference. These notes follow Leonard Susskind's lecture, with supporting board-frame evidence curated by LazyingArt LLC and Video2Book.

8.1 From Classical Flow To The Hamiltonian

The opening classical recap is deliberately brief. For a discrete system, time evolution amounts to jumps from one state to another, and those jumps naturally come in sampled time intervals. Continuous time becomes natural only for systems with continuous variables. A particle moving on a line is the simplest example: at each instant it has a position x and a momentum p, and its motion is described not by hops on a finite graph but by a flow in phase space.

The Hamiltonian is the object that describes that flow. For a particle on a line it is a function of x and p, and in the standard mechanical example it takes the form

$$H(x,p) = \frac{p^2}{2m} + U(x). \tag{8.1}$$

In the board frame the equality sign and punctuation are not completely secure, but the printed expression is the natural completion of the visible formula.

The flow is determined by Hamilton's equations,

$$\dot{x} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial x}. \tag{8.2}$$

In particular, for the Hamiltonian above we have $\partial H/\partial p = p/m$, which is the velocity, while $-\partial H/\partial x$ is the force.

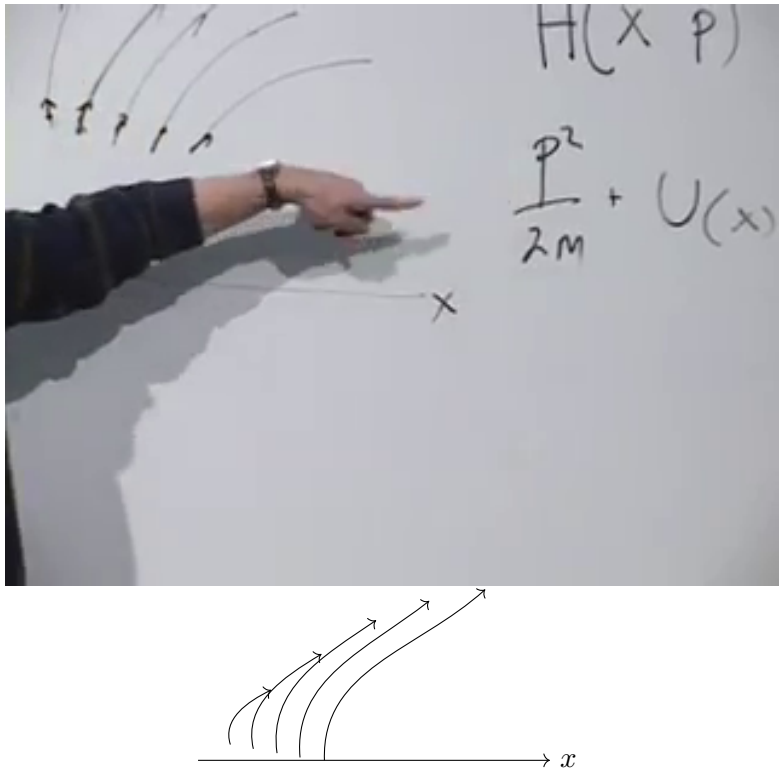


Figure 8.1: Classical Hamiltonian and flow-line sketch. The redraw is intentionally conservative: the lecture frame clearly secures the curved flow and the x label, but not a full axis system.

If $f(x, p)$ is any observable quantity, its time derivative is obtained from the chain rule:

$$\dot{f} = \frac{\partial f}{\partial x}\dot{x} + \frac{\partial f}{\partial p}\dot{p} \tag{8.3}$$

$$= \frac{\partial f}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial H}{\partial x}. \tag{8.4}$$

This combination is the Poisson bracket with the Hamiltonian:

$$\dot{f} = \{f, H\}_{\mathrm{PB}}. \tag{8.5}$$

Susskind immediately compresses all of this into the sentence that matters: the Hamiltonian generates time evolution. The Poisson-bracket formalism is not being reviewed for its own sake. It is there to prepare the quantum replacement.

8.2 From Preserved Relations To Unitary Evolution

The pivot is sharp. Classical mechanics, Susskind says, always felt more obscure than it should have; something simpler must underlie it, and what underlies it is quantum mechanics. So instead of asking for trajectories in phase space, we ask what quantum time evolution preserves.

The answer is: the relations between states are preserved. If two states are orthogonal, they remain orthogonal. If they overlap by a certain amount, that overlap is preserved. One may picture the whole space of states as rotating, but without changing the angles between vectors.

Suppose a state $|A\rangle$ evolves over a time interval T. We write

$$|A\rangle \mapsto U(T)|A\rangle. \tag{8.6}$$

If a second state $|B\rangle$ evolves over the same interval, then

$$|B\rangle \mapsto U(T)|B\rangle, \qquad \langle B| \mapsto \langle B|U^\dagger(T). \tag{8.7}$$

Now require that the inner product be preserved:

$$\langle B|A\rangle = \langle B|U^\dagger(T)U(T)|A\rangle \tag{8.8}$$

for arbitrary states $|A\rangle$ and $|B\rangle$. Therefore

$$U^\dagger(T)U(T) = 1. \tag{8.9}$$

So time evolution is governed by a unitary operator.

In the lecture this is presented almost as the whole principle in one line: things which are orthogonal stay orthogonal. That is enough to force unitarity.

It is useful now to write the evolution law in the form used throughout the rest of the lecture:

$$|\psi(t+T)\rangle = U(T)|\psi(t)\rangle. \tag{8.10}$$

This says nothing more mysterious than that if we want to push a state ahead by an amount of time T, we do it by acting on it with the evolution operator $U(T)$.

8.3 Infinitesimal Time Steps And The Hermitian Generator

Once time evolution is known to be unitary for finite intervals, the next natural question is local: what is $U(\epsilon)$ for a very short interval ϵ? Since no evolution at all should do nothing, we must have

$$U(0) = 1. \tag{8.11}$$

Therefore the leading small-ϵ form must be

$$U(\epsilon) = 1 - \frac{i\epsilon}{\hbar}H + O(\epsilon^2). \tag{8.12}$$

At this stage the notation is partly anticipatory. The factors of i and $\hbar$ are chosen so that the operator H will later be recognized as the Hamiltonian. Susskind is explicit about this: the formula has been written in a way that will eventually make contact with classical mechanics.

A student immediately raises the classical-limit question: if $\hbar$ appears in the denominator, what can it mean to let $\hbar \to 0$? The answer is that the meaningful quantity is not $\hbar$ in isolation, but the dimensionless ratio $\epsilon H/\hbar$, or more broadly an action measured in units of $\hbar$. At this stage the placement of $\hbar$ is bookkeeping; later it will be precisely what allows H to have a sensible classical meaning.

Now take the Hermitian conjugate:

$$U^\dagger(\epsilon) = 1 + \frac{i\epsilon}{\hbar}H^\dagger + O(\epsilon^2). \tag{8.13}$$

Imposing unitarity to first order,

$$1 = U^\dagger(\epsilon)U(\epsilon) \tag{8.14}$$

$$= \left(1 + \frac{i\epsilon}{\hbar}H^\dagger\right)\left(1 - \frac{i\epsilon}{\hbar}H\right) + O(\epsilon^2) \tag{8.15}$$

$$= 1 + \frac{i\epsilon}{\hbar}(H^\dagger - H) + O(\epsilon^2), \tag{8.16}$$

so the first-order term must vanish:

$$H^\dagger = H. \tag{8.17}$$

That is the first genuinely new conclusion. The generator of infinitesimal time evolution must be Hermitian. It follows that it is an observable: it has real eigenvalues and an orthonormal basis of eigenvectors. Only after deriving that fact does Susskind give the operator its physical name. We call it the Hamiltonian, and we call its eigenvalues energies. He is careful about the logical order: we are not yet proving that this is the same energy as in classical mechanics, only that this is the observable whose eigenvalues deserve the name.

8.4 The Abstract Schrödinger Equation

Now we turn the infinitesimal update rule into a differential equation. Starting from

$$|\psi(t+\epsilon)\rangle = \left(1 - \frac{i\epsilon}{\hbar}H\right)|\psi(t)\rangle, \tag{8.18}$$

we bring the identity term to the left:

$$|\psi(t+\epsilon)\rangle - |\psi(t)\rangle = -\frac{i\epsilon}{\hbar}H|\psi(t)\rangle. \tag{8.19}$$

Dividing by ϵ and letting $\epsilon \to 0$ gives

$$\frac{\partial}{\partial t}|\psi\rangle = -\frac{i}{\hbar}H|\psi\rangle. \tag{8.20}$$

Multiplying by $i\hbar$, we obtain the canonical form

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle. \tag{8.21}$$

This is the abstract Schrödinger equation. Susskind pauses at exactly this point to stress what it is and what it is not. It is the general law of quantum time evolution. It is not yet the original wave equation Schrödinger wrote down for a particle moving in space. In this abstract form, it applies to any quantum system whatever, provided we know its Hamiltonian.

The lecture later reuses the same equation in the simplified $\hbar = 1$ convention. The board snapshot below captures that stripped-down ket form:

The cleaned board equation is

$$|\dot{\psi}\rangle = -iH|\psi\rangle, \tag{8.22}$$

which is exactly the same statement as the previous equation with $\hbar$ suppressed.

At this point the lecture takes a deliberate turn. A differential equation only earns its keep when it predicts the future. So Susskind asks how this formalism is actually used.

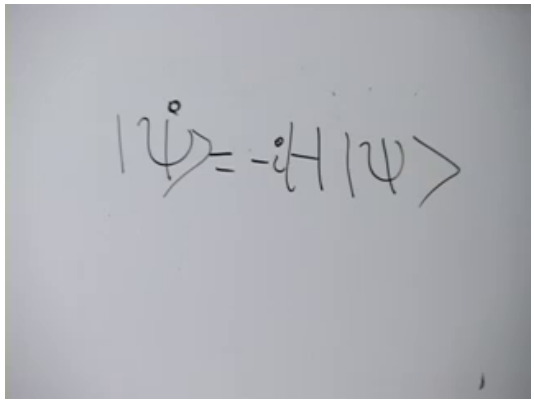


Figure 8.2: Ket form of the Schrödinger equation in the lecture's temporary $\hbar = 1$ convention.

8.5 Polarization Example: Energy Eigenstates And Phase Evolution

The first worked example returns to the familiar two-state system of photon polarization. Imagine a material in which a photon of fixed wavelength has different energies depending on its polarization. Let the x polarization have energy E_1 and the y polarization have energy E_2. Then the Hamiltonian has the basis states $|x\rangle$ and $|y\rangle$ as eigenvectors:

$$H|x\rangle = E_1|x\rangle, \qquad H|y\rangle = E_2|y\rangle. \tag{8.23}$$

A general state of polarization is written as

$$|\psi(t)\rangle = \begin{pmatrix} \alpha(t) \\ \beta(t) \end{pmatrix}, \tag{8.24}$$

where $\alpha(t)$ is the amplitude for x polarization and $\beta(t)$ is the amplitude for y polarization.

From this point on in the live lecture Susskind temporarily sets $\hbar = 1$. We will restore $\hbar$ in the printed formulas, but it is worth remembering that the corresponding board equations are written in the simplified convention.

In the $(|x\rangle, |y\rangle)$ basis the Hamiltonian is diagonal:

$$H = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}. \tag{8.25}$$

The lecture briefly writes the diagonal entries in the wrong order and then corrects them; the matrix above is the corrected form.

Now apply the Schrödinger equation:

$$i\hbar \frac{d}{dt} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{8.26}$$

Because the Hamiltonian is diagonal, the two amplitudes do not mix. We simply get

$$i\hbar\, \dot{\alpha} = E_1 \alpha, \qquad i\hbar\, \dot{\beta} = E_2 \beta. \tag{8.27}$$

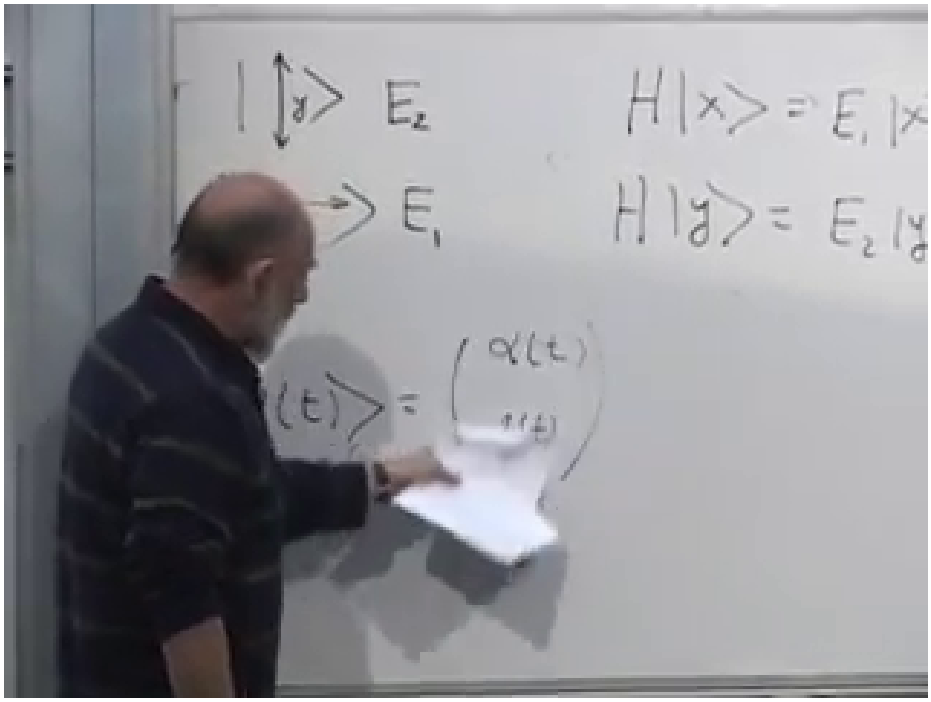


Figure 8.3: Polarization energy eigenstates and time-dependent state. The lower state vector is partly blocked in the board frame, so the printed form follows the transcript rather than the image alone.

These are the simplest possible first-order differential equations, and their solutions are pure exponentials:

$$\alpha(t) = \alpha(0)e^{-iE_1 t/\hbar}, \qquad \beta(t) = \beta(0)e^{-iE_2 t/\hbar}. \tag{8.28}$$

Therefore

$$|\psi(t)\rangle = \begin{pmatrix} \alpha(0)e^{-iE_1 t/\hbar} \\ \beta(0)e^{-iE_2 t/\hbar} \end{pmatrix}. \tag{8.29}$$

Susskind now asks the obvious experimental question. What happens if we measure the polarization in the x/y basis itself? The answer is almost perversely simple:

$$|\alpha(t)|^2 = |\alpha(0)|^2, \qquad |\beta(t)|^2 = |\beta(0)|^2. \tag{8.30}$$

The phases drop out. At first sight it looks as if time evolution has done nothing at all.

8.5.1 Question & Answer

Question. If the two components only pick up phases, and if the x/y probabilities do not change, has anything observable happened?

Answer. Yes, but we have to look in the right basis. The phases disappear only when we measure in the basis in which the Hamiltonian is diagonal. If we rotate the measurement basis, the relative phase becomes visible through interference.

Take the 45° polarization state

$$|/\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}. \tag{8.31}$$

The amplitude to pass a 45° polarizer is

$$\langle /|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(\alpha(0)e^{-iE_1t/\hbar} + \beta(0)e^{-iE_2t/\hbar}\right). \tag{8.32}$$

Now specialize to the experimental arrangement emphasized in the lecture: the photon first passes through a 45° polarizer, so at $t = 0$ we have

$$\alpha(0) = \beta(0) = \frac{1}{\sqrt{2}}. \tag{8.33}$$

Then the amplitude becomes

$$A_/(t) = \frac{1}{2}\left(e^{-iE_1t/\hbar} + e^{-iE_2t/\hbar}\right). \tag{8.34}$$

The corresponding probability is

$$\begin{aligned} P(t) &= |A_/(t)|^2 && (8.35)\\ &= \frac{1}{4}\left(e^{-iE_1t/\hbar} + e^{-iE_2t/\hbar}\right)\left(e^{iE_1t/\hbar} + e^{iE_2t/\hbar}\right) && (8.36)\\ &= \frac{1}{4}\left(2 + e^{i(E_2-E_1)t/\hbar} + e^{-i(E_2-E_1)t/\hbar}\right). && (8.37) \end{aligned}$$

Defining

$$\Delta E = E_2 - E_1, \tag{8.38}$$

we arrive at

$$P(t) = \frac{1}{2}\left(1 + \cos\frac{\Delta E\, t}{\hbar}\right). \tag{8.39}$$

This is the real payoff of the example. The state has certainly changed. The relative phase between the two components becomes a physically measurable oscillation when we project onto the rotated basis. At $t = 0$ we get $P(0) = 1$, as we should: two parallel 45° polarizers with no time evolution between them transmit with certainty. Later the probability can fall all the way to zero. That means the photon certainly does *not* pass the second polarizer in the original direction. The lecture interprets this physically as a rotation of the linear polarization as the photon moves downstream through the material.

The central point is broader than this one example. Only the energy difference matters. If we add the same constant to both energies, then the state acquires only an overall phase, and that common phase cancels from every probability. Thus physics depends on ΔE, not on the arbitrary zero of energy.

This also resolves the student's worry that the choice of x and y seems arbitrary. It is the material itself that picks preferred axes. When our measurement basis is aligned with those axes, the Hamiltonian is diagonal and the probabilities in that basis stay fixed. When the basis is misaligned, the relative phase shows up as rotation and interference.

8.6 Expectation Values, Commutators, And Conserved Quantities

Before taking a break, Susskind does one more thing. He steps back from the polarization example and asks for a general theorem. Suppose K is some observable. How does its expectation value

$$\langle K \rangle = \langle \psi | K | \psi \rangle \tag{8.40}$$

change with time?

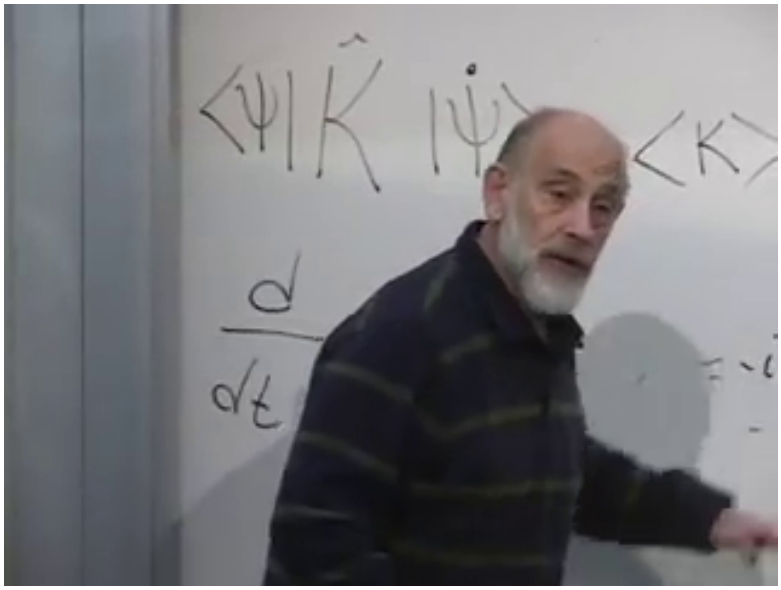


Figure 8.4: Product-rule step toward the commutator formula. The board image secures the intermediate structure, while the full displayed formula below is reconstructed from the transcript.

The first step is utterly ordinary: the time derivative of a product is the sum of two terms. Since K itself is held fixed, only the bra and ket change:

$$\frac{d}{dt}\langle \psi | K | \psi \rangle = \langle \psi | K | \dot{\psi} \rangle + \langle \dot{\psi} | K | \psi \rangle. \tag{8.41}$$

Now use the Schrödinger equation for the ket and its Hermitian conjugate for the bra:

$$|\dot{\psi}\rangle = -\frac{i}{\hbar} H |\psi\rangle, \qquad \langle \dot{\psi}| = \frac{i}{\hbar} \langle \psi | H. \tag{8.42}$$

Substituting, we find

$$\frac{d}{dt}\langle K \rangle = -\frac{i}{\hbar}\langle \psi | KH | \psi \rangle + \frac{i}{\hbar}\langle \psi | HK | \psi \rangle \tag{8.43}$$

$$= -\frac{i}{\hbar}\langle \psi | (KH - HK) | \psi \rangle. \tag{8.44}$$

With the convention

$$[K, H] = KH - HK, \tag{8.45}$$

the result becomes

$$\frac{d}{dt}\langle K \rangle = -\frac{i}{\hbar}\langle [K, H] \rangle. \tag{8.46}$$

That is the general law. In the lecture Susskind is pleased by how short the derivation is. Time-differentiate the sandwich, use the ket equation once and the bra equation once, and the opposite signs conspire to produce a commutator.

This is the direct quantum analog of the classical rule

$$\dot{f} = \{f, H\}_{\mathrm{PB}}. \tag{8.47}$$

In both theories, it is the Hamiltonian that controls time evolution. In classical mechanics the bracket is the Poisson bracket; in quantum mechanics it is the commutator.

Proposition 8.1. *If an observable K commutes with the Hamiltonian, then its expectation value is conserved:*

$$[K, H] = 0 \qquad \Longrightarrow \qquad \frac{d}{dt}\langle K\rangle = 0. \tag{8.48}$$

Proof. If $[K, H] = 0$, then the right-hand side of

$$\frac{d}{dt}\langle K\rangle = -\frac{i}{\hbar}\langle [K, H]\rangle$$

vanishes. □

Susskind remarks that one can prove a stronger statement about the full probability distribution, but he does not pursue it here. For present purposes the expectation-value theorem is enough. In particular, the Hamiltonian commutes with itself, so the Hamiltonian is conserved. Since the Hamiltonian is the energy, energy conservation appears here not as an additional postulate, but as a direct consequence of the formalism.

8.7 Particle On A Line And The Free-Particle Schrödinger Equation

Only after the two-state example and the commutator theorem does the lecture turn to Schrödinger's original setting: a particle moving on a line. The state can be described by a wave function $\psi(x, t)$, or equivalently by its Fourier transform $\tilde{\psi}(p, t)$. Nothing fundamental has changed. We have only traded an abstract ket for its coordinate representation. Functions of x are vectors too.

The rule of time evolution is the same rule as before:

$$i\hbar \frac{\partial}{\partial t}\psi(x, t) = H\psi(x, t). \tag{8.49}$$

What remains is to identify the Hamiltonian for a free particle.

We know that in classical mechanics the energy is

$$E = \frac{p^2}{2m}. \tag{8.50}$$

Susskind is careful here: classical mechanics does not prove the quantum Hamiltonian. At best it suggests a guess. The natural guess is

$$\hat{H} = \frac{\hat{p}^2}{2m}. \tag{8.51}$$

The momentum operator is

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}. \tag{8.52}$$

Therefore

$$\hat{H} = \frac{1}{2m}\left(-i\hbar \frac{\partial}{\partial x}\right)^2 = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}. \tag{8.53}$$

The live lecture then goes through exactly the kind of sign-and-factor tangle one expects in real time, notices a missing $\hbar$, loses track of a minus sign, and finally decides to do the calculation over carefully. The cleaned result is the standard free-particle Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x,t). \tag{8.54}$$

Now solve it for a momentum eigenstate. We look for a wave function that remains an eigenfunction of momentum at every instant:

$$\psi_p(x,t) = f(t)e^{ipx/\hbar}. \tag{8.55}$$

Differentiating with respect to time hits only $f(t)$:

$$\frac{\partial \psi_p}{\partial t} = \dot{f}(t)e^{ipx/\hbar}. \tag{8.56}$$

Differentiating twice with respect to x hits only the exponential:

$$\frac{\partial^2 \psi_p}{\partial x^2} = -\frac{p^2}{\hbar^2} f(t)e^{ipx/\hbar}. \tag{8.57}$$

Substituting into the Schrödinger equation, we get

$$i\hbar\, \dot{f}(t)e^{ipx/\hbar} = \frac{p^2}{2m} f(t)e^{ipx/\hbar}. \tag{8.58}$$

Cancel the common factor $e^{ipx/\hbar}$:

$$i\hbar\, \dot{f} = \frac{p^2}{2m} f, \qquad \text{or} \qquad \dot{f} = -\frac{ip^2}{2m\hbar} f. \tag{8.59}$$

Hence

$$f(t) = e^{-ip^2 t/(2m\hbar)} \tag{8.60}$$

up to normalization, and therefore

$$\psi_p(x,t) = \exp\left[\frac{i}{\hbar}\left(px - \frac{p^2}{2m}t\right)\right] \tag{8.61}$$

up to an overall constant factor.

Once again the time dependence is a pure phase. For a single momentum eigenstate there is no change in the momentum probability distribution; the state simply accumulates the phase appropriate to its energy. The lecture closes by pointing out the simple exercise

$$[p, H] = 0 \qquad \text{for} \qquad H = \frac{p^2}{2m}, \tag{8.62}$$

so momentum is conserved for the free particle. The next natural questions, which Susskind leaves for later, concern what happens for superpositions: does the probability for a given momentum change, and what about the probability for a given position?

8.8 Summary

The lecture has unfolded in a very specific order. We began with classical flow only long enough to isolate the structural role of the Hamiltonian. We then replaced phase-space flow by preservation of inner products, and from that preservation recovered unitary time evolution. Studying an infinitesimal time step forced the generator to be Hermitian, and from the infinitesimal update rule we obtained the abstract Schrödinger equation.

The polarization example then showed why the equation matters. A diagonal Hamiltonian makes the individual components pick up phases, and at first sight those phases seem invisible. But once we rotate the measurement basis, the relative phase becomes measurable through interference, and only energy differences survive. From there the commutator law for expectation values follows almost at once, together with the criterion for conserved quantities. Finally, by replacing kets with wave functions and guessing the Hamiltonian from the classical free-particle energy, we recover Schrödinger's original differential equation and its plane-wave solutions. The whole lecture is guided by one persistent idea: the Hamiltonian is the generator of time evolution.

Chapter 9

Free Particles, Momentum Space, and the Classical Limit

These notes follow Leonard Susskind's lecture closely. We begin where the lecture begins, not with a polished differential equation, but with the historical crisis that made quantum mechanics unavoidable. The transcript and frame material curated by LazyingArt LLC and Video2Book are used only to keep faith with the visible board work and the order of the spoken argument. What matters throughout is the lecture's logical direction: quantum mechanics is fundamental, and classical mechanics must be recovered from it, not the other way around.

9.1 From Quantum Crisis to Quantum First Principles

The lecture opens by clearing the ground. Michelson–Morley belongs to that story, but the mathematically sharper crisis for the present chapter is blackbody radiation. The old wave theory of light said that electromagnetic radiation could occur at arbitrary wavelength, and each wavelength behaved like a harmonic oscillator. Statistical mechanics then said that in thermal equilibrium each oscillator should carry the same average energy,

$$E_{\text{mode}} = kT. \tag{9.1}$$

Now put radiation in a cavity. The cavity supports infinitely many modes. If each allowed mode carries the same nonzero energy, the total energy diverges. That is the ultraviolet catastrophe.

Planck's intervention was to write

$$E = nh\nu, \tag{9.2}$$

but the lecture is careful here: Planck did not yet correctly interpret this as a statement about the cavity radiation itself. In his original picture, the quantization belonged to the oscillators in the walls that emitted the radiation. Einstein's step was deeper. He treated the radiation itself as quantized. Then the high-frequency modes are suppressed: if

$$h\nu > kT, \tag{9.3}$$

the classical equipartition estimate would assign less than one quantum to that mode, and quantum mechanics does not permit that. Less than one quantum means no quanta.

Susskind then walks through the long hesitation over photons, the eventual force of Compton scattering, and the slow collapse of mechanistic rescue attempts. At the same time he emphasizes a second lesson. Einstein, well before the final formalism of quantum mechanics, already understood that spontaneous emission and radioactive decay point toward a fundamentally probabilistic law. The old picture of atoms as little solar systems, with particles in definite orbits, could be patched for a while, but only for a while.

That entire prelude has one function in the lecture. Before we write down the Schrödinger equation again, we are meant to see why the direction of explanation matters. Quantum mechanics is not to be derived from classical mechanics. At most we may borrow a classical Hamiltonian as a guess, and then test what the quantum theory does with it.

9.2 Recalling the Schrödinger Equation and Guessing the Free Hamiltonian

Only after that clearing of the ground does the lecture pivot back: let us get back to the Schrödinger equation and really go through it now. The general time-evolution law is

$$i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle, \tag{9.4}$$

and the momentum operator for motion on a line is

$$\hat{P} = -i\hbar\frac{\partial}{\partial x}. \tag{9.5}$$

The lecture lingers over the structure of these formulas. Time derivatives are tied to the Hamiltonian. Space derivatives are tied to momentum. The minus sign in the momentum operator is a convention, and the lecture even remarks on that convention, but the point is that these are quantum definitions, not deductions from classical mechanics.

Now we specialize to a nonrelativistic particle on a line. Here Susskind says plainly that we are making a guess, not a derivation:

$$\hat{H} = \frac{\hat{P}^2}{2m}. \tag{9.6}$$

Why $\hat{P}^2$ rather than $\hat{P}$? Because momentum has a direction, but the free-particle energy should not care whether the particle moves left or right. A particle with momentum p and a particle with momentum $-p$ should have the same energy. That already tells us the dependence must be even in p, and the simplest nonrelativistic choice is p^2. The factor $2m$ is then the familiar normalization, and in that sense it is what makes m deserve the name mass.

This is one of the lecture's persistent motifs. We borrow a classical-looking expression, but only as a tentative bridge. Then we ask, what does it mean quantum mechanically? In particular, what does $\hat{P}^2$ mean? It simply means that the same operator acts twice.

9.3 From Kets to Wave Functions: The Free-Particle Equation

At this point the lecture shifts from abstract ket language to the wave function on a line. The board frame below records that shift in layout. What is visible on the board is only an isolated ket and a

rightward arrow. The lecture, however, is already speaking of the wave function $\psi(x)$ as living in a vector space. So the figure should be read as evidence for the transition in board language, not as a complete written equation.

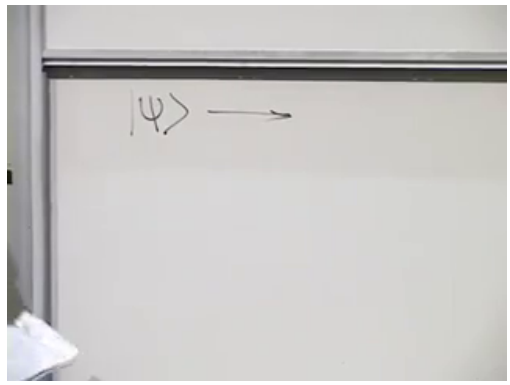


Figure 9.1: Ket state with rightward arrow

The lecture slows down here for a reason. ψ depends on position because position is an observable. But ψ also changes with time, and that dependence is logically different. We do not ask for the probability that the time is such-and-such. We ask for the probability that the particle is at position x at a given time t. Thus $\psi(x,t)$ is the state written in the position representation, evolving in time.

Let us now see what the guess $\hat{H} = \hat{P}^2/2m$ means in this representation. Substitute it into the general Schrödinger equation:

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = \frac{1}{2m}\hat{P}^2\psi(x,t) \tag{9.7}$$

$$= \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial x}\right)\left(-i\hbar\frac{\partial}{\partial x}\right)\psi(x,t). \tag{9.8}$$

Since $(-i)^2 = -1$, we obtain

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x,t). \tag{9.9}$$

This is the free-particle Schrödinger equation.

Let us do the substitution the way the lecture does it. First write the time derivative on the left. Then insert the Hamiltonian $\hat{P}^2/2m$ on the right. Then remember that each $\hat{P}$ means $-i\hbar\partial_x$. Acting once gives a first derivative; acting twice gives a second derivative. Only after that step-by-step substitution does the familiar differential equation appear. The lecture is careful not to let us skip the passage from operator language to wave-function language.

9.4 Plane Waves, Momentum Eigenstates, and the Meaning of Energy

Now comes the next question: what are the solutions? Again the lecture proceeds by trial rather than declaration. Let us look for wave-like solutions. Let the x-dependence be that of a momentum eigenfunction, and let the time-dependence be oscillatory:

$$\psi(x,t) = e^{ipx/\hbar}e^{i\omega t}. \tag{9.10}$$

The first factor is chosen because momentum eigenstates in the position representation have precisely that form. The second factor is a guess; we shall see whether the Schrödinger equation permits it and, if so, what ω must be.

Differentiate with respect to time:

$$\frac{\partial \psi}{\partial t} = i\omega\psi. \tag{9.11}$$

Therefore the left-hand side of the Schrödinger equation is

$$i\hbar\frac{\partial \psi}{\partial t} = i\hbar(i\omega)\psi = -\hbar\omega\,\psi. \tag{9.12}$$

Differentiate twice with respect to x:

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2}\psi. \tag{9.13}$$

So the right-hand side is

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} = -\frac{\hbar^2}{2m}\left(-\frac{p^2}{\hbar^2}\right)\psi = \frac{p^2}{2m}\psi. \tag{9.14}$$

Cancelling the common factor ψ, we find

$$-\hbar\omega = \frac{p^2}{2m}. \tag{9.15}$$

Hence

$$\omega = -\frac{p^2}{2m\hbar}, \tag{9.16}$$

and the plane-wave solution becomes

$$\psi_p(x,t) = e^{ipx/\hbar - ip^2 t/(2m\hbar)}. \tag{9.17}$$

This is the first real payoff of the lecture. The x-dependence labels the state by momentum p, while the time phase shows that the Hamiltonian eigenvalue is

$$E = \frac{p^2}{2m}. \tag{9.18}$$

So the free-particle energy takes the expected nonrelativistic form. The lecture even pauses to note that there is no extra $\hbar$ in the relation between energy and momentum. The $\hbar$ lives in the oscillatory phases, not in the classical-looking formula $E = p^2/2m$.

There is, however, another solution with exactly the same energy:

$$\psi_{-p}(x,t) = e^{-ipx/\hbar - ip^2 t/(2m\hbar)}. \tag{9.19}$$

The momentum has been reversed, but the time phase is unchanged because only p^2 enters the energy.

9.4.1 Question & Answer

If the energy is E, why do two momentum states contribute, and how do we compute the probability for that energy?

Because free-particle energy depends on p^2, not on p. If we reverse the momentum, the energy does not change. In the simplest two-state example,

$$\psi(x,t) = \alpha\,\psi_p(x,t) + \beta\,\psi_{-p}(x,t), \tag{9.20}$$

the probability of right-moving momentum is $\alpha^*\alpha$, and the probability of left-moving momentum is $\beta^*\beta$. But both components have the same energy $E = p^2/(2m)$. So for this example the energy probability is

$$P(E) = \alpha^*\alpha + \beta^*\beta. \tag{9.21}$$

The lecture then states the same idea in the more general language we shall need later. If a fixed energy E is given, we solve

$$E = \frac{p^2}{2m}, \tag{9.22}$$

which gives two momenta,

$$p_\pm = \pm\sqrt{2mE}. \tag{9.23}$$

The probability associated with that energy is obtained by collecting the contributions from both momentum values. The conceptual lesson is the important one: momentum labels and energy labels are not the same labels.

9.5 General Solution in Momentum Space and the Fourier Transform at $t = 0$

The lecture now asks the next natural question in exactly these words: what is the general solution? The answer is not another special wave, but a superposition of all such waves. Better yet, since momentum is continuous, an integral:

$$\psi(x,t) = \int_{-\infty}^{\infty} \frac{dp}{\sqrt{2\pi}}\, e^{ipx/\hbar - ip^2t/(2m\hbar)}\, \tilde{\psi}(p). \tag{9.24}$$

Here $\tilde{\psi}(p)$ is the momentum-space amplitude. Nothing mysterious is fixed except the constants. We integrate over all p, positive and negative, so the formula already includes both right-moving and left-moving momenta. In general it is a superposition of different momenta and, because $E = p^2/(2m)$, of different energies as well.

At this point the lecture pauses over the interpretation of $\tilde{\psi}(p)$. What is the probability density for momentum? Because momentum is continuous, we ask for a density. The board frame below is especially useful here. Its upper integral is only partially visible, so the full superposition formula should be reconstructed from the transcript. But the two lower equations are legible and directly support the local point being made.

The cleaned equations are

$$P(q) = \tilde{\psi}^*(q)\tilde{\psi}(q), \tag{9.25}$$

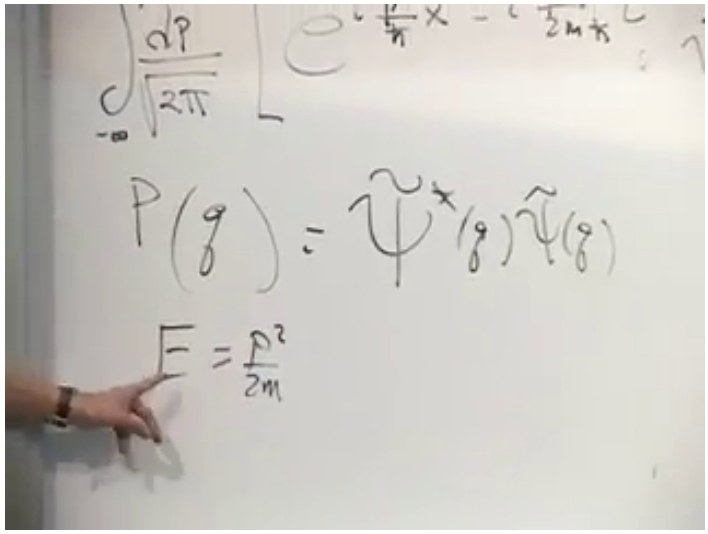


Figure 9.2: Momentum-space probability and free-particle energy

and

$$E = \frac{p^2}{2m}. \tag{9.26}$$

The notation q matters. The lecture corrects the symbol precisely because p is already serving as the dummy variable of integration in the general superposition formula.

So far $\tilde{\psi}(p)$ is still only an amplitude function in momentum space. The lecture now asks the right operational question: what does it mean in terms of the original wave function in x-space? To answer that, let us go to time $t = 0$. This is exactly the move the lecture makes.

The next board frame records the rewrite step. The left-hand side is heavily occluded by Susskind's body, so the full equation must be reconstructed from the transcript. What the frame does securely preserve is the rhythm of the board: the earlier momentum integral on the left, and the new ψ-expression on the right.

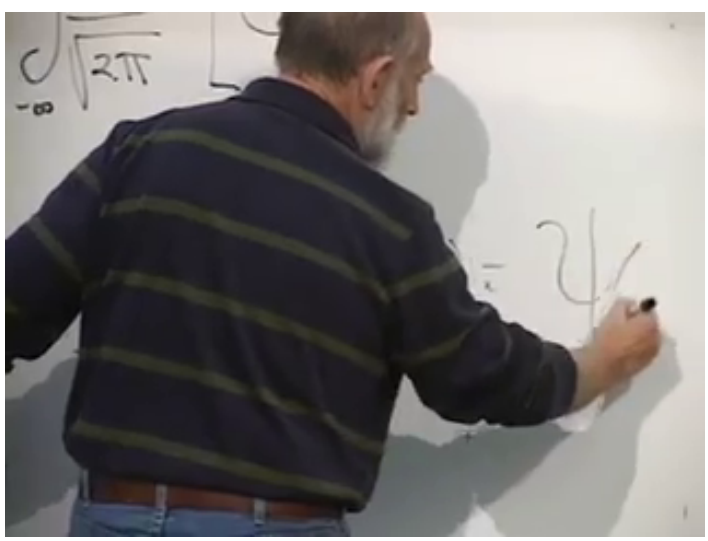


Figure 9.3: Momentum integral rewritten as the initial wave function

At $t = 0$, the time phase disappears, and on the board the lecture temporarily suppresses $\hbar$ in the Fourier formulas. In that shorthand,

$$\psi(x, 0) = \int_{-\infty}^{\infty} \frac{dp}{\sqrt{2\pi}}\, e^{ipx}\, \tilde{\psi}(p). \tag{9.27}$$

Immediately afterward, the lecture abbreviates $\psi(x, 0)$ to $\psi(x)$ when the discussion is restricted to the initial data. The inverse transform is

$$\tilde{\psi}(p) = \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi}}\, e^{-ipx}\, \psi(x, 0). \tag{9.28}$$

Thus $\tilde{\psi}(p)$ is not an arbitrary bookkeeping device. It is the Fourier transform of the initial wave function.

9.5.1 Question & Answer

Could this free-particle equation describe a photon?

No. The lecture answers this immediately and without hedging. This is a nonrelativistic theory. Its Hamiltonian is built from

$$E = \frac{p^2}{2m}, \tag{9.29}$$

and photons do not satisfy that relation. In fact they do not come with a mass parameter m of this sort at all. So one would not use this equation in this form for a photon. It is the appropriate description for particles moving much slower than the speed of light.

9.6 Dispersive Time Evolution from Initial Data

Once we know what $\tilde{\psi}(p)$ means, we also know how to solve an initial-value problem. This is the practical heart of the lecture. Suppose someone tells us the wave function at time $t = 0$. What do we do?

We proceed exactly in the order Susskind lays out:

1. Start with the initial wave function $\psi(x, 0)$.
2. Compute its Fourier transform $\tilde{\psi}(p)$.
3. Put that $\tilde{\psi}(p)$ into the time-dependent superposition formula.
4. Let each momentum component acquire its own phase.
5. Reconstruct $\psi(x, t)$, and then compute $|\psi(x, t)|^2$.

Equivalently, one may say that the Fourier transform itself evolves by a phase:

$$\tilde{\psi}(p, t) = e^{-ip^2 t/(2m\hbar)} \tilde{\psi}(p, 0). \tag{9.30}$$

Therefore

$$\psi(x, t) = \int_{-\infty}^{\infty} \frac{dp}{\sqrt{2\pi}}\, e^{ipx/\hbar} e^{-ip^2 t/(2m\hbar)} \tilde{\psi}(p, 0). \tag{9.31}$$

The lecture then makes the physical point. A general wave packet will not preserve its shape. Different Fourier components carry different momenta, hence different energies, hence different time phases. The result is a complicated evolution in which the packet deforms. This is precisely what the student in the lecture calls a dispersive wave, and Susskind agrees: the different components propagate with different speeds. He mentions phase velocity and group velocity, but postpones that subject. Here the essential statement is simpler and already sufficient. Free evolution is dispersive because the time phase depends on p^2, so different momentum components do not stay locked together.

9.7 Expectation Values, Commutators, Potentials, and the Classical Limit

At this point the lecture deliberately backs off from the special free-particle calculation. Let us put a little variety into this, Susskind says. Before studying the more complicated Schrödinger equation with a potential, let us remind ourselves how expectation values evolve. That reminder is the bridge to the classical limit.

Expectation values and their time evolution. Take an observable $\hat{K}$, assumed not to depend explicitly on time. Its expectation value is

$$\langle K\rangle = \langle\psi|\hat{K}|\psi\rangle. \tag{9.32}$$

Only the bra and ket change with time. Differentiating and using the Schrödinger equation gives

$$\frac{d}{dt}\langle K\rangle = \langle\dot{\psi}|\hat{K}|\psi\rangle + \langle\psi|\hat{K}|\dot{\psi}\rangle \tag{9.33}$$

$$= \frac{i}{\hbar}\langle\psi|\hat{H}\hat{K}|\psi\rangle - \frac{i}{\hbar}\langle\psi|\hat{K}\hat{H}|\psi\rangle \tag{9.34}$$

$$= \frac{i}{\hbar}\langle[\hat{H},\hat{K}]\rangle. \tag{9.35}$$

On the board the lecture sets $\hbar = 1$ for convenience and hesitates openly over signs and order. The clean formula, with $\hbar$ restored, is the one above.

The corresponding classical law is

$$\dot{K} = \{K, H\}. \tag{9.36}$$

The lecture pauses over the order here too, but that pause is part of the pedagogy. The final point is that classical evolution is governed by Poisson brackets, quantum evolution of expectation values by commutators.

How Poisson brackets and commutators line up. Let us begin with a function of position and the momentum. Classically,

$$\{f(x), p\} = f'(x), \qquad \{x, p\} = 1. \tag{9.37}$$

Quantum mechanically, let $f(\hat{X})$ act on an arbitrary wave function $\psi(x)$. Then

$$[f(\hat{X}), \hat{P}]\psi(x) = f(x)\left(-i\hbar\frac{d\psi}{dx}\right) - \left(-i\hbar\frac{d}{dx}\right)(f(x)\psi(x)) \tag{9.38}$$

$$= f(x)\left(-i\hbar\frac{d\psi}{dx}\right) + i\hbar\left(f(x)\frac{d\psi}{dx} + f'(x)\psi(x)\right) \tag{9.39}$$

$$= i\hbar f'(x)\psi(x). \tag{9.40}$$

Hence

$$[f(\hat{X}), \hat{P}] = i\hbar f'(\hat{X}). \tag{9.41}$$

By the same reciprocity between x and p, the lecture also records

$$[\hat{X}, f(\hat{P})] = i\hbar f'(\hat{P}), \tag{9.42}$$

and in particular

$$[\hat{X}, \hat{P}] = i\hbar. \tag{9.43}$$

The lecture then checks the algebra. Both Poisson brackets and commutators are antisymmetric:

$$\{A, B\} = -\{B, A\}, \qquad [A, B] = -[B, A]. \tag{9.44}$$

Both satisfy a product rule. The corrected commutator identity, after the classroom correction on the board, is

$$\{AB, C\} = A\{B, C\} + B\{A, C\}, \tag{9.45}$$

$$[AB, C] = A[B, C] + [A, C]B. \tag{9.46}$$

So in the simple situations relevant here,

$$[A, B] \approx i\hbar\{A, B\}. \tag{9.47}$$

The lecture does warn that operator-ordering ambiguities can spoil a naive one-to-one identification in more complicated expressions. But for the present purpose the parallel is close enough to reveal how the classical equations are going to emerge.

The Schrödinger equation with a potential. Now we come back to a particle on a line, but no longer free. The Hamiltonian is

$$\hat{H} = \frac{\hat{P}^2}{2m} + U(\hat{X}), \tag{9.48}$$

where the force is

$$F(x) = -\frac{dU}{dx}. \tag{9.49}$$

In the position representation, this becomes

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + U(x)\right]\psi(x,t). \tag{9.50}$$

The lecture emphasizes a simple but important point. $U(\hat{X})$ is an operator. In position space it acts by multiplication. That is why the equation contains $U(x)\psi(x,t)$, not merely $U(x)$.

How the average position moves. Now apply the expectation-value formula with $\hat{K} = \hat{X}$:

$$\frac{d}{dt}\langle X\rangle = \frac{i}{\hbar}\left\langle\left[\frac{\hat{P}^2}{2m} + U(\hat{X}), \hat{X}\right]\right\rangle. \tag{9.51}$$

Since any function of $\hat{X}$ commutes with $\hat{X}$,

$$[U(\hat{X}), \hat{X}] = 0. \tag{9.52}$$

So only the kinetic term survives:

$$\frac{d}{dt}\langle X\rangle = \frac{i}{\hbar}\left\langle\left[\frac{\hat{P}^2}{2m}, \hat{X}\right]\right\rangle. \tag{9.53}$$

Now use the corrected product rule:

$$[\hat{P}^2, \hat{X}] = \hat{P}[\hat{P}, \hat{X}] + [\hat{P}, \hat{X}]\hat{P}. \tag{9.54}$$

Since $[\hat{P}, \hat{X}] = -i\hbar$, this becomes

$$[\hat{P}^2, \hat{X}] = -2i\hbar\hat{P}. \tag{9.55}$$

Therefore

$$\frac{d}{dt}\langle X\rangle = \frac{\langle P\rangle}{m}. \tag{9.56}$$

This is exactly the result the lecture wants to see: the average velocity is the average momentum divided by the mass.

How the average momentum moves. Now take $\hat{K} = \hat{P}$:

$$\frac{d}{dt}\langle P\rangle = \frac{i}{\hbar}\left\langle\left[\frac{\hat{P}^2}{2m} + U(\hat{X}), \hat{P}\right]\right\rangle. \tag{9.57}$$

This time $\hat{P}$ commutes with $\hat{P}^2$, so the interesting term is the potential:

$$\frac{d}{dt}\langle P\rangle = \frac{i}{\hbar}\langle[U(\hat{X}), \hat{P}]\rangle. \tag{9.58}$$

Using the identity already derived,

$$[U(\hat{X}), \hat{P}] = i\hbar U'(\hat{X}), \tag{9.59}$$

we obtain

$$\frac{d}{dt}\langle P\rangle = -\langle U'(\hat{X})\rangle. \tag{9.60}$$

In other words,

$$\frac{d}{dt}\langle P\rangle = \langle F(\hat{X})\rangle. \tag{9.61}$$

So the exact quantum equations for the expectation values are

$$\frac{d}{dt}\langle X\rangle = \frac{\langle P\rangle}{m}, \tag{9.62}$$

and

$$\frac{d}{dt}\langle P\rangle = -\langle U'(\hat{X})\rangle. \tag{9.63}$$

Combining them,

$$m\frac{d^2}{dt^2}\langle X\rangle = -\langle U'(\hat{X})\rangle. \tag{9.64}$$

This is where the lecture cashes its promise. If the wave packet is well localized and the potential is smooth across the region where the wave function has appreciable support, then the force does not vary much across the packet. In that circumstance,

$$\langle F(\hat{X})\rangle \approx F(\langle X\rangle), \tag{9.65}$$

and so

$$m\frac{d^2}{dt^2}\langle X\rangle \approx F(\langle X\rangle). \tag{9.66}$$

That is the classical limit the lecture wants. But the exact quantum statement is still the one involving $\langle U'(\hat{X})\rangle$. The replacement by $F(\langle X\rangle)$ is an approximation, justified only for a narrow packet in a smooth potential.

9.8 Summary

The lecture begins by insisting that quantum mechanics was not born as a decorative correction to classical mechanics. Blackbody radiation, photons, atomic structure, and probabilistic decay already forced a more radical change of viewpoint. That is why the historical prelude matters: it prepares the quantum-first stance of the whole lecture.

Once we return to the Schrödinger equation, the mathematics unfolds in a very specific order. We guess the free Hamiltonian $\hat{H} = \hat{P}^2/2m$, translate from kets to $\psi(x,t)$, derive the free-particle Schrödinger equation, and solve it first by plane waves. From there we pass to the general momentum-space superposition, identify $\tilde{\psi}(p)$ as the Fourier transform of the initial wave function, and see why free evolution is dispersive.

Finally the lecture steps back and asks how averages move. The exact quantum equations

$$\frac{d}{dt}\langle X\rangle = \frac{\langle P\rangle}{m}, \qquad \frac{d}{dt}\langle P\rangle = -\langle U'(\hat{X})\rangle \tag{9.67}$$

show why momentum and Hamiltonian deserve their classical names in the appropriate limit. Classical mechanics appears only at the end, and only as the motion of a narrow wave packet in a smooth potential. That is exactly the direction of explanation the lecture set out to establish.

Chapter 10

The Quantum Harmonic Oscillator

Susskind opens this lecture by calling it the final class in quantum mechanics and then immediately taking that back: quantum mechanics will return when the course reaches quantum field theory. Still, as he says, the harmonic oscillator is one of the last indispensable pieces of the elementary subject. It is worth doing with some care, because it is not merely a spring with a mass attached to it. The same mathematics reappears whenever a system has a stable equilibrium and we disturb it only slightly. The lecture unfolds in that order: broad motivation first, then the classical oscillator, then quantization, then the question of why only special energies are allowed, and finally the algebra that generates the whole spectrum.

Credit. These notes follow Leonard Susskind's lecture closely. The preserved blackboard frames are kept as documentary support for the derivation, with credit to Leonard Susskind, LazyingArt LLC, and Video2Book.

10.1 Why This System Matters

The first task is not to write down an equation, but to understand why this system deserves attention at all. The harmonic oscillator is, in Susskind's words, basically a particle moving on a line under the influence of a restoring force. A spring is the obvious example, but the lecture immediately broadens the point. A point on a passing wave oscillates. A sound wave oscillates. Molecules vibrate about equilibrium configurations. In fact, almost any system with a stable equilibrium behaves this way when it is disturbed only a little.

That is the real reason for quantizing the oscillator. There is no practical urgency in quantizing a visible laboratory spring with a heavy mass hanging from it; for such a system the quantum corrections are utterly negligible. But the same mathematics does apply to molecular motion and to many other genuinely microscopic systems. We therefore use the spring as a model, not because it is the only oscillator, but because it is the cleanest entry point.

There is also a more precise way to state the generality. Suppose a one-dimensional potential has a stable minimum at x_0. Then near that minimum we may expand

$$V(x) \approx V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2. \tag{10.1}$$

A local minimum therefore looks, to quadratic order, like a parabola. Near a stable point, the world becomes harmonic.

10.2 Classical Oscillator From Spring Sketch to Equation of Motion

The lecture begins concretely with a weight hanging from a spring. Gravity pulls downward, the spring pulls upward, and there is an equilibrium position at which the two balance. Once that equilibrium has been found, the sensible coordinate is not the absolute height but the displacement from equilibrium. Susskind chooses to call that displacement x, even though in the picture it is drawn vertically.

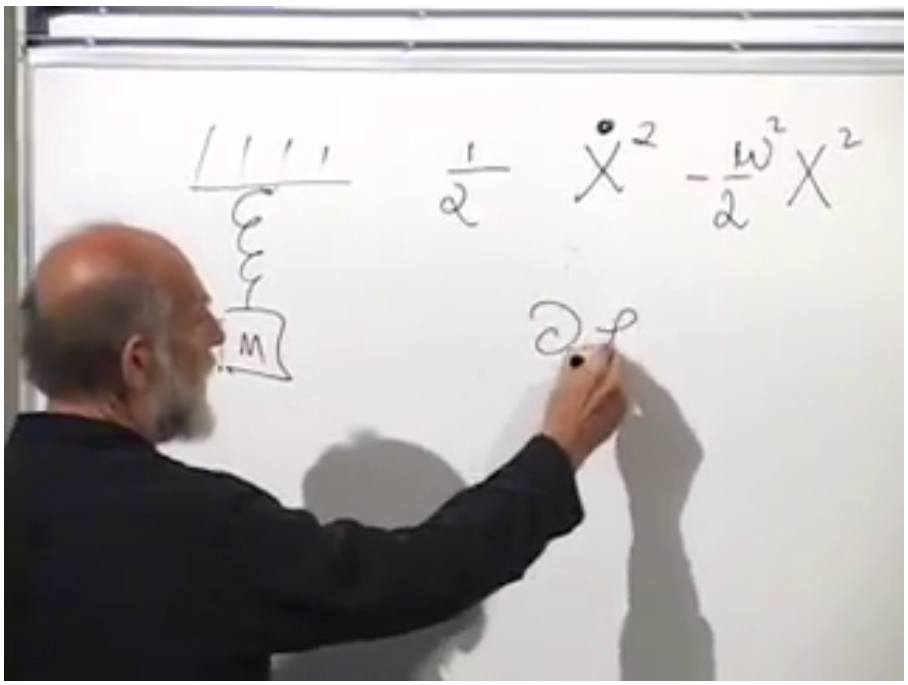


Figure 10.1: Spring-mass oscillator and the quadratic Lagrangian term.

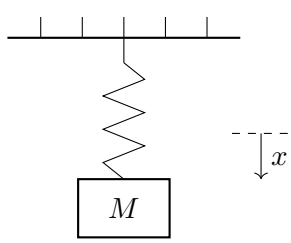


Figure 10.2: A clean redraw of the spring-mass sketch used in the classical setup.

Once the equilibrium position has been absorbed into the definition of x, gravity disappears from the local dynamics. The only remaining ingredients are kinetic energy and the restoring potential. Susskind then makes a second simplification: he absorbs the mass into the coordinate. If the original displacement is x_{old}, then one may define

$$y = \sqrt{m}\, x_{\text{old}}, \qquad \dot{y}^{\,2} = m\, \dot{x}_{\text{old}}^{2}. \tag{10.2}$$

After exhibiting this rescaling explicitly, he simply renames the new coordinate x. The mass has thus been scaled out, and the spring constant is renamed ω^2. With those choices the Lagrangian becomes

$$L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \tag{10.3}$$

The potential energy must be proportional to x^2, not to x, because the energy should increase whether the system is displaced to the right or to the left. That is the local signature of a stable equilibrium. The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = \dot{x}. \tag{10.4}$$

Applying the Euler–Lagrange equation,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0, \tag{10.5}$$

we find

$$\ddot{x} = -\omega^2 x. \tag{10.6}$$

The minus sign is the restoring sign. If x is positive, the force points back toward the origin; if x is negative, it again points back toward the origin.

The general classical solution is therefore

$$x(t) = a\cos\omega t + b\sin\omega t. \tag{10.7}$$

So much for the classical oscillator. The point of doing this first is that the quantum system will inherit the same Hamiltonian structure.

10.3 Quantization Setup: States, Operators, and the Hamiltonian

To go from the classical oscillator to the quantum oscillator, Susskind insists that we proceed in the usual order. First we specify the space of states. Only then do we construct the Hamiltonian operator.

For a particle moving on a line, the state space is the space of wave functions $\psi(x)$. The probability density is

$$|\psi(x)|^2 = \psi^*(x)\psi(x). \tag{10.8}$$

In the x-representation the position operator acts by multiplication:

$$\hat{x}\,\psi(x) = x\,\psi(x). \tag{10.9}$$

The momentum operator, temporarily setting $\hbar = 1$, acts as

$$\hat{p}\,\psi(x) = -i\frac{\partial \psi}{\partial x}. \tag{10.10}$$

The lecture pauses here to emphasize that x and p are Hermitian operators acting on the vector space of wave functions.

On the classical side, the Hamiltonian comes from the Lagrangian in the standard way:

$$H = p\dot{x} - L \tag{10.11}$$
$$= \frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2 \tag{10.12}$$
$$= \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2. \tag{10.13}$$

This three-line progression is visible on the board and is worth preserving because it is exactly how the lecture moves.

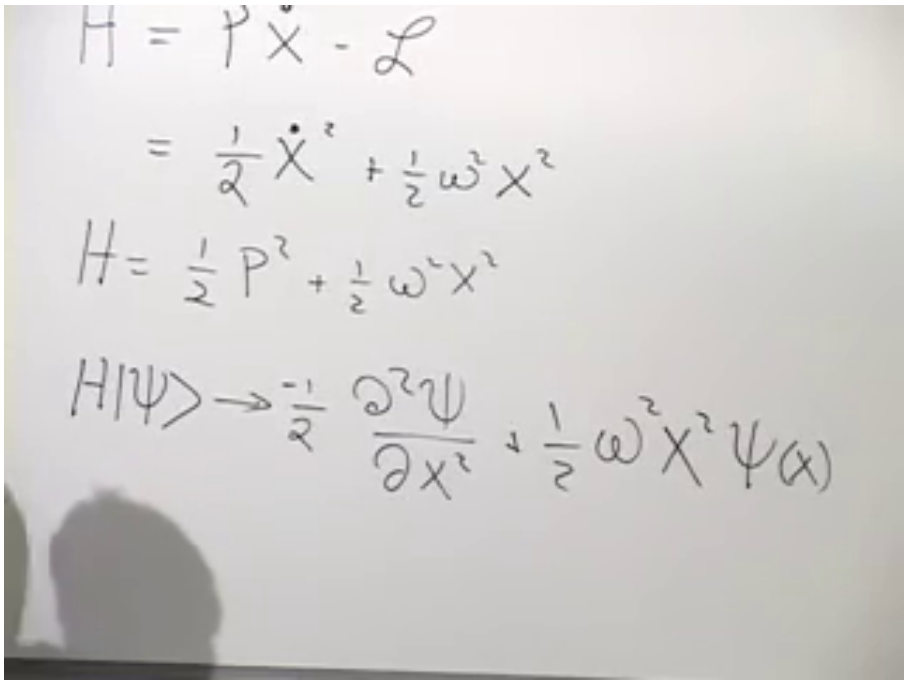


Figure 10.3: Hamiltonian rewritten as an operator on the wave function.

Once we quantize, the Hamiltonian acts on a wave function as

$$H\psi(x) = \frac{1}{2}\hat{p}^2\psi(x) + \frac{1}{2}\omega^2 x^2 \psi(x) \tag{10.14}$$
$$= -\frac{1}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}\omega^2 x^2 \psi(x). \tag{10.15}$$

The bottom line of the board uses a quick correspondence between ket notation and the coordinate representation; here we normalize it into the cleaner operator notation (10.15).

10.4 Two Uses of the Hamiltonian and the Energy Eigenvalue Problem

At this point the lecture makes an important distinction. The Hamiltonian has two jobs. One is to govern the time evolution of the state. The other is to determine the allowed energy levels.

The time-dependent Schrödinger equation is

$$i\frac{\partial\psi(x,t)}{\partial t} = H\psi(x,t). \tag{10.16}$$

Substituting (10.15), we obtain

$$i\frac{\partial\psi}{\partial t} = -\frac{1}{2}\frac{\partial^2\psi}{\partial x^2} + \frac{1}{2}\omega^2 x^2\psi. \tag{10.17}$$

Because of the factor of i, even a real wave function at $t = 0$ will generically acquire an imaginary part as time evolves. So the wave function must be allowed to be complex.

Susskind briefly notes that one could solve this dynamics directly, for example by updating the wave function in small time steps. At the schematic level one would write

$$\psi(x, t + \delta t) \approx \psi(x,t) - i\,\delta t\, H\psi(x,t). \tag{10.18}$$

That is the dynamical problem. But he then narrows the focus quite deliberately: let us instead look for the energy eigenvectors themselves.

Those satisfy

$$H\psi_E(x) = E\,\psi_E(x), \tag{10.19}$$

which in explicit form becomes

$$-\frac{1}{2}\psi_E''(x) + \frac{1}{2}\omega^2 x^2\psi_E(x) = E\,\psi_E(x). \tag{10.20}$$

This equation does two things at once. It is an equation for the wave function $\psi_E(x)$, but it also selects the allowed values of E.

Why should not every value of E be allowed? The lecture answers this in the most direct way possible. Equation (10.20) is a second-order differential equation, so for a generic energy there are two linearly independent solutions. At large $|x|$, the quadratic potential dominates, and at the rough level needed here the solutions behave like Gaussian-type exponentials,

$$\psi(x) \sim \exp\left(-\frac{\omega x^2}{2}\right) \quad \text{or} \quad \psi(x) \sim \exp\left(+\frac{\omega x^2}{2}\right), \qquad |x| \to \infty, \tag{10.21}$$

up to subleading factors.

The lecture is careful not to leave this as a slogan. The generic second-order situation is actually worse than merely having one good global solution and one bad global solution. More typically, a solution that decays in one direction blows up in the other, and vice versa. Only for very special energies does a single solution decay in both directions. Those are the energy eigenvalues.

The physical criterion is square-integrability:

$$\int_{-\infty}^{\infty} |\psi(x)|^2\, dx = 1. \tag{10.22}$$

A wave function that blows up exponentially pushes all its probability out to infinity, and in a confining potential that also means infinite potential energy. So the discrete spectrum is not being imposed by hand; it emerges from the demand that the state be admissible.

10.5 Ground-State Guess and Zero-Point Energy

Having explained why only certain energies can work, the lecture next looks for one concrete solution. This is the first real payoff. Susskind guesses a Gaussian concentrated near the origin. In the live lecture there is a moment of hesitation about the factor of 1/2 in the exponent; the cleaned result is

$$\psi_0(x) \propto e^{-\frac{\omega}{2}x^2}. \tag{10.23}$$

We leave the normalization constant aside for now, exactly as the lecture does.

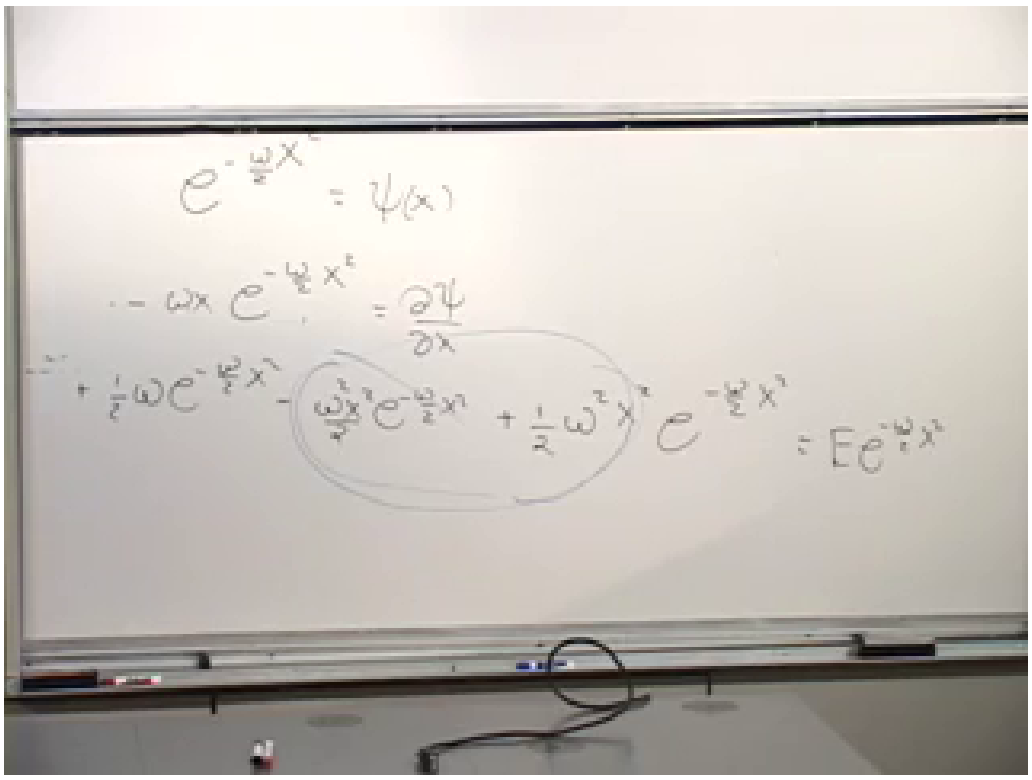


Figure 10.4: Gaussian trial state checked as an energy eigenvector.

The blackboard calculation is partly hard to read in the middle, but its logical structure is clear: differentiate, differentiate again, substitute into the Hamiltonian, and watch the x^2 terms cancel. In cleaned notation,

$$\psi_0'(x) = -\omega x\, \psi_0(x), \tag{10.24}$$

and then

$$\psi_0''(x) = (-\omega + \omega^2 x^2)\psi_0(x). \tag{10.25}$$

Now act with the Hamiltonian:

$$H\psi_0 = -\frac{1}{2}\psi_0'' + \frac{1}{2}\omega^2 x^2 \psi_0 \tag{10.26}$$

$$= -\frac{1}{2}(-\omega + \omega^2 x^2)\psi_0 + \frac{1}{2}\omega^2 x^2 \psi_0 \tag{10.27}$$

$$= \left(\frac{\omega}{2} - \frac{\omega^2 x^2}{2} + \frac{\omega^2 x^2}{2}\right)\psi_0 \tag{10.28}$$

$$= \frac{\omega}{2}\psi_0. \tag{10.29}$$

Thus the Gaussian really is an eigenfunction, and its energy is

$$E_0 = \frac{\omega}{2}. \tag{10.30}$$

This is the ground-state energy. Restoring $\hbar$ at the end gives

$$E_0 = \frac{\hbar\omega}{2}. \tag{10.31}$$

The lecture then pauses to underline the point: classically the minimum energy would be zero, but quantum mechanically the lowest state already has nonzero energy.

10.5.1 Question & Answer

Why is there no zero-energy state? This is the natural conceptual obstacle raised by the lecture, and it is answered in three mutually reinforcing ways.

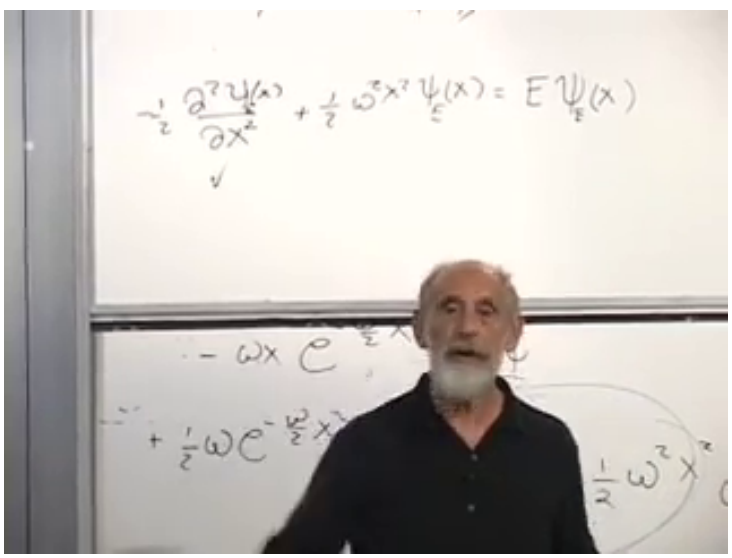


Figure 10.5: Harmonic-oscillator Schrödinger equation during the $\psi = 1$ discussion.

The equation under discussion is still

$$-\frac{1}{2}\psi''(x) + \frac{1}{2}\omega^2 x^2 \psi(x) = E\psi(x). \tag{10.32}$$

1. The first answer is the uncertainty principle. Classically we would minimize the positive-definite Hamiltonian
$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2 \tag{10.33}$$
by setting both x and p equal to zero. Quantum mechanically that is too much to ask. We can compromise, but we cannot make both vanish sharply. Therefore the lowest possible energy cannot be zero.

2. The second answer is the explicit test $\psi(x) = 1$. A constant wave function is indeed a zero-momentum state, because the derivative term vanishes on it. But it does not solve the eigenvalue equation:
$$H(1) = \frac{1}{2}\omega^2 x^2 \neq 0. \tag{10.34}$$
So $\psi = 1$ kills the kinetic term but leaves the potential term untouched. It does not represent $x = 0$; it only represents zero momentum.

3. The third answer is the asymptotic one. If we set $E = 0$, we obtain
$$-\frac{1}{2}\psi'' + \frac{1}{2}\omega^2 x^2 \psi = 0. \tag{10.35}$$
This equation does have formal solutions, but none of them is acceptable. One may tune initial conditions so that the solution decays in one direction, but then it blows up in the other. Typically it blows up in both directions. So $E = 0$ is excluded at the level of normalizable states.

The lecture ends this exchange by stating the admissibility condition plainly: the state vectors of a properly defined quantum system must be normalizable. That can be posed as an axiom, but the lecture also offers the more satisfying physical explanation: otherwise all the probability would sit infinitely far away.

10.6 Ladder Operators and the Algebraic Reformulation

After the break, the method changes. Up to this point the discussion has been in the language of differential equations. Now the lecture introduces the special algebraic trick that makes the harmonic oscillator soluble in a completely different way.

Susskind begins with the unnormalized operators
$$b_- = p - i\omega x, \qquad b_+ = p + i\omega x. \tag{10.36}$$
The first reason for writing them is the elementary identity $a^2 + b^2 = (a + ib)(a - ib)$. Classically one would simply rewrite the Hamiltonian as a product. Quantum mechanically the same manipulation is nearly right, but not quite right, because x and p do not commute.

The live lecture contains some hesitation over signs and naming. The safest way to stabilize the notation is to demand that the operator with the minus sign annihilate the ground state; that is the clue Susskind himself uses. With $[x, p] = i$ and $\hbar = 1$, we compute
$$b_+ b_- = (p + i\omega x)(p - i\omega x) \tag{10.37}$$
$$= p^2 + \omega^2 x^2 + i\omega(xp - px) \tag{10.38}$$
$$= p^2 + \omega^2 x^2 - \omega. \tag{10.39}$$

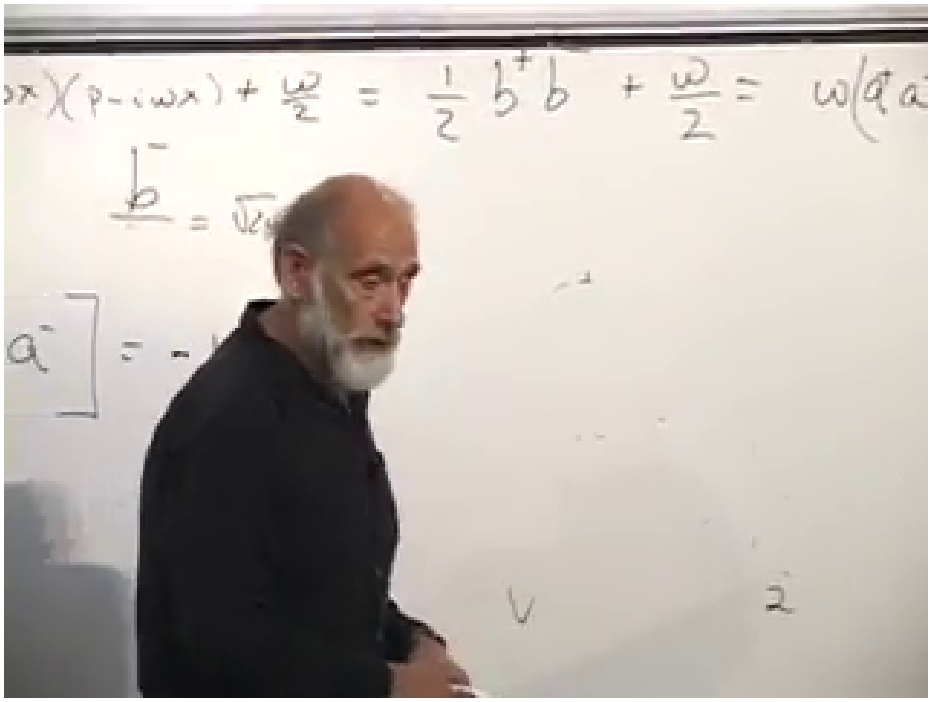


Figure 10.6: Hamiltonian in ladder-operator form.

So the Hamiltonian may be written as

$$H = \frac{1}{2}(p^2 + \omega^2 x^2) = \frac{1}{2} b_+ b_- + \frac{\omega}{2}. \tag{10.40}$$

The additional constant is precisely the zero-point energy already found by the Gaussian calculation. The lecture insists on this point: adding an ordinary number to an operator shifts all eigenvalues by the same amount and does nothing more mysterious than that.

Next we compute the commutator:

$$[b_-, b_+] = [p - i\omega x,\, p + i\omega x] \tag{10.41}$$

$$= 2\omega. \tag{10.42}$$

It is then convenient to divide by $\sqrt{2\omega}$ and introduce the cleaned operators

$$a = \frac{1}{\sqrt{2\omega}}\, b_- = \frac{1}{\sqrt{2\omega}}(p - i\omega x), \qquad a^\dagger = \frac{1}{\sqrt{2\omega}}\, b_+ = \frac{1}{\sqrt{2\omega}}(p + i\omega x). \tag{10.43}$$

By construction,

$$[a, a^\dagger] = 1. \tag{10.44}$$

Equation (10.40) becomes

$$H = \omega \left(a^\dagger a + \frac{1}{2} \right). \tag{10.45}$$

Now the ground state reveals itself algebraically. Since

$$a = \frac{1}{\sqrt{2\omega}}(p - i\omega x), \tag{10.46}$$

we find

$$a\,\psi_0 = \frac{1}{\sqrt{2\omega}}\left(-i\frac{d}{dx} - i\omega x\right)e^{-\omega x^2/2} \tag{10.47}$$

$$= -\frac{i}{\sqrt{2\omega}}\left(\frac{d}{dx} + \omega x\right)e^{-\omega x^2/2} \tag{10.48}$$

$$= 0, \tag{10.49}$$

because $\frac{d}{dx}e^{-\omega x^2/2} = -\omega x e^{-\omega x^2/2}$. So the operator a annihilates the ground state. That is why it is the lowering operator.

10.7 Spectrum, Excited States, and What the Wave Functions Mean

Once the Hamiltonian has the form (10.45), the problem becomes algebraic. Let

$$N = a^\dagger a \tag{10.50}$$

and suppose

$$N|n\rangle = n|n\rangle. \tag{10.51}$$

Then

$$N(a^\dagger|n\rangle) = a^\dagger a\, a^\dagger|n\rangle \tag{10.52}$$

$$= a^\dagger(aa^\dagger)|n\rangle \tag{10.53}$$

$$= a^\dagger(a^\dagger a + 1)|n\rangle \tag{10.54}$$

$$= (n+1)a^\dagger|n\rangle. \tag{10.55}$$

So $a^\dagger$ raises the eigenvalue of N by one. Similarly, a lowers it by one until the ground state is reached, at which point it gives zero. The whole spectrum is therefore

$$E_n = \omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{10.56}$$

or, with $\hbar$ restored,

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right). \tag{10.57}$$

This is exactly the equally spaced ladder that the lecture is aiming toward.

The first excited wave function may be found directly by acting once with the raising operator:

$$\psi_1(x) \propto a^\dagger\psi_0(x) \tag{10.58}$$

$$= \frac{1}{\sqrt{2\omega}}\left(-i\frac{d}{dx} + i\omega x\right)e^{-\omega x^2/2} \tag{10.59}$$

$$\propto x\, e^{-\omega x^2/2}. \tag{10.60}$$

Up to an overall constant and phase, the first excited state is the Gaussian multiplied by x. It has one node at the origin.

This is the pattern the lecture wants us to notice. Repeated action of $a^\dagger$ produces a Gaussian multiplied by higher and higher polynomials:

$$\psi_n(x) = P_n(x)e^{-\omega x^2/2}, \tag{10.61}$$

where $P_n(x)$ is a degree-n polynomial. The lecture does not turn this into a full Hermite-polynomial discussion, and it is better not to overstate what was actually motivated. The important qualitative points are these:

- each new excitation introduces one more node;
- the parity alternates from symmetric to antisymmetric and back again;
- the wave function oscillates more rapidly as the energy increases, indicating larger momentum content;
- the wave function also spreads farther from the origin, reflecting larger excursions in position.

The lecture adds one further observation. An equally spaced spectrum is not by itself enough to characterize the harmonic oscillator. One can invent other operators with equally spaced eigenvalues, but they may extend upward and downward without bound. What makes the oscillator special, within the scope of this lecture, is the combination of equal spacing and a spectrum bounded from below.

10.7.1 Question & Answer

What does a quantum harmonic oscillator look like? Late in the lecture, a student asks what such a system physically looks like. Susskind's answer is that this is not quite the right question. A quantum system is not something we simply stare at or photograph without disturbing it. The more meaningful question is what its probability distributions look like.

For the ground state, the probability density is Gaussian. It is concentrated near the minimum and has a width controlled by ω. The larger ω is, the tighter the restoring force, and the narrower the wave function becomes. This is exactly the interpretation Susskind gives in words: a stiffer spring holds the system more tightly to the origin.

Above the ground state, the probability density develops nodes. There are points at which the probability is exactly zero. The lecture treats this as one of the distinctly quantum features of the oscillator: the particle is still, in an appropriate sense, an oscillator, but the wave function can have exact zeros produced by interference.

This returns us to the broad motivation with which the lecture began. A quantum harmonic oscillator is not a tiny visible spring. It is the local quantum description of motion near a stable minimum. A diatomic molecule is a good example. More generally, whenever a system has a stable configuration and we study small deviations from it, the potential near the minimum is approximately quadratic, and the quantized motion is approximately that of the harmonic oscillator.

10.8 Summary

The lecture develops the harmonic oscillator in the same order in which one would discover it if one were being careful. We begin with a concrete spring and a classical Lagrangian, shift the

coordinate to equilibrium, scale out the mass, and arrive at $\ddot{x} = -\omega^2 x$. We then quantize the system by specifying the space of states, representing x and p as operators, and writing the Schrödinger equation. The discrete spectrum appears because acceptable wave functions must be normalizable.

The first explicit payoff is the Gaussian ground state $\psi_0(x) \propto e^{-\omega x^2/2}$, with zero-point energy $E_0 = \hbar\omega/2$. The second is the algebraic reformulation $H = \omega(a^\dagger a + \frac{1}{2})$, from which the whole spectrum follows:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right). \tag{10.62}$$

The larger lesson is the same one with which the lecture began: near stability, the world is locally quadratic, and therefore locally harmonic. Quantizing the oscillator is not the study of a single mechanical contrivance. It is the study of the quantum mechanics of small oscillations wherever nature admits them.